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S. R. S. VARADHAN

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Stochastic Processes

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Stochastic Processes

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Dedication

To Gopal

I had planned to complete this book within a short time of the publication of the volume on probability theory. But the events of September 11, 2001, intervened. We lost our son Gopal that day, a victim of violence in the name of God. I dedicate this volume to his memory.

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Preface

This is a continuation of the volume on probability theory and likewise covers the contents of courses given at the Courant Institute. This volume deals with certain elementary continuous-time processes. We start with a description of the Poisson process and related processes with independent increments. After a brief look at Markov processes with a finite number of jumps we proceed to study Brownian motion. We then go on to develop stochastic integrals and Itô's theory in the context of one-dimensional diffusion processes. It ends with a brief survey of the general theory of Markov processes.

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CHAPTER 1

Introduction

1.1. Continuous Time Processes

A stochastic process $\{x(t) : t \in \mathbb{T}\}$ is a collection $x(t) = x(t, \omega)$ of random variables defined on some probability space $(\Omega, \mathcal{F}, P)$ indexed by $t \in \mathbb{T}$. The random variable $x(t)$ can take its values in an arbitrary measurable space which is often $\mathbb{R}$ or $\mathbb{R}^d$ with its respective Borel σ -fields $\mathcal{B}(\mathbb{R})$ or $\mathcal{B}(\mathbb{R}^d)$. The variable t usually represents time. When $\mathbb{T}$ is a countable set, for instance the set of integers or a subset of it, we call the process a discrete parameter or a discrete time process. In case $\mathbb{T}$ is a finite or infinite interval in $\mathbb{R}$, it is called a continuous-time process.

There is one essential difference between the two cases. In either case, to begin with, we only know that $x(t, \cdot)$ is measurable for each t . If, for example, $x(t, \omega)$ is real valued and $\mathbb{T}$ is countable this automatically implies the measurability of sets of the form

$$(1.1) \quad A = \{\omega : x(t, \omega) \in [a, b] \forall t \in \mathbb{T}\}$$

If $\mathbb{T}$ is uncountable this is not necessarily so and for good reason. Random variables $x(t)$ are objects that are only defined almost everywhere and we are free to change them on a set of measure zero. If we do change $x(t)$ on a set $E_t \subset \Omega$ with $P(E_t) = 0$, events like (1.1) that depend on $x(t)$ for all $t \in \mathbb{T}$, could be changed on $E = \bigcup_{t \in \mathbb{T}} E_t$. If $\mathbb{T}$ is uncountable $P(E)$ need not be a set of measure zero and therefore for the set A given by (1.1), $P(A)$ may no longer be uniquely defined. Since we are often interested in calculating probabilities of events like (1.1), we need to fix this.

Let us assume that $x(t, \omega)$ is real valued. One can think of $x(t, \omega)$ as a map of $\Omega \rightarrow \mathcal{X}$ where $\mathcal{X} = \prod_{t \in \mathbb{T}} \mathbb{R}$ is the space of $\mathbb{R}$ -valued functions on $\mathbb{T}$. On $\mathcal{X}$ there is the natural σ -field $\mathcal{K}$, generated by sets of the form $\{x(\cdot) : x(t) \in A\}$ with $t \in \mathbb{T}$ and $A \in \mathcal{B}(\mathbb{R}^d)$. It is not hard to see that if $x(t, \omega)$ is measurable for each $t \in \mathbb{T}$, then $\omega \rightarrow x(\cdot, \omega)$ is a measurable map of $(\Omega, \mathcal{F}) \rightarrow (\mathcal{X}, \mathcal{K})$. Hence there is an induced measure Q on $(\mathcal{X}, \mathcal{K})$. Such measures Q are uniquely determined by the collection $\{Q_{t_1, t_2, \dots, t_n}\}$ of joint distributions on $\mathbb{R}^n$ of $(x(t_1), x(t_2), \dots, x(t_n))$ as n varies over all positive integers and $(t_1, t_2, \dots, t_n)$ varies over all subsets of $\mathbb{T}$ of cardinality n .

The construction of a probability measure Q on $(\mathcal{X}, \mathcal{K})$ from a consistent family of finite-dimensional distributions $\{Q_{t_1, t_2, \dots, t_n}\}$ when $\mathbb{T}$ is uncountable is no different from the situation when $\mathbb{T}$ is countable. After all, the crucial step is to check countable additivity on the field of finite-dimensional cylinder sets with a measurable base. If $\{A_n\}$ is from this field then each A_n , being a finite-dimensional

cylinder set, depends only on the values of $x(t)$ with t from a finite subset $F_n \subset \mathbb{T}$. We can always replace $\mathbb{T}$ by the set $\mathbb{T}_0 = \bigcup_n F_n$. Since $\mathbb{T}_0$ is countable, checking the countable additivity for the sequence A_n reduces to the countable or discrete-parameter case proved in [7].

While in the countable case events like (1.1) are always in $\mathcal{K}$, when $\mathbb{T}$ is uncountable this is not necessarily so. One has to cut down the space Ω to a suitable subset Ω_0 , such that when restricted to Ω_0 these sets are measurable, i.e. belong to the natural restriction of $\mathcal{K}$ to Ω_0 . When $\mathbb{T}$ is a subset of $\mathbb{R}$, if Ω_0 were to consist of all continuous maps from $\mathbb{T} \rightarrow \mathbb{R}$, then events like (1.1) are clearly representable as countable intersections by restricting t to the rationals in $\mathbb{T}$. According to standard results in measure theory, even when $\Omega_0 \subset \Omega$ is not in $\mathcal{K}$, if the outer measure, defined for arbitrary sets E by

$$P^*(E) = \inf_{\substack{A \supset E \\ A \in \mathcal{K}}} P(A)$$

satisfies $P^*(\Omega_0) = 1$, then the restriction of P^* to the σ -field $\mathcal{K}_0$ of subsets of Ω_0 which are intersections of sets in $\mathcal{K}$ with Ω_0 , i.e. sets of the form $A \cap \Omega_0$ with $A \in \mathcal{K}$, is a countably additive probability measure. With a suitable choice of Ω_0 this could provide a more convenient model $(\Omega_0, \mathcal{K}_0, P^*)$ with the same finite-dimensional distributions for $\{x(t, \omega)\}$ as the original one. The condition $P^*(\Omega_0) = 1$ is to be interpreted as P being supported on Ω_0 is not inconsistent with the given collection of finite-dimensional distributions. Of course it is often not inconsistent with P being supported on Ω_0^c either. We have the option to choose a ‘good’ model or a ‘bad’ one. We will now prove the result concerning restriction of the outer measure to full sets.

Let $(\Omega, \mathcal{F}, P)$ be a probability space. We can define for any $E \subset \Omega$ the outer measure

$$P^*(E) = \inf_{\substack{B \supset E \\ B \in \mathcal{F}}} P(B)$$

It is obvious that if $A \in \mathcal{F}$, then $P^*(A) = P(A)$. Let $\Omega_0 \subset \Omega$. Consider sets of the form $\Omega_0 \cap A$ where $A \in \mathcal{F}$. They constitute a σ -field $\mathcal{F}_0$ of subsets of Ω_0 .

THEOREM 1.1 *If $P^*(\Omega_0) = 1$, then the restriction of P^* to $\mathcal{F}_0$ is a countably additive probability measure such that $P^*(\Omega_0 \cap A) = P(A)$ for $\forall A \in \mathcal{F}$.*

PROOF: Outer measures are always countably sub-additive. This is easily seen. Let $E = \bigcup_j E_j$. Let $\epsilon > 0$ be given. By definition we can find $B_j \in \mathcal{F}$ such that $B_j \supset E_j$ and $P(B_j) \leq P^*(E_j) + \frac{\epsilon}{2^j}$. Clearly $B = \bigcup_j B_j \supset \bigcup_j E_j = E$ and

$$P^*(E) \leq P(B) \leq \sum_j P(B_j) \leq \sum_j P^*(E_j) + \sum_j \frac{\epsilon}{2^j} = \sum_j P^*(E_j) + \epsilon$$

Since $\epsilon > 0$ is arbitrary the assertion follows. In order to prove that P^* is countably additive on $\mathcal{F}_0$, we need to prove that if $A_1, A_2 \in \mathcal{F}$ are such that $A_1 \cap \Omega_0$ and $A_2 \cap \Omega_0$ are disjoint then

$$P^*[(\Omega_0 \cap A_1) \cup (\Omega_0 \cap A_2)] = P^*[\Omega_0 \cap A_1] + P^*[\Omega_0 \cap A_2]$$

which amounts to proving

$$(1.2) \quad P(A_1 \cup A_2) = P(A_1) + P(A_2)$$

Since $\Omega_0 \cap A_1$ and $\Omega_0 \cap A_2$ are disjoint, $A_1 \cap A_2 \subset \Omega_0^c$ and $P(A_1 \cap A_2) = 0$. It follows that $P(A_1 \cup A_2) = P(A_1) + P(A_2)$, i.e. (1.2) holds. $\square$

We will not pursue this but will adopt a slightly different point of view. Most of the processes we deal with will be almost surely either continuous or continuous except for jumps. We will therefore find it more convenient to put the probability measure directly on a nice space consisting of such functions. The space $C[a, b]$ will consist of all continuous functions on the interval $[a, b]$. On the other hand the space $D[a, b]$ will consist of functions that have left and right limits at every point. These functions $x(t)$ are normalized so that the value at any t is equal to the limit $x(t + 0)$ from the right, except when the point t is the right end point b , in which case the value is equal to the left limit $x(b - 0)$. Since these functions are all determined by their values on rationals, there are no problems regarding the measurability of events like (1.1).

The spaces $C[a, b]$ and $D[a, b]$ can be provided with metrics turning them into nice Polish (i.e. complete separable metric) spaces. While $C[a, b]$ is a standard Banach space with its norm $\|f\| = \sup_{t \in [a, b]} |f(t)|$, the metric for $D[a, b]$ is unusual and the corresponding topology is known as the *Skorohod topology*. See [5] for a description and properties of $D[a, b]$ under this metric. The σ -field of Borel sets in either case is generated by $\{x(t) : t \in [a, b]\}$. Of course the same σ -field is generated by $x(t)$ with t restricted to $\mathbb{Q} \cap [a, b]$, the rationals in $[a, b]$.

1.2. Continuous Parameter Martingales

Martingales that depend on a continuous parameter will play an important role in the study of continuous-parameter Markov processes just as discrete-parameter martingales play a very useful role in the theory of Markov chains. We start with some definitions.

DEFINITION 1.1 $(\Omega, \mathcal{F})$ is a space with a σ -field. A family $\{\mathcal{F}_t : t \geq 0\}$ of sub- σ -fields of $\mathcal{F}$ is called a *filtration* if $\mathcal{F}_s \subset \mathcal{F}_t$ if $s \leq t$.

If Ω is a space of functions on $\mathbb{T} \subset \mathbb{R}^+$, it comes with a natural filtration $\mathcal{F}_t = \sigma\{x(s) : s \leq t\}$.

DEFINITION 1.2 Given a probability space $(\Omega, \mathcal{F}, P)$ and a filtration $\mathcal{F}_t \subset \mathcal{F}$ indexed by $t \geq 0$, a family $M(t) = M(t, \omega)$ of random variables is called a *martingale relative to* $(\Omega, \mathcal{F}_t, P)$ if the following are true.

- (i) For almost all ω , $M(t, \omega)$ has left and right limits at every t and is continuous from the right, i.e. $M(t, \omega) = M(t + 0, \omega)$
- (ii) For each $t \geq 0$, $M(t)$ is $\mathcal{F}_t$ measurable and integrable.
- (iii) For $0 \leq s \leq t$,

$$(1.3) \quad E\{M(t) \mid \mathcal{F}_s\} = M(s) \text{ a.e.}$$

Most of the results proved for discrete-parameter martingales extend to the continuous-parameter case. One notes that a continuous-parameter martingale $M(t)$ evaluated at increasing times $\{t_j\}$ yields a discrete-parameter martingale $X_j = M(t_j)$ with respect to the σ -fields $\{\mathcal{F}_{t_j}\}$.

EXERCISE 1.1. Extend to continuous-parameter martingales Doob's inequality. That is, if $M(t)$ is a martingale and E_ℓ is the event

$$E_\ell = \left\{ \omega : \sup_{0 \leq s \leq t} |M(s, \omega)| \geq \ell \right\}$$

then,

$$(1.4) \quad P[E_\ell] \leq \frac{1}{\ell} \int_{E_\ell} |M(t)| dP \leq \frac{1}{\ell} \int |M(t)| dP$$

$$(1.5) \quad P[E_\ell] \leq \frac{1}{\ell^2} \int_{E_\ell} |M(t)|^2 dP \leq \frac{1}{\ell^2} \int |M(t)|^2 dP$$

DEFINITION 1.3 Let $\mathcal{F}_t$ be a filtration and $x(t)$ a right continuous process with left limits, defined for t in $\mathbb{R}^+$ or an interval $[0, T]$. The process $x(t)$ is assumed to be $\mathcal{F}_t$ measurable and integrable for each t . It is called a *submartingale* if it satisfies

$$(1.6) \quad E[x(t) \mid \mathcal{F}_s] \geq x(s) \text{ a.e.}$$

and a *supermartingale* if it satisfies

$$(1.7) \quad E[x(t) \mid \mathcal{F}_s] \leq x(s) \text{ a.e.}$$

EXERCISE 1.2. Show that if $M(t)$ is a martingale with respect to a filtration then $|M(t)|$ is a submartingale with respect to the same filtration. In fact, if $\phi(x)$ is a convex function of x and $\phi(M(T))$ is integrable, then $\phi(M(t))$ is integrable for $0 \leq t \leq T$ and is a submartingale.

We can define stopping times and their associated σ -fields in a manner analogous to the discrete case.

DEFINITION 1.4 A *stopping time* for the filtration $\{\mathcal{F}_t\}$ is a random variable $\tau(\omega) : \Omega \rightarrow [0, +\infty]$ that satisfies for every $t \geq 0$,

$$(1.8) \quad E_t = \{\omega : \tau(\omega) \leq t\} \in \mathcal{F}_t$$

DEFINITION 1.5 Given a stopping time τ , the corresponding σ -field $\mathcal{F}_\tau$ is defined as consisting of those sets $A \in \mathcal{F}$ that satisfy for each $t \geq 0$,

$$(1.9) \quad A \cap \{\omega : \tau(\omega) \leq t\} \in \mathcal{F}_t$$

The following properties of stopping times are left as exercises.

EXERCISE 1.3. If τ_1 and τ_2 are stopping times so are $\tau_1 \wedge \tau_2$ and $\tau_1 \vee \tau_2$.

EXERCISE 1.4. If $f(\cdot)$ is a measurable function and $f(t) \geq t$ for all t , then for any stopping time τ , $f(\tau)$ is a stopping time as well. In particular, if τ is a stopping time so is $\tau_k = \frac{[k\tau]+1}{k}$ for any positive integer k , where $[y]$ is the largest integer not exceeding y . This shows that any bounded stopping time τ can be approximated

by a sequence $\{\tau_k\}$ of stopping times with $\tau_k \geq \tau$ and each τ_k taking only a finite set of values.

EXERCISE 1.5. If $0 \leq \tau_1 \leq \tau_2$ are two stopping times then $\mathcal{F}_{\tau_1} \subset \mathcal{F}_{\tau_2}$.

We can extend Doob's optional stopping theorem to continuous-time martingales.

THEOREM 1.2 *If $0 \leq \tau_1 \leq \tau_2 \leq C$ are two bounded stopping times and $M(t)$ is a martingale then*

$$(1.10) \quad E[M(\tau_2) \mid \mathcal{F}_{\tau_1}] = M(\tau_1) \text{ a.e.}$$

First we establish a lemma that we will need.

LEMMA 1.3 *$X(\omega)$ is an integrable random variable on a probability space $(\Omega, \mathcal{F}, P)$ and $X_\Sigma = E[X \mid \Sigma]$ where $\Sigma \subset \mathcal{F}$ is a sub- σ -field of $\mathcal{F}$. The collection $\{X_\Sigma\}$ as Σ varies over all sub- σ -fields of $\mathcal{F}$ is uniformly integrable.*

PROOF: From Jensen's inequality

$$E[|X| \mid \Sigma] \geq |X_\Sigma| \text{ a.e.}$$

and therefore

$$P[|X_\Sigma| \geq \ell] \leq \frac{E[|X_\Sigma|]}{\ell} \leq \frac{E[|X|]}{\ell}$$

On the other hand the set $\{\omega : X_\Sigma(\omega) \geq \ell\} \in \Sigma$ and

$$\int_{|X_\Sigma| \geq \ell} |X_\Sigma| dP \leq \int_{|X_\Sigma| \geq \ell} |X| dP$$

can be made small if $P[|X_\Sigma| \geq \ell]$ is small, because $|X|$ is integrable. $\square$

PROOF: (of Theorem 1.2). It is enough to prove that for any stopping time τ bounded by C

$$(1.11) \quad E[M(C) \mid \mathcal{F}_\tau] = M(\tau)$$

It would then follow that

$$E[M(\tau_2) \mid \mathcal{F}_{\tau_1}] = E[E[M(C) \mid \mathcal{F}_{\tau_2}] \mid \mathcal{F}_{\tau_1}] = E[M(C) \mid \mathcal{F}_{\tau_1}] = M(\tau_1)$$

To prove (1.11) we note that it suffices to prove it for some approximating sequence $\{\tau_k\}$ that satisfies $\tau_k \downarrow \tau$ and $\tau_k \leq C$. Then if $A \in \mathcal{F}_\tau$, it will also be true that $A \in \mathcal{F}_{\tau_k}$ and

$$\int_A M(\tau_k) dP = \int_A M(C) dP$$

We can let $k \rightarrow \infty$. By lemma 1.3 $\{M(\tau_k)\}$ is uniformly integrable and it follows that

$$\int_A M(\tau) dP = \int_A M(C) dP$$

We can take $\tau_k = \frac{[k\tau]+1}{k}$ where $[x]$ is the largest integer not exceeding x . Each stopping time τ_k in our approximating sequence takes only a finite set of possible

values. For τ that takes only a finite set $t_1 < t_2 < \dots < t_k$ of values, if $E_i = \{\omega : \tau(\omega) = t_i\}$ and $A \in \mathcal{F}_\tau$, then $A \cap E_i \in \mathcal{F}_{t_i}$ and

$$\int_{A \cap E_i} M(\tau) dP = \int_{A \cap E_i} M(t_i) dP = \int_{A \cap E_i} M(C) dP$$

Summing over i gives

$$\int_A M(\tau) dP = \int_A M(C) dP$$

□

COROLLARY 1.4 *In particular, if τ is any stopping time and $M(t)$ is a martingale with respect to $(\Omega, \mathcal{F}_t, P)$, then so is $M(\tau \wedge t)$.*

REMARK 1.1. Often it is necessary to apply Doob's optional stopping theorem to unbounded stopping times. We can always approximate τ by $\tau \wedge T$ which is bounded and then let $T \rightarrow \infty$. It is now a question of establishing the uniform integrability of $\{M(\tau \wedge T)\}$. If we have a martingale that is uniformly bounded this is not a problem.

REMARK 1.2. In a similar manner one can establish analogues of the upcrossing inequality and martingale convergence theorem for the continuous case without much additional work.

Doob's optional stopping theorem works for sub and supermartingales too.

THEOREM 1.5 *If $M(t)$ is a submartingale and $\tau_1 \leq \tau_2 \leq C$ are two bounded stopping times then*

$$(1.12) \quad E[M(\tau_2) \mid \mathcal{F}_{\tau_1}] \geq M(\tau_1)$$

The proof of Theorem 1.2 will not work. It will only show that

$$E[M(\tau_C) \mid \mathcal{F}_{\tau_i}] \geq M(\tau_i)$$

for $i = 1, 2$, which will not do.

Actually a slightly different proof of Theorem 1.2 is more useful. Suppose $\tau_1 \leq \tau_2 \leq C$ are two bounded stopping times that take only a finite set $t_1 < t_2 < \dots < t_k$ of values. Let us denote by $E_i = \{\omega : \tau_1 = t_i\}$ and $F_i = \{\omega : \tau_2 = t_i\}$. It is clear that $E_i \cap F_j = \emptyset$ unless $i \leq j$. We can then write E_i as $E_i = \bigcup_{j \geq i} (E_i \cap F_j)$. Now if $A \in \mathcal{F}_{\tau_1}$, for any i we begin by writing E_i as the disjoint union of $E_i \cap F_i$ and $E_i \cap F_i^c$.

$$\begin{aligned} \int_{A \cap E_i} M(\tau_1) dP &= \int_{A \cap E_i} M(t_i) dP \\ &= \int_{A \cap E_i \cap F_i} M(t_i) dP + \int_{A \cap E_i \cap F_i^c} M(t_i) dP \\ &= \int_{A \cap E_i \cap F_i} M(t_i) dP + \int_{A \cap E_i \cap F_i^c} M(t_{i+1}) dP \end{aligned}$$

Here we have used the fact that $A \cap E_i \cap F_i^c \in \mathcal{F}_{t_i}$. The process can be repeated to yield

$$\int_{A \cap E_i} M(\tau_1) dP = \sum_{j \geq i} \int_{A \cap E_i \cap F_j} M(t_j) dP = \int_{A \cap E_i} M(\tau_2) dP$$

and summing over i we get

$$\int_A M(\tau_1) dP = \int_A M(\tau_2) dP$$

or

$$E[M(\tau_2) \mid \mathcal{F}_{\tau_1}] = M(\tau_1) \text{ a.e.}$$

The advantage of this proof is that if $M(t)$ were a submartingale instead of a martingale we would have the inequality at every step and end up with

$$E[M(\tau_2) \mid \mathcal{F}_{\tau_1}] \geq M(\tau_1) \text{ a.e.}$$

for stopping times $\tau_1 \leq \tau_2$ that take only a finite set of values. It is then easy to extend this inequality to arbitrary bounded stopping times and thereby prove Theorem 1.5.

PROOF: If $\tau_1 \leq \tau_2 \leq C$ are bounded stopping times, replacing τ_2 by $\tau_2 + \delta$ one can assume that we have approximations $\tau_{1,r}, \tau_{2,k}$ with the following properties: for any two sufficiently large choices of k and r we have $\tau_{2,k} \geq \tau_{1,r}$ and $\tau_{1,r}, \tau_{2,k}$ are nonincreasing and tend to τ_1, τ_2 , respectively. From the inequality

$$E[M(\tau_{2,k}) \mid \mathcal{F}_{\tau_{1,r}}] \geq M(\tau_{1,r}) \text{ a.e.}$$

by letting $r \rightarrow \infty$ while keeping k fixed, using the standard martingale convergence theorem, we can deduce that

$$E[M(\tau_{2,k}) \mid \mathcal{F}_{\tau_1}] \geq M(\tau_1) \text{ a.e.}$$

In order to let $k \rightarrow \infty$ and conclude that

$$E[M(\tau_2) \mid \mathcal{F}_{\tau_1}] \geq M(\tau_1) \text{ a.e.}$$

we will need the uniform integrability of the positive parts $\{M(\tau_{2,k})^+\}$ which would imply that for $A \in \mathcal{F}_{\tau_1}$,

$$\int_A M(\tau_2) dP \geq \limsup_{k \rightarrow \infty} \int_A M(\tau_{2,k}) dP \geq \int_A M(\tau_1) dP$$

This uniform integrability follows from the inequality

$$M(\tau_{2,k}) \leq E[M(C) \mid \mathcal{F}_{\tau_{2,k}}]$$

and Lemma 1.3. The positive parts of $M(\tau_{2,k})$ are dominated by a uniformly integrable family and hence they are themselves uniformly integrable. A similar argument allows us to let $\delta \rightarrow 0$ at the end. $\square$

1.3. Semimartingales

Let $(\Omega, \mathcal{F}, P)$ be a probability space and $\mathcal{F}_j$ a filtration. If $\{X_j : j \geq 0\}$ is any discrete-parameter process, X_j being $\mathcal{F}_j$ measurable and integrable, we can define

$$a_j(\omega) = E[X_j - X_{j-1} \mid \mathcal{F}_{j-1}]$$

and for $n \geq 1$

$$A_n(\omega) = \sum_{j=1}^n a_j(\omega)$$

with $A_0(\omega) = 0$. It is easy to verify that

$$Y_n = X_n - A_n$$

is $\mathcal{F}_n$ measurable and is a martingale. The process X_n is called a semimartingale (any process of integrable random variables is a semimartingale if time is discrete) and A_n is called the compensator. The decomposition

$$X_n = Y_n + A_n$$

is called the Doob decomposition. Note that A_n is $\mathcal{F}_{n-1}$ measurable. This is important. Otherwise one can take $A_n = X_n$ and $Y_n = 0$. If X_n is a submartingale, then $a_n \geq 0$ or A_n is increasing. For supermartingales X_n we will have $a_n \leq 0$ and A_n decreasing.

EXERCISE 1.6. Let $(\Omega, \mathcal{F}_j, P)$ be a filtration and $\{X_j : j \geq 0\}$ any integrable $\mathcal{F}_j$ measurable process. Show that a decomposition of the form $X_j = Y_j + A_j$, with $A_0 = 0$ and A_j being $\mathcal{F}_{j-1}$ measurable, is unique.

The continuous analogue of this result which is called the Doob-Meyer decomposition has several problems. Most but not all submartingales and supermartingales can be written as $X(t) = M(t) + A(t)$, with $A(t)$ being either increasing or decreasing. The natural definition of a semimartingale would be that the process can be written as $X(t) = M(t) + A(t)$ where $A(t)$ is a process which is of bounded variation. But in the discrete case A_n was $\mathcal{F}_{n-1}$ measurable and the difference was important. While it is natural for $A(t)$ to be $\mathcal{F}_t$ measurable what is the analogue of $\mathcal{F}_{n-1}$ in the continuous case?

This is not a simple question and we will not try to answer it here. However if we can find a decomposition of the form $X(t) = M(t) + A(t)$, with all processes adapted to the filtration $(\Omega, \mathcal{F}_t, P)$, where $M(t)$ is a martingale and $A(t)$ is an almost surely *continuous process of bounded variation*, then such a decomposition is unique. A continuous version of the compensator may not always exist. But, if it exists, it is unique. Proving uniqueness is essentially the following lemma.

LEMMA 1.6 *If $\{A(t) : t \in [0, T]\}$ is continuous-parameter martingale satisfying*

- (i) *It is almost surely continuous,*
- (ii) *it is almost surely of bounded variation, and*
- (iii) $A(0) = 0$,

then $A(t) \equiv 0$.

PROOF: Without loss of generality we can assume that the total variation of $A(t)$ is uniformly bounded by a constant C . Otherwise, if $A^*(t)$ is the variation of $A(\cdot)$ on $[0, t]$, $A^*(t)$ is continuous and we can define the stopping time

$$\tau = \inf\{t : A^*(t) \geq C\}$$

and by Corollary 1.4, $A(\tau \wedge t)$ would be a continuous martingale with total variation bounded uniformly by C . This would make $A(\tau \wedge t) \equiv 0$ and we can let $C \rightarrow \infty$ to establish that $A(t) \equiv 0$.

If the total variation of $A(\cdot)$ is bounded by C on $[0, T]$, we divide $[0, T]$ into n equal parts $0 = t_0 < t_1 < \dots < t_n = T$, with $t_j = \frac{j}{n}T$.

$$\begin{aligned} \sum_{j=1}^n [A(t_j) - A(t_{j-1})]^2 &\leq \left[\sup_{1 \leq j \leq n} |A(t_j) - A(t_{j-1})| \right] \left[\sum_{j=1}^n |A(t_j) - A(t_{j-1})| \right] \\ &\leq C \left[\sup_{1 \leq j \leq n} |A(t_j) - A(t_{j-1})| \right] \end{aligned}$$

We note that $A(\cdot)$ is bounded by C and continuous on $[0, T]$. Taking limits as $n \rightarrow \infty$ we conclude

$$\lim_{n \rightarrow \infty} E \left[\sum_{j=1}^n [A(t_j) - A(t_{j-1})]^2 \right] = 0$$

On the other, because $A(\cdot)$ is a martingale, for any n ,

$$\begin{aligned} E[A^2(T)] &= E \left[\sum_{j=1}^n [A^2(t_j) - A^2(t_{j-1})] \right] \\ &= E \left[\sum_{j=1}^n [A(t_j) + A(t_{j-1})][A(t_j) - A(t_{j-1})] \right] \\ &= E \left[2 \sum_{j=1}^n A(t_{j-1})[A(t_j) - A(t_{j-1})] \right] \\ &\quad + E \left[\sum_{j=1}^n [A(t_j) + A(t_{j-1})][A(t_j) - A(t_{j-1})] \right] \\ &= E \left[\sum_{j=1}^n [A(t_j) - A(t_{j-1})]^2 \right] \end{aligned}$$

If we let $n \rightarrow \infty$ we conclude that $E[A^2(T)] = 0$. □

REMARK 1.3. There are martingales $M(t)$ that are almost surely of bounded variation. They need not be constant if they are not almost surely continuous. The natural decomposition for them of course is to take the compensator as 0. We could take $M(t)$ itself as compensator and we need to rule it out. Continuity will rule it out. While compensators need not be continuous, since they are of bounded variation they can have simple jumps. The time of first jump of a certain size both for a

compensator and a martingale are stopping times. But they are qualitatively different. The jump times for martingales are like accidents without warning. You only notice them after they have occurred. They are never foreseen. On the other hand jump times of compensators while still random and unpredictable are nevertheless accessible in the sense that one can see them about to happen like an inevitable accident on the street.

While we will not appeal to the existence of the decomposition, we will see many instances of it. The semimartingales may have jumps, but the compensators will be always almost surely continuous functions of bounded variation. The following lemma is often useful.

LEMMA 1.7 *Let $M(t)$ be a martingale with respect to $(\Omega, \mathcal{F}_t, P)$. Let $A(t)$ be an adapted process which is almost surely a continuous function of bounded variation of t in any interval $[0, T]$ with a uniform bound $C(T)$ on the total variation. Then*

$$X(t) = M(t)A(t) - M(0)A(0) - \int_0^t M(s)dA(s)$$

is a martingale with respect to $(\Omega, \mathcal{F}_t, P)$

PROOF: Since $A(\cdot)$ has no jumps we can approximate

$$\begin{aligned} \int_s^t M(\sigma)dA(\sigma) &= \lim_{n \rightarrow \infty} \sum_{j=1}^n M(t_j)[A(t_j) - A(t_{j-1})] \\ &= M(t)A(t) - M(s)A(s) \\ &\quad - \lim_{n \rightarrow \infty} \sum_{j=1}^n A(t_{j-1})[M(t_j) - M(t_{j-1})] \end{aligned}$$

where $s = t_0 < t_1 < \dots < t_n = t$ is a partition of $[s, t]$ into n equal parts. The rest is an elementary calculation if we use

$$E[A(t_{j-1})[M(t_j) - M(t_{j-1})] \mid \mathcal{F}_{t_{j-1}}] = 0$$

□

The simple intuition behind the lemma is that

$$dX(t) = d[M(t)A(t)] = M(t)dA(t) + A(t)dM(t)$$

and $\int_0^t A(s)dM(s)$ is a martingale.

1.4. Martingales and Stochastic Integrals

If $\{X_j : j \geq 1\}$ is a sequence of independent random variables on $(\Omega, \mathcal{F}, P)$ with mean 0 and variance $\{\sigma_j^2\}$, in order to sum the series

$$S = \sum_j a_j X_j$$

in $L_2(P)$, where $\{a_j\}$ is a sequence of constants, we only need the summability of the series $\sum_j |a_j|^2 \sigma_j^2$. Then S will be well defined as a square integrable random

variable with mean 0 and variance $\sigma^2 = \sum_j |a_j|^2 \sigma_j^2$. The infinite sum S will be the limit of the partial sums S_n , in mean square and by a theorem of Kolmogorov almost surely as well. In fact, $\{a_j\}$ do not have to be constants. They can be random variables. While a_0 has to be a constant, for $j \geq 1$, a_j can be a function of $X_1, X_2, \dots, X_j$. The partial sums $S_0 = 0$ and for $n \geq 1$,

$$S_n = \sum_{j=0}^{n-1} a_j(X_1, X_2, \dots, X_j) X_{j+1}$$

define a martingale with respect to the filtration $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$. A simple calculation yields

$$E[S_n^2] = E\left[\sum_{j=0}^{n-1} \sigma_{j+1}^2 |a_j(\omega)|^2\right]$$

The convergence of the series

$$\sum_{j=1}^{\infty} \sigma_{j+1}^2 E[|a_j(\omega)|^2]$$

is enough to guarantee both mean square and almost sure convergence of the series

$$S = \sum_{j=0}^{\infty} a_j(X_1, X_2, \dots, X_j) X_{j+1}$$

In fact one does not need the independence of $\{X_j\}$. It is enough if

$$M_n = \sum_{j=1}^n X_j$$

is a martingale. We state all of this as a theorem. The proof is elementary and will be omitted.

THEOREM 1.8 *Let $\{\mathcal{F}_n : n \geq 0\}$ be a filtration on some probability space $(\Omega, \mathcal{F}, P)$ and M_n a square integrable martingale with respect to this filtration, with $M_0 = 0$. Let $\sigma_n^2(\omega) = E[(M_{n+1} - M_n)^2 | \mathcal{F}_n]$. Let $\{a_n(\omega)\}$ be a sequence with a_n being measurable with respect to $\mathcal{F}_n$ and satisfying $E[|a_n(\omega)|^2 \sigma_n^2(\omega)] < \infty$. Then $\{S_n : n \geq 0\}$, defined by $S_0 = 0$, and*

$$S_n = \sum_{j=0}^{n-1} a_j(\omega) [M_{j+1} - M_j]$$

for $n \geq 1$, is again a square integrable martingale with

$$E[S_n^2] = E\left[\sum_{j=0}^{n-1} |a_j(\omega)|^2 \sigma_j^2(\omega)\right]$$

There is of course an analogue in the continuous case. Let $M(t)$ be a square integrable martingale. By Jensen's inequality $M^2(t)$ is a submartingale. We assume that there is a continuous increasing function $A(t, \omega)$ such that

$$M^2(t) - A(t)$$

is a martingale with respect to $(\Omega, \mathcal{F}_t, P)$. We assume that $E[M(t)^2] = E[A(t)] < \infty$ for every t . Our aim is to define the stochastic integral

$$X(t) = \int_0^t c(s, \omega) dM(s)$$

as another square integrable martingale in a manner quite parallel to Theorem 1.8. Let us assume that $c(s, \omega)$ is piece-wise constant i.e. $c(s, \omega) = c_i(\omega)$ for $s \in [t_i, t_{i+1})$ with $c_i(\omega)$ being uniformly bounded and $\mathcal{F}_{t_i}$ measurable. Then it is not hard to see that the natural definition of the stochastic integral

$$X(t) = \int_0^t c(s, \omega) dM(s)$$

is

$$(1.13) \quad X(t) = \sum_{i=0}^{k-1} c(t_i, \omega)[M(t_{i+1}) - M(t_i)] + c(t_k, \omega)[M(t) - M(t_k)]$$

for $t_k \leq t < t_{k+1}$ and with this definition $X(t)$ is again a martingale. Moreover

$$X^2(t) - \int_0^t c^2(s, \omega) dA(s)$$

is a martingale as well. We can complete the space of integrands $\{c(s, \omega)\}$ with respect to the norms

$$E \left[\int_0^t |c(s, \omega)|^2 dA(s) \right]$$

and the square integrable martingales

$$X(t) = \int_0^t c(s, \omega) dM(s)$$

can be defined for $c(\cdot, \cdot)$ in the completion.

$$X^2(t) - \int_0^t |c(s, \omega)|^2 dA(s)$$

will again be a martingale. If the original martingale $M(t)$ is almost surely continuous in t so will $X(t)$ be. This is obvious for c that are piecewise constant. For general c it follows from Doob's inequality

$$E \left[\sup_{0 \leq t \leq T} |X(t)|^2 \right] \leq 4E[|X(T)|^2]$$

which ensures that the convergence of the martingales is almost surely uniform in t . The question of describing the class of functions $c(s, \omega)$ that can be approximated in this manner remains to be answered. We will revisit this issue when we discuss stochastic integrals with respect to Brownian motion later in Chapter 5.

CHAPTER 2

Processes with Independent Increments

2.1. The Basic Poisson Process

The Poisson process is a counting process. A Poisson process on the interval $[0, \infty)$ counts the number of times some primitive event has occurred during the time interval $[0, t]$. The following assumptions are made about the ‘process’ $N(t)$ or the measure P .

- (i) With probability 1, $N(0) = 0$, $N(t)$ is integer valued, right continuous with left limits and nondecreasing in t , i.e. the probability measure P lives on the subset of $D[0, \infty)$ consisting of functions $N(\cdot)$ that are non-decreasing, integer valued and satisfy $N(0) = 0$.
- (ii) For each $h > 0$, the distribution of $N(t+h) - N(t)$ is independent of t , i.e. the same for all t .
- (iii) The random variables $N(t'_j) - N(t_j)$ are mutually independent if the intervals $[t_j, t'_j]$ are nonoverlapping.
- (iv) $P[N(t+h) - N(t) \geq 2] = P[N(h) \geq 2] = o(h)$ as $h \rightarrow 0$.

THEOREM 2.1 *Under the above assumptions, the process $N(\cdot)$ has the following additional properties.*

- (i) With probability 1, $N(t)$ is a step function that increases in steps of size 1.
- (ii) There exists a number $\lambda \geq 0$ such that the distribution of $N(t+s) - N(s)$ has a Poisson distribution with parameter λt .
- (iii) The gaps $\tau_1, \tau_2, \dots$ between successive jumps are independent and identically distributed exponential random variables.

$$(2.1) \quad P\{\tau_j \geq x\} = \begin{cases} \exp[-\lambda x] & \text{for } x \geq 0 \\ 1 & \text{for } x \leq 0 \end{cases}$$

PROOF: Let us divide the interval $[0, T]$ into n equal parts and denote by ξ_n the number of intervals of the form $[\frac{k}{n}, \frac{k+1}{n}]$ which have an increment $N(\frac{(k+1)T}{n}) - N(\frac{kT}{n})$ that is at least 2. Since the existence a jump of size 2 implies that at least one interval has an increment of at least 2, it is enough to control $P[\xi_n \geq 1]$ and show that $\lim_{n \rightarrow \infty} P[\xi_n \geq 1] = 0$. We estimate

$$P[\xi_n \geq 1] \leq E[\xi_n] = nP\left[N\left(\frac{T}{n}\right) \geq 2\right] = n \cdot o\left(\frac{1}{n}\right) = o(1) \quad \text{as } n \rightarrow \infty$$

thereby proving property (i). Because of right continuity we have

$$P[N(t) \geq 1] \rightarrow 0 \quad \text{as } t \rightarrow 0$$

proving that the distribution of $N(t)$ is infinitesimal as $t \rightarrow 0$. By the independence of the increments over disjoint intervals, $N(t)$ is the sum of n independent copies of $N(\frac{t}{n})$.

$$(2.2) \quad E\{\exp[-\sigma N(t)]\} = \lim_{n \rightarrow \infty} \left[E\left\{\exp\left[-\sigma N\left(\frac{t}{n}\right)\right]\right\} \right]^n = \exp[-tg(\sigma)]$$

where

$$(2.3) \quad g(\sigma) = \lim_{h \rightarrow 0} \frac{1}{h} [1 - E\{\exp[-\sigma N(h)]\}]$$

Since the limit in (2.2) exists and is not 0, it follows that the limit in (2.3) exists and is finite.

We can compute

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{1}{h} [1 - E\{\exp[-\sigma N(h)]\}] &= \lim_{h \rightarrow 0} \frac{1}{h} [E\{1 - \exp[-\sigma N(h)]\}] \\ &= \lim_{h \rightarrow 0} \frac{1}{h} [(1 - e^{-\sigma})P[N(h) = 1] + o(h)] \\ (2.4) \quad &= (1 - e^{-\sigma}) \lim_{h \rightarrow 0} \frac{1}{h} [P[N(h) = 1]] \end{aligned}$$

$$(2.5) \quad = \lambda(1 - e^{-\sigma})$$

where

$$(2.6) \quad \lambda = \lim_{h \rightarrow 0} \frac{1}{h} P[N(h) = 1].$$

The limit in (2.6) must necessarily exist because the limit in (2.3) clearly exists. The positivity of $E\{\exp[-\sigma N(t)]\}$ and the identity (2.5) guarantees the finiteness of the limit in (2.6). The formulas (2.5) and (2.6) together identify the distribution of $N(t)$ as the Poisson distribution with parameter λt , thereby proving property (ii).

Finally, we turn to the proof of property (iii). First let us prove that τ_1 has the right distribution.

$$P[\tau_1 > x] = P[N(x) = 0] = e^{-\lambda x}$$

because of the Poisson distribution. We will prove τ_1 is a renewal time in the sense that $N(\tau_1 + t) - N(\tau_1) = N(\tau_1 + t) - 1$ is again a Poisson process independent of τ_1 . This will prove that τ_2 will have the same distribution as τ_1 and will be independent of it. Repeating the step inductively on n will complete the proof. Let τ be a stopping time that takes a countable set of values $\{t_j\}$. Since the set $\tau = t_j$ is measurable with respect to $\mathcal{F}_{t_j} = \sigma\{N(t) : t \leq t_j\}$, the process $N(t + t_j) - N(t_j)$ is again a Poisson process independent of $\mathcal{F}_{t_j}$. If $A \in \mathcal{F}_\tau$ then $A \cap \{\tau = t_j\} \in \mathcal{F}_{t_j}$ and therefore $N(\cdot + t_j) - N(t_j)$ is independent of $A \cap \{\tau = t_j\}$. In other words

$$P[\{[N(\cdot + \tau) - N(\tau)] \in E\} \cap A \cap \{\tau = t_j\}] = P[N(\cdot) \in E]P[A \cap \{\tau = t_j\}]$$

Summing over j

$$P[\{[N(\cdot + \tau) - N(\tau)] \in E\} \cap A] = P[N(\cdot) \in E]P[A]$$

Therefore for such a τ , $N(\tau + t) - N(\tau)$ is again a Poisson process independent of $\mathcal{F}_\tau$. Finally, τ_1 is a stopping time and for any k , $\tau^{(k)} = \frac{[k\tau_1] + 1}{k}$ is a stopping time that takes only a countable number of values. Therefore $N(\tau^{(k)} + t) - N(\tau^{(k)})$ is a Poisson process with parameter λ that is independent of $\tau^{(k)}$. We let $k \rightarrow \infty$. Because $\tau^{(k)} \geq \tau_1$, and the process is right continuous we are done. This property known generally as the *strong Markov property* is quite similar to the optional stopping theorem for martingales in spirit as well as substance. $\square$

REMARK 2.1. What we have proved is that if P is a measure on the space $D[0, \infty)$ satisfying the properties listed in the assumptions of Theorem 2.1, then $P = P_\lambda$ is determined by a single parameter λ and the additional properties (i)–(iii) of Theorem 2.1 will be valid for it. The process P_λ is referred to as the Poisson process with parameter or ‘rate’ λ . Such a process clearly exists, by defining $N(t)$ either through finite-dimensional distributions or through successive waiting times.

EXERCISE 2.1. Let τ_i be a sequence of independent identically distributed random variables with a common exponential distribution $e^{-\tau} d\tau$. Define

$$N(t) = \begin{cases} 0 & \text{if } 0 \leq t < \tau_1 \\ k & \text{if } \tau_1 + \cdots + \tau_k \leq t < \tau_1 + \cdots + \tau_{k+1} \end{cases}$$

Show that $N(t)$ is the Poisson process with rate $\lambda = 1$

EXERCISE 2.2. Alternate proof of property (iii) of Theorem 2.1. Let $\sigma \leq 0$ be arbitrary. The processes

$$M_\sigma(t) = \exp[\sigma N(t) - \lambda t(e^\sigma - 1)]$$

are martingales with respect to $(\Omega, \mathcal{F}_t, P)$ and Doob’s stopping theorem provides the relation

$$E\{M_\sigma(\tau + t) \mid \mathcal{F}_\tau\} = M_\sigma(\tau) \text{ a.e.}$$

Turn it into a proof that τ_1 is a renewal time.

EXERCISE 2.3. Verify that for any Poisson process with parameter λ

$$N(t) - \lambda t \quad \text{and} \quad [N(t) - \lambda t]^2 - \lambda t$$

are martingales with respect to $(\Omega, \mathcal{F}_t, P)$.

In other words $N(t)$ is semimartingale with compensator λt . Notice that $N(t)$ is an increasing function and we could have tried to take $N(t)$ for its compensator. But $N(t)$ is not continuous and as we said earlier the natural choice for a compensator is a continuous one if it exists. Here λt is that choice.

EXERCISE 2.4. The distribution of $\tau_1 + \cdots + \tau_k$ is a Gamma distribution with density $\frac{\lambda^k}{(k-1)!} e^{-\lambda x} x^{k-1}$. Why does it look a little bit like the Poisson probability for k jumps? What is the exact relation?

2.2. Compound Poisson Processes

Suppose $X_1, X_2, \dots, X_n, \dots$ is a sequence of independent and identically distributed random variables with a common distribution α . S_n are the partial sums $S_n = X_1 + X_2 + \dots + X_n$ and $N(t)$ is a Poisson process with rate λ . We define a process $X(t)$ by $X(t) = S_{N(t)}$. Each time the Poisson process $N(t)$ jumps by 1, the process $X(t)$ jumps by a random amount which has distribution α . The jumps at different instances are independent. Such a process $X(t)$ inherits the independent increments property from $N(t)$. For any collection of nonoverlapping intervals $[t_j, t'_j]$ the increments $X(t'_j) - X(t_j)$ are independent random variables. The distribution of any increment $X(t+s) - X(s)$ is that of $X_{N(t)}$, and is determined as the distribution of S_n where n is random and has a Poisson distribution with parameter λt .

$$\begin{aligned} E\{\exp[i y X(t)]\} &= \sum_{n=0}^{\infty} e^{-\lambda t} \frac{(\lambda t)^n}{n!} [\hat{\alpha}(y)]^n \\ &= e^{-\lambda t} e^{\lambda t \hat{\alpha}(y)} = e^{\lambda t [\hat{\alpha}(y) - 1]} = \exp \left[\lambda t \int (e^{i y x} - 1) d\alpha(x) \right] \end{aligned}$$

In other words $X(t)$ has an infinitely divisible distribution with a Lévy measure given by $\lambda t \alpha$. If we denote by $M = \lambda \alpha$ then

$$E\{\exp[i y X(t)]\} = \exp \left[t \int (e^{i y x} - 1) dM(x) \right]$$

EXERCISE 2.5. Assuming that α has two moments, show that $X(t) - At$ and $[X(t) - At]^2 - Bt$ are martingales with $A = \int x dM = \lambda \int x d\alpha$ and $B = \int x^2 dM = \lambda \int x^2 d\alpha$

EXAMPLE 2.1. Let us suppose that there is a bank account in which deposits are made at random times, each deposit being a dollar and the process $D(t)$ of total number deposits is a Poisson process with parameter λ /unit time. If the withdrawal from the account is at a steady rate of 1 dollar per unit time, and the initial balance is k dollars what is the probability that the balance will ever become zero?

The current balance is $x(t) = D(t) - t$. For any constant c

$$M(t) = \exp[cx(t) + ct - \lambda t(e^c - 1)]$$

is a martingale. If τ is the first time the balance hits 0,

$$E[M(\tau \wedge t)] = E[\exp[cx(\tau \wedge t) + (\tau \wedge t)[c + \lambda - \lambda e^c]]] = e^{ck}$$

If $\lambda < 1$, then the balance process will almost surely hit 0, and if we choose $c < 0$ and let $t \rightarrow \infty$ we obtain

$$E[\exp[(c + \lambda - \lambda e^c)\tau]] = e^{ck}$$

If we solve $c + \lambda - \lambda e^c = -\sigma$ to get $c = c(\sigma) < 0$ then

$$E[\exp[-\sigma \tau]] = e^{c(\sigma)k}$$

which gives us the Laplace transform of the time it takes to hit 0. On the other hand if $\lambda > 1$, $c + \lambda - \lambda e^c = 0$ has a unique nonzero negative root $c = c(\lambda)$ and $e^{c(\lambda)k}$ gives us the probability of the balance ever reaching 0.

2.3. Infinite Number of Small Jumps

Whereas a Poisson process cannot have an infinite number of jumps in a finite interval, once we start considering compound Poisson processes we can, in principle, add an infinite number of small jumps so that we still have a finite sum. Let us suppose that $X_k(t)$ is a compound Poisson process which corresponds to $M_k = \lambda_k \alpha_k$. Let

$$E[X_k(t)] = t \int x dM_k(x)$$

$$Var[X_k(t)] = t \int x^2 dM_k(x)$$

Let us take $X_k(t)$ to be mutually independent processes with independent increments and try to sum up

$$X(t) = \sum_k X_k(t)$$

for each t . If the sum exists then it is a process with independent increments. In fact, we can do a little better. We may center these processes with suitable constants $\{a_k t\}$, and write

$$X(t) = \sum_k [X_k(t) - a_k t]$$

to help in convergence. We shall assume that

$$\sum_k \int \frac{x^2}{1+x^2} dM_k(x) < \infty$$

which amounts to two conditions

$$(2.7) \quad \sum_k M_k[|x| \geq 1] < \infty$$

and

$$(2.8) \quad \sum_k \int_{|x| \leq 1} x^2 dM_k(x) < \infty$$

We can always decompose any M as $M_1 + M_2$ and this will result in a decomposition of the process $X(t) = X_1(t) + X_2(t)$, into two mutually independent processes

with independent increments corresponding to M_1 and M_2 , respectively. It is better for us to decompose each M_k as $M_k^{(1)} + M_k^{(2)}$ corresponding to jumps of size $|x| \leq 1$ and $|x| > 1$.

$$M = \sum_k M_k^{(2)}$$

sums to a finite measure and offers no difficulty. The process

$$X^{(2)}(t) = \sum_k X_k^{(2)}(t)$$

exists very nicely because from (2.7)

$$\begin{aligned} \sum_k P\left\{\sup_{0 \leq s \leq t} |X_k^{(2)}(s)| \neq 0\right\} &\leq \sum_k 1 - e^{-t M_k^{(2)}[\mathbb{R}]} \leq \sum_k t M_k^{(2)}[\mathbb{R}] \\ &= t \sum_k M_k[|x| > 1] < \infty \end{aligned}$$

and by Borel-Cantelli lemma, in any finite interval, the sum is almost surely a finite sum. As for the convergence of $\sum_k X_k^{(1)}(t)$, we can assume that $M_k^{(1)}[|x| > 1] = 0$ for every k . If we take $a_k = E\{X_k(1)\} = \int x dM_k^{(1)}(x)$,

$$E\{[X_k^{(1)}(t) - a_k t]^2\} = t \int x^2 dM_k^{(1)}(x)$$

From (2.8) and the two-series theorem

$$\sum_k [X_k^{(1)}(t) - a_k t]$$

converges almost surely. A simple application of Doob's inequality yields in fact almost sure uniform convergence in finite time intervals. If we now reassemble the pieces we see that

$$E\{e^{iyX(t)}\} = \exp\left[t \int_{|x| \leq 1} (e^{iyx} - 1 - iyx) dM(x) + t \int_{|x| > 1} (e^{iyx} - 1) dM(x)\right]$$

which is essentially the same as Lévy-Khintchine representation for infinitely divisible distributions except for the missing Gaussian term. That is the continuous part, which cannot be built from jumps and needs the Brownian motion or Wiener process.

The usual Lévy-Khintchine representation theorem for infinitely divisible distributions takes the form (ignoring the Gaussian part)

$$(2.9) \quad E\{e^{iyX(t)}\} = \exp\left[it\gamma + t \int \left(e^{iyx} - 1 - \frac{iyx}{1+x^2}\right) dM(x)\right]$$

There is nothing special about $\frac{x}{1+x^2}$. We can replace it by any bounded function $\theta(x)$ with the property $\theta(x) = x + o(x^2)$ as $x \rightarrow 0$. We would then have

$$E\{e^{iyX(t)}\} = \exp\left[it\gamma' + t \int \left(e^{iyx} - 1 - iy\theta(x)\right) dM(x)\right]$$

with a γ' different from γ . This is just a reflection of the fact that there is a lot of choice in the centering needed for the convergence. It can always be compensated by altering the term $i t \gamma$ that corresponds to a deterministic process $X(t) = \gamma t$. Our splitting of M led to the choice of $\theta(x) = \mathbf{1}_{\{|x| \leq 1\}}$, which has the disadvantage of being discontinuous and in the Lévy-Khintchine representation it is usually replaced by a more gentle $\theta(x)$. Note that while the Lévy measure M always integrates x^2 near 0, it need not necessarily integrate $|x|$. If it does, the centering can be pulled out (i.e. the choice of $\theta(x) \equiv 0$ works) and the representation can be rewritten as

$$E[e^{i \gamma X(t)}] = \exp \left[i t \gamma + t \int (e^{i y x} - 1) dM(x) \right]$$

with a different γ .

REMARK 2.2. When $\int_{|x| \leq 1} |x| dM < \infty$, we can write $X(t) = \gamma t + X^+(t) + X^-(t)$ where $X^\pm$ correspond to the restrictions of M , respectively, to $\mathbb{R}^\pm$. This makes $X^+(t)$ an increasing process consisting entirely of positive jumps and $X^-(t)$ a decreasing process made out of negative jumps. In particular, the process $X(t)$ is of bounded variation.

EXERCISE 2.6. Show that if a process with independent increments is almost surely of bounded variation then $\int_{|x| \leq 1} |x| dM(x) < \infty$.

REMARK 2.3. One can have M concentrated on $(0, 1]$. But at the same time one can have $\int_{0 \leq x \leq 1} x dM(x) = \infty$. In such a case one needs $\theta(x)$ and it can not be pulled out i.e., if we try we end up with $\gamma = -\infty$. This means that although the process $X(t)$ has only positive jumps, all of them from $(0, 1]$, there are too many small ones and the sum of the jumps is infinite in any interval. However, if compensated by the infinite term $-\gamma t$, we have a finite answer. This is not very different from trying to sum $\sum_j X_j$ of random variables with values in $[0, 1]$, such that $\sum_j E[X_j] = \infty$ but $\sum_j \text{Var}(X_j) < \infty$. This makes $\sum_j (X_j - E[X_j])$ convergent, but the centering constants cannot be pulled out of the sum. Such a process, however, fails to be of bounded variation. Moreover, since the compensatory adjustment is infinite and a fair number of the jumps on the positive side can with positive probability fail to materialize, the random variable $X(t)$ is not always positive and in fact, can take arbitrarily large negative values with positive probability.

REMARK 2.4. The symmetric stable laws correspond to $dM(x) = \frac{cdx}{|x|^{1+\alpha}}$ for $0 < \alpha < 2$. They are of bounded variation in the range $0 < \alpha < 1$. $\alpha = 1$ leads to the Cauchy distribution. The corresponding processes with independent increments are called stable processes. They have the scaling property

$$X(\lambda t) \simeq \lambda^{\frac{1}{\alpha}} X(t)$$

in the sense that, as processes, they have the same distribution.

REMARK 2.5. The process with independent increments $X(t)$ with distribution given by (2.9) has paths in $D[0, \infty)$, i.e. it has almost surely right continuous paths

with left limits. The proof is not hard. The process can be written as $X(t) = X^1(t) + X^2(t)$. The part $X(t)$ from large jumps causes no problem because it is a compound Poisson process with a finite number of jumps and we take the right continuous version. The part $X^1(t)$ from small jumps is an infinite series

$$(2.10) \quad X^1(t) = \sum_j Y_j(t)$$

of independent processes with independent increments. Since $E[Y_j(t)] \equiv 0$, $Y_j(t)$ is a martingale. In addition $\sum_j \text{Var}(Y_j(t)) < \infty$. Therefore for the tail

$$T_n(t) = \sum_{j \geq n} Y_j(t)$$

which is a martingale, one has from Doob's inequality (1.5)

$$P\left[\sup_{0 \leq s \leq t} |T_n(s)| \geq \delta\right] \leq \frac{1}{\delta^2} \sum_{j \geq n} \text{Var}(Y_j(t)) \rightarrow 0 \quad \text{as } n \rightarrow \infty$$

proving almost surely the uniform convergence of the series (2.10) in any finite time interval. This is clearly enough.

Any time we have a one-parameter convolution semigroup of probability measure μ_t , that satisfy $\mu_t * \mu_s = \mu_{t+s}$ we can construct a process with independent increments, the distribution of the increment $X(t+s) - X(t)$ being μ_s . Such processes where the distribution of $X(t+s) - X(t)$ is independent of t are called processes with stationary independent increments. This need not be the case for all processes with independent increments. For instance one can think of a Poisson process where the distribution of the increment over an interval $[s, t]$ is Poisson with parameter $\int_s^t \lambda(\sigma) d\sigma$. The Poisson rate or intensity depends on the time σ . However processes with stationary independent increments correspond to one-parameter convolution semigroups μ_t of probability distributions.

EXERCISE 2.7. For the following semigroups find the Lévy measures.

$$\begin{aligned} \mu_t(dx) &= \frac{1}{\Gamma(t)} e^{-x} x^{t-1} dx \quad x \geq 0 \\ \mu_t(dx) &= \frac{1}{\pi} \frac{t dx}{t^2 + x^2} \quad -\infty < x < \infty \end{aligned}$$

2.4. Infinitesimal Generators

If $X(t)$ is a process with stationary independent increments then we have a convolution semigroup that corresponds to it. It can be used to define a semigroup T_t of linear transformations on functions $f \in C(\mathbb{R})$, the space of bounded continuous functions, by

$$(T_t f)(x) = \int f(x+y) \mu_t(dy)$$

It is not hard to check that T_t maps $C(\mathbb{R})$ into itself and satisfies $T_{t+s} f = T_t T_s f = T_s T_t f$ for all $f \in C(\mathbb{R})$.

The infinitesimal generator $\mathcal{A}$ of T_t is formally defined as the derivative

$$(\mathcal{A}f)(x) = \lim_{t \rightarrow 0} \frac{T_t f - f}{t}$$

For a Poisson process it is easy to calculate.

$$(\mathcal{A}f)(x) = \lambda[f(x+1) - f(x)]$$

For a translation process $(T_t f)(x) = f(x+ct)$

$$(\mathcal{A}f)(x) = cf'(x)$$

One can combine the two and check that for a process that has a Lévy-Khintchine representation

$$E[e^{iyX(t)}] = \exp \left[t \int (e^{izy} - 1) dM(x) + i\gamma t \right]$$

with a finite Lévy measure $M(dx)$, the generator is

$$(\mathcal{A}f)(x) = cf'(x) + \int [f(x+z) - f(x)] M(dz)$$

An easy way to verify the relation for a process with a general Lévy measure is to note that if $f(x) = e^{ixy}$, then

$$(T_t f)(x) = f(x) E[e^{iyX(t)}] = f(x) \exp \left[i\gamma t + \int (e^{izy} - 1) - \frac{izy}{1+z^2} \right] M(dz)$$

and it is easy to check that

$$(\mathcal{A}f)(x) = \gamma f'(x) + \int \left[f(x+z) - f(x) - \frac{zf'(x)}{1+z^2} \right] M(dz)$$

It is not hard to extend this formula from functions of the form $f(x) = e^{ixy}$ to smooth functions f . Two bounded continuous derivatives are sufficient to make the integrals behave well.

2.5. Some Associated Martingales

Given a Poisson process $N(t)$ with rate λ we saw that $N(t) - \lambda t$ is a martingale with respect to the natural filtration $\mathcal{F}_t$. But there are many other natural martingales that are associated with the Poisson process $N(t)$. If f is any continuous function with at most polynomial growth on $\mathbb{R}$

$$M_f(t) = f(N(t)) - f(0) - \int_0^t [f(N(s)+1) - f(N(s))] ds$$

is a martingale with respect to $\mathcal{F}_t$. If τ is the time of the first jump, then until then $N(s) = 0$, and we can verify first that

$$f(N(\tau \wedge t)) - f(0) - [f(1) - f(0)](\tau \wedge t)$$

is a martingale. Since only $f(0)$ and $f(1)$ are involved we can assume f is linear and it is clearly true for $f(x) = x$. The next step is to use the renewal property and prove it until time τ_2 the time of the second jump. Then by induction till the time

of the n -th jump. Finally, we let $n \rightarrow \infty$ and deduce it for all t . The induction step needs the following technical lemma.

LEMMA 2.2 *Let $M(t)$ be martingale with respect to a filtration $\mathcal{F}_t$ on a probability space $(\Omega, \mathcal{F}, P)$. Let $X_\theta(t)$ be a family of martingales, again with respect to the same filtration $\mathcal{F}_t$ on $(\Omega, \mathcal{F}, P)$, but depending continuously also on a parameter $\theta \in \Theta \subset \mathbb{R}^k$ for some k . τ is a stopping time and $Y(t)$ is defined as*

$$Y(t) = M(\tau \wedge t) + X_{\theta(\omega)}(t \vee \tau) - X_{\theta(\omega)}(\tau)$$

i.e. $Y(t) = M(t)$ for $t \leq \tau(\omega)$ and $Y(t) = M(\tau) + X_{\theta(\omega)}(t) - X_{\theta(\omega)}(\tau)$ for $t > \tau(\omega)$. If $\theta(\omega)$ is $\mathcal{F}_\tau$ measurable then $Y(t)$ is again a martingale provided the following natural integrability condition is satisfied:

$$\sup_{\theta \in \Theta} |Y_\theta(t)| = Y^*(t)$$

is integrable for every t .

PROOF: The lemma has a natural interpretation. We can think of $M(t)$ as well as $X_\theta(t)$ as a family of fair games. At a stopping time τ we are free to switch from the current game $M(t)$ to a new game X_θ , with the choice depending on currently available information $\mathcal{F}_\tau$. No matter what, the game is still fair! We can assume that $\theta(\omega)$ takes only finitely many values. We can approximate arbitrary $\theta(\omega)$ by those that take only finitely many values and then use the integrability condition to pass to the limit.

Let $A \in \mathcal{F}_{t_1}$ and $\theta(\omega) = \theta_i$ on $E_i \in \mathcal{F}_\tau$. Let $0 \leq t_1 \leq t_2 < \infty$. The sets $B_1 = \{\tau \leq t_1\}$, $B_2 = \{t_1 < \tau \leq t_2\}$ and $B_3 = \{\tau > t_2\}$ determine the location of $\tau(\omega)$ relative to t_1 and t_2 . We need to show

$$\int_A [Y(t_2) - Y(t_1)] dP = 0$$

$$Y(t_2) - Y(t_1) = \begin{cases} M(t_2) - M(t_1) & \text{on } B_3 \\ X_{\theta(\omega)}(t_2) - X_{\theta(\omega)}(\tau) + M(\tau) - M(t_1) & \text{on } B_2 \\ X_{\theta(\omega)}(t_2) - X_{\theta(\omega)}(t_1) & \text{on } B_1 \end{cases}$$

Let $E_i = \{\omega : \theta(\omega) = \theta_i\}$. We observe that the set $B_1 \cap E_i$ is in $\mathcal{F}_{t_1}$ and so is $A \cap B_1 \cap E_i$. On B_1 we have $\tau \leq t_1 \leq t_2$.

$$\int_{A \cap B_1 \cap E_i} [Y(t_2) - Y(t_1)] dP = \int_{A \cap B_1 \cap E_i} [X_{\theta_i}(t_2) - X_{\theta_i}(t_1)] dP = 0$$

On B_2 we have $t_1 < \tau \leq t_2$ and $A \cap B_2 \cap E_i \in \mathcal{F}_{\tau \wedge t_2}$. Therefore

$$\int_{A \cap B_2 \cap E_i} [X_{\theta(\omega)}(t_2) - X_{\theta(\omega)}(\tau)] dP = \int_{A \cap B_2 \cap E_i} [X_{\theta_i}(\tau \vee t_2) - X_{\theta_i}(\tau \wedge t_2)] dP = 0$$

It remains to show that

$$\begin{aligned} \int_{A \cap B_2} [M(\tau) - M(t_1)] dP + \int_{A \cap B_3} [M(t_2) - M(t_1)] dP = \\ \int_{A \cap B_1^c} [M((\tau \wedge t_2) \vee t_1) - M(t_1)] dP = 0. \end{aligned}$$

Since $A \cap B_1^c$ is in $\mathcal{F}_{t_1}$ and $M(t)$ is a martingale this is now obvious. $\square$

There is a better way of generating a whole class of martingales associated with processes with independent increments. Let

$$\psi(y) = i\gamma y + \int \left[e^{izy} - 1 - \frac{izy}{1+z^2} \right] M(dz)$$

Then

$$E[e^{iy(X(t)-X(s))} \mid \mathcal{F}_s] = e^{(t-s)\psi(y)}$$

This can be restated as

$$X_y(t) = \exp[iyX(t) - t\psi(y)]$$

is a martingale for every $y \in \mathbb{R}$. A complex valued martingale is just a pair of real valued martingales, comprising of its real and imaginary parts. We can now apply Lemma 1.7 with $A(t) = e^{t\psi(y)}$ to conclude that

$$e^{iyX(t)} - \int_0^t \psi(y) e^{iyX(s)} ds$$

is a martingale. It is easy to check that if $f(x) = e^{iyx}$ then

$$(\mathcal{A}f)(x) = i\gamma f'(x) + \int \left[f(x+z) - f(x) - \frac{zf'(x)}{1+z^2} \right] M(dz) = \psi(y) f(x)$$

Hence we have shown that

$$(2.11) \quad M_f(t) = f(X(t)) - f(X(0)) - \int_0^t (\mathcal{A}f)(X(s)) ds$$

is a martingale when $f = e^{ixy}$ with $y \in \mathbb{R}$. Since any smooth f can be approximated by linear combinations of exponentials, it is not difficult to conclude that $M_f(t)$ are martingales for “nice” (having two bounded continuous derivatives) f . The converse is also true, by essentially retracing the steps. Any measure on the space $D[0, \infty)$ for which $M_f(t)$ are martingales has to be the process with independent increments with Lévy measure M . Starting with the martingales $M_f(t)$ with $f(x) = e^{ixy}$, we appeal to Lemma 1.7 again, this time with $A(t) = e^{-t\psi(y)}$. This allows us to conclude that with respect to $(\Omega, \mathcal{F}_t, P)$

$$X_y(t) = \exp[iyX(t) - t\psi(y)]$$

are martingales. This in turn implies

$$E[e^{iy(X(t)-X(s))} \mid \mathcal{F}_s] = e^{(t-s)\psi(y)}$$

which is all that we need. We have therefore proved the following theorem.

THEOREM 2.3 *The measure P corresponding to a process with independent increments with Lévy measure $M(dz)$ and characteristic function*

$$E[e^{iyX(t)}] = \exp \left[it\gamma y + \int \left(e^{iyz} - 1 - \frac{iyz}{1+z^2} \right) M(dz) \right]$$

is characterized as the unique probability measure on $\Omega = D[0, \infty)$ that satisfies

- (i) $P[X(0) = 0] = 1$
- (ii) *For any function f which is bounded and has two bounded continuous derivatives, the expression $M_f(t)$ given by (2.11) is a martingale with respect to $(\Omega, \mathcal{F}_t, P)$.*

CHAPTER 3

Poisson Point Processes

3.1. Point Processes

A point process is a random countable subset of some space X . We often encounter situations in which they arise. A point process is described by counting the number of points of this random set that belong to various measurable subsets of X . In other words $(X, \mathcal{B})$ is a measurable space and we describe a random subset of X by specifying the random variables $\lambda(A, \omega)$ that count the number of members of the random set that are in A . It is convenient to assume that X is a countable union of compact sets. We shall think of it as $\mathbb{R}^n$ for some n . Another way to describe the same random set is to think of a random measure on X that consists of unit point mass at every point of the random set. Then such a measure $\lambda(A, \omega)$ will clearly count the number of points of the random set that belong to A .

If we assume that our random sets do not have points of accumulation, i.e. there are only finitely many points in any compact set, then the random variables

$$\langle f, \lambda \rangle = \int_X f(x) \lambda(dx, \omega) = \sum_i f(x_i)$$

are well defined for functions f that are compactly supported and continuous. There are various ways of identifying the distribution of the random measure λ . The Fourier transform

$$\psi(f) = E[e^{i\langle f, \lambda \rangle}]$$

is one such. One can also calculate various moments of $\langle f, \lambda \rangle$ provided they exist. The mean

$$\mu(A) = E[\lambda(A, \omega)]$$

if it exists, defines a countably additive (possibly only σ -finite) measure μ and it is called the intensity measure. It is clear that for a large class of functions f we should have

$$(3.1) \quad E[\langle f, \lambda \rangle] = \langle f, \mu \rangle = \int f d\mu$$

Fubini's theorem will tell us that if $f \in L_1(\mu)$, then $f \in L_1(\lambda)$ for almost all ω and (3.1) will hold. If there is a natural reference measure like the Lebesgue measure on $\mathbb{R}^d$ and μ is absolutely continuous with density $m(x)$ then $m(\cdot)$ is called the density of the point process.

The higher moments require the calculation of expectations

$$E\left[\sum f(x_1, x_2, \dots, x_n)\right] = \Lambda(f)$$

for any symmetric function $f(x_1, x_2, \dots, x_n)$ of n variables. The summation is carried over subsets of cardinality n of our random set. One has a representation

$$\Lambda_n(f) = \int f(x_1, \dots, x_n) \mu_n(dx_1, \dots, dx_n)$$

with a uniquely defined symmetric measure μ_n . One can recover the moments of $\langle f, \lambda \rangle$ from μ_n . For instance

$$E[\langle f, \lambda \rangle^2] = 2 \int f(x_1) f(x_2) d\mu_2 + \int f^2(x) d\mu(x)$$

3.2. Poisson Point Process

A point process requires a self consistent specification of joint distributions of $\{\lambda(A_i, \omega)\}$ for all finite collections $\{A_i\}$ from $\mathcal{B}$. In general, $\lambda(A, \omega)$ is an integer and the probability distribution of a single $\lambda(A, \omega)$ is not fully specified by its expectation. However if $\lambda(A, \omega)$ has a Poisson distribution, then it is fully specified by its mean $\mu(A)$. It is enough to specify the joint distributions for collections $\{A_i\}$ that are disjoint subsets. We can declare them to be independent and since the sum of two independent random variables with Poisson distributions again has a Poisson distribution we have a self consistent definition. The Fourier transform

$$\psi(f) = E[e^{\langle f, \lambda \rangle}]$$

is easily calculated for $f = \sum_i f_i \mathbf{1}_{A_i}$ as

$$\begin{aligned} E[e^{\langle f, \lambda \rangle}] &= E[e^{\sum f_i \lambda(A_i, \omega)}] \\ &= \prod_i E[e^{f_i \lambda(A_i, \omega)}] \\ &= \prod_i \exp[\mu(A_i)(e^{f_i} - 1)] \\ &= \exp \left[\sum_i \mu(A_i)(e^{f_i} - 1) \right] \\ &= \exp \left[\int (e^{f(x)} - 1) \mu(dx) \right] \end{aligned}$$

and extended to all f with compact support.

DEFINITION 3.1 A *Poisson random measure* or a *Poisson point process* with intensity μ is a random measure $\lambda(A, \omega)$ such that for any $A \in \mathcal{B}$ it has a Poisson distribution with parameter $\mu(A)$ and $\lambda(A_j, \omega)$ are mutually independent for any finite collection of disjoint sets $\{A_j\}$ in $\mathcal{B}$.

REMARK 3.1. It can be specified by its Fourier transform

$$\psi(f) = \exp \left[\int (e^{f(x)} - 1) \mu(dx) \right]$$

Point processes provide a useful and convenient way of looking at processes with jumps, particularly processes with independent increments that lead to Poisson point processes. The simplest example of a Poisson point process is the random set of times at which a Poisson process $N(t)$ has jumps. Here X is $\mathbb{R}^+$ and $d\mu = \rho dt$ where ρ is the intensity. The number of jumps in an interval $[a, b] \subset [0, \infty)$ is the increment of $N(t)$ over that interval and has the Poisson distribution with parameter $\rho(b - a)$. These increments are mutually independent if the intervals are disjoint. The definition of Poisson random measure is therefore satisfied when the sets in question are intervals. It is left as an exercise to extend it to arbitrary Borel sets.

EXERCISE 3.1. If $\{A_i\}$ are a finite collection disjoint Borel sets with finite measure, for any given $\epsilon > 0$, show that there are sets B_j that are disjoint, each B_j being itself a finite union of disjoint intervals such that $|B_j \Delta A_j| \leq \epsilon$. ($|A|$ is the Lebesgue measure of A .)

EXERCISE 3.2. If $\lambda(A, \omega)$ is the number of jumps of $N(t)$ with $t \in A$, show that $E[\lambda(A, \omega)] = \rho|A|$

EXERCISE 3.3. Show that $\lambda(A)$ defines a Poisson point process.

A process with independent increments with Lévy measure $M(dx)$ has many jumps of different sizes at different times. One can put a marker at $(t, x) \in [0, \infty) \times \mathbb{R}$ if there is a jump of size x at time t . This produces a random measure on $[0, \infty) \times \mathbb{R}$ where $\lambda([a, b] \times E)$ is the number of jumps during the time interval $[a, b]$ that lie in the set E .

THEOREM 3.1 $\lambda(A)$ is Poisson point process on the space $[0, \infty) \times \mathbb{R}$ with intensity $dt \times M(dx)$

PROOF: Let $\{A_j : 1 \leq j \leq k\}$ be a finite collection of disjoint subsets of $\mathbb{R} \setminus \{0\}$ with $M(A_j) < \infty$ for each j . From the way that processes with independent increments were constructed in Chapter 2, it is clear that if we decompose M into a finite sum $M = M_0 + \sum_{j=1}^k M_j$ where M_j for $j \geq 1$ is the restriction of M to A_j and M_0 is the restriction to $(\bigcup_j A_j)^c$ then the process $X(t)$ with Lévy measure M is the independent sum $X(t) = X_0(t) + \sum_{j=1}^k X_j(t)$ where $X_j(t)$ is the process consisting entirely of jumps corresponding to M_j .

$$E[e^{iyX_j(t)}] = \exp \left[t \int (e^{iyz} - 1) M_j(dz) \right]$$

The process $X_0(t)$ satisfies

$$E[e^{iyX_0(t)}] = \exp \left[t \int \left(e^{iyz} - 1 - \frac{iyz}{1+z^2} \right) M_0(dz) - \sum_{j=1}^k t \int \frac{iyz}{1+z^2} M_j(dz) \right]$$

It is clear if we are given a path $X(t)$ we can reconstruct $\{X_j(t) : j \geq 1\}$ and by removing the jumps that belong to $\bigcup_{j=1}^k A_j$ we will be left with $X_0(t)$. This proves that the point process $\lambda(A, \omega)$ on $[0, \infty) \times \mathbb{R}$ is indeed a Poisson point process with intensity $dt \times M(dx)$. $\square$

Finally, we can reconstruct the process with independent increments $X(t)$ from the marked Poisson point process $\lambda(\cdot, \omega)$. Roughly speaking, we have to add all the jumps up to time t . This means

$$X(t) = \int_{[0,t] \times \mathbb{R}} x \lambda(dx, \omega)$$

However we may need to center it.

$$X(t) = \int_{[0,t] \times \mathbb{R}} x \lambda(dx, \omega) - t \int_{\mathbb{R}} \frac{x}{1+x^2} M(dx)$$

Now we have the problem of canceling infinities arising from small jumps.

$$(3.2) \quad X(t) = \lim_{\epsilon \rightarrow 0} \left[\int_{[0,t] \times \{x: |x| \geq \epsilon\}} x \lambda(dx, \omega) - t \int_{|x| \geq \epsilon} \frac{x}{1+x^2} M(dx) \right]$$

can work. Finally, we need to add γt and take

$$X(t) = \lim_{\epsilon \rightarrow 0} \left[\int_{[0,t] \times \{x: |x| \geq \epsilon\}} x \lambda(dx, \omega) - t \int_{|x| \geq \epsilon} \frac{x}{1+x^2} M(dx) \right] + \gamma t$$

if the original representation for $X(t)$ was

$$E[e^{iyX(t)}] = \exp \left[i\gamma t + t \int_{\mathbb{R}} \left(e^{iyz} - 1 - \frac{iyz}{1+z^2} \right) M(dz) \right]$$

EXERCISE 3.4. Prove that limit (3.2) is valid.

CHAPTER 4

Jump Markov Processes

4.1. Simple Examples

Let $(\mathcal{X}, \mathcal{B})$ be a space with a σ -field. A Markov chain in discrete time with $\mathcal{X}$ as state space is fully specified by the one step transition probability $\Pi(x, A)$. Defined for $x \in \mathcal{X}$ and $A \in \mathcal{B}$, it is a probability measure on $(\mathcal{X}, \mathcal{B})$ for each $x \in \mathcal{X}$ and a $\mathcal{B}$ -measurable function of x for each $A \in \mathcal{B}$. The multistep transition probabilities $\{\Pi^n(x, A)\}$ are easily computed by iteration.

$$\Pi^{(n+1)}(x, A) = \int_{\mathcal{X}} \Pi^{(n)}(y, A) \Pi(x, dy)$$

They will then automatically satisfy the Chapman-Kolmogorov equations

$$(4.1) \quad \Pi^{(n+m)}(x, A) = \int_{\mathcal{X}} \Pi^{(n)}(y, A) \Pi^{(m)}(x, dy)$$

In the continuous-time case such a simple description in terms of a single transition probability is not possible.

In general, if we want to construct a continuous-time Markov process defined for $t \geq 0$ on a state space $\mathcal{X}$ with its σ -field $\mathcal{B}$, we need the transition probabilities $p(t, x, A)$ defined for $t > 0$, $x \in \mathcal{X}$ and $A \in \mathcal{B}$. It is natural to define $p(0, x, A) = \chi_A(x)$. For each $t > 0$, $p(t, \cdot, \cdot)$ will be a transition probability from $(\mathcal{X}, \mathcal{B}) \rightarrow (\mathcal{X}, \mathcal{B})$ and the collection will have to satisfy the continuous version of Chapman-Kolmogorov equations

$$(4.2) \quad p(t+s, x, A) = \int_{\mathcal{X}} p(s, y, A) p(t, x, dy)$$

Given such a collection, for each starting point $x \in \mathcal{X}$, we can define a consistent family of finite-dimensional distributions on the space Ω of trajectories $x(\cdot)$ that map $[0, \infty) \rightarrow \mathcal{X}$. On the σ -field $\mathcal{F}(t_1, \dots, t_k)$ that corresponds to the times $t_1 < \dots < t_k$, we define it first for rectangles $\{x(\cdot) : x(t_j) \in A_j, \forall j = 1 \dots k\}$ by the formula

$$P_x \left\{ \bigcap_{j=1}^k [x(t_j) \in A_j] \right\} = \int_{A_1} \dots \int_{A_k} p(t_1, x, dy_1) \dots p(t_k - t_{k-1}, x, dy_k)$$

The measure is then extended from rectangles to arbitrary sets in the product σ -field $\mathcal{F}(t_1, \dots, t_k)$. The consistency of these finite-dimensional distributions is a consequence of the Chapman-Kolmogorov property. Once we have consistency, a

theorem of A. Tulcea and I. Tulcea that can be found in [6], guarantees the existence of a measure P_x on the σ -field $\mathcal{F}$ generated by all the $\mathcal{F}(t_1, \dots, t_k)$.

REMARK 4.1. Kolmogorov consistency theorem proved in [7] about the existence of a probability measure on product spaces, requires the underlying component space to be a Polish space, i.e. a complete separable metric space. However when the conditional probabilities are given, as in the current situation, the underlying spaces can be arbitrary and this is the content of the theorem of A. and I. Tulcea.

The construction of a family $\{p(t, x, A)\}$ of transition probabilities that satisfy (4.2) is not obvious. There is however a simple way of generating examples. If we have a Markov chain $\{X_n\}$ on a state space $\mathcal{X}$, with one step transition probability $\Pi(x, dy)$, and a Poisson process $N(t)$ with intensity λ , we can combine the two to define a continuous-time Markov process $x(t)$ with $\mathcal{X}$ as state space by the formula

$$x(t) = X_{N(t)}$$

The transition probabilities of this Markov process are given by

$$\begin{aligned} P_x\{x(t) \in A\} &= P\{x(t) \in A \mid x(0) = x\} \\ &= p(t, x, A) \\ (4.3) \qquad &= \sum_{k=0}^{\infty} e^{-\lambda t} \frac{(\lambda t)^k}{k!} \Pi^{(k)}(x, A) \end{aligned}$$

Because of the Markov property of $\{X_n\}$ and the independence of the increments of $N(t)$, it is not difficult to see that

$$\begin{aligned} P\{x(t) \in A \mid \mathcal{F}_s\} &= P\{X_{N(t)-N(s)} \in A \mid X_0 = X_{N(s)}\} \\ &= P\{X_{N(t-s)} \in A \mid X_0 = x(s)\} \\ &= p(t-s, x(s), A) \end{aligned}$$

thereby proving the Markov property of $x(t)$. Moreover we can use (4.1) and the relation (4.3) to establish (4.2). The process itself will live on the space of step functions with values in $\mathcal{X}$. The trajectories wait at each site for a random exponential time with parameter λ and if they are at x , jump to a new location y chosen randomly according to the probability distribution $\Pi(x, dy)$ and start afresh.

REMARK 4.2. If the Markov chain is a random walk on $\mathbb{R}$ with transition probability $\Pi(x, A) = \alpha(A - x)$ then the Markov process is the compound Poisson process with Lévy measure $\lambda \alpha$.

EXERCISE 4.1. If $X(t)$ is the compound Poisson process with a finite Lévy measure $M(dz)$, i.e.

$$E[e^{iyX(t)}] = \int (e^{iyz} - 1)M(dz)$$

then what can you say about the process of successive states of the process, ignoring the waiting times?

4.2. Semigroups of Operators

Any transition probability $\Pi(x, dy)$ on $\mathcal{X}$ defines a certain bounded linear operator Π . We denote by $\mathbf{B}$, the Banach space of all bounded functions $f(x) : \mathcal{X} \rightarrow \mathbb{R}$ that are $\mathcal{B}$ measurable with the norm

$$\|f\| = \sup_{x \in \mathcal{X}} |f(x)|$$

Then

$$(4.4) \quad \begin{aligned} (\Pi f)(x) &= \int_{\mathcal{X}} f(y) \Pi(x, dy) \\ &= E[f(X_1) \mid X_0 = x] \end{aligned}$$

In particular, the transition functions $p(t, x, dy)$, introduced in the previous section, that satisfy (4.2), define a family of bounded linear operators

$$(4.5) \quad (T_t f)(x) = \int_{\mathcal{X}} f(y) p(t, x, dy)$$

mapping the Banach space $\mathbf{B}$ into itself. The collection $\{T_t\}$ has the following properties.

- (i) For each $t > 0$, T_t maps nonnegative functions in $\mathcal{B}$ into nonnegative functions.
- (ii) For each $t > 0$ and $f \in \mathbf{B}$, $\|T_t f\| \leq \|f\|$ and therefore T_t is a bounded linear operator from $\mathbf{B} \rightarrow \mathbf{B}$ with $\|T_t\| \leq 1$. In fact, $T_t 1 = 1$ and this implies $\|T_t\| = 1$.
- (iii) With T_0 defined naturally as the identity operator I , for any $t, s \geq 0$, $T_{t+s} = T_t T_s = T_s T_t$.

In addition if $p(t, x, dy)$ is given by (4.3), for any $t \geq 0$,

$$T_t = \sum_{k=0}^{\infty} e^{-\lambda t} \frac{(\lambda t)^k}{k!} \Pi^k$$

where Π is given by (4.4). It is clear that in this case we can write

$$(4.6) \quad T_t = e^{-\lambda t} \exp[\lambda t \Pi] = \exp[\lambda t [\Pi - I]] = \exp[tL]$$

where $L = \lambda[\Pi - I]$.

Such families are called semigroups of operators and L is called the *infinitesimal generator* of the semigroup. In our case we can recover L from T_t by the formula

$$(4.7) \quad L = \lim_{t \downarrow 0} \frac{T_t - I}{t}$$

From the expansion of T_t in powers of Π this relationship is easy to derive. It is valid in the strongest sense. As operators one can check that $\|\frac{T_t - I}{t} - L\| \rightarrow 0$ as

$t \downarrow 0$. The operator L contains all the relevant information regarding the Markov process. We can consider without centering a processes with independent increments that has a Lévy measure M that integrates $|x|$ near 0 (i.e. $\int_{|x| \leq 1} |x| dM(x) < \infty$, but still with lots of small jumps).

$$(Lf)(x) = \int_{\mathbb{R}} [f(x+y) - f(x)] M(dy)$$

where $M(dy)$ is an infinite measure that integrates $|x|$ near 0. For such an L to be well defined we need a Lipschitz condition on f . Otherwise Lf may not be defined. In any case L is not a bounded operator and the expansion in powers of L is very suspect.

Although we have only looked at operators of the form

$$(4.8) \quad (Lf)(x) = \lambda \int_{\mathbb{R}} [f(y) - f(x)] \Pi(x, dy)$$

as potential candidates of generators of Markov processes one can see that operators of the form

$$(4.9) \quad (Lf)(x) = a(x) \int_{\mathbb{R}} [f(y) - f(x)] \Pi(x, dy)$$

where λ is replaced by a bounded nonnegative function $a(x)$ will work just as well. The general bounded generator L of the form (4.9) is really not all that different from (4.8). We pick $\lambda > 0$ such that $\lambda \geq a(x)$ for all x . Redefine

$$\widehat{\Pi}(x, A) = \frac{a(x)}{\lambda} \Pi(x, A) + (1 - \frac{a(x)}{\lambda}) \chi_A(x)$$

so that

$$(Lf)(x) = a(x) \int_{\mathbb{R}} [f(y) - f(x)] \Pi(x, dy) = \lambda \int_{\mathbb{R}} [f(y) - f(x)] \widehat{\Pi}(x, dy)$$

We have allowed for the possibility of not jumping from the current position x (or jumping from x back to x again) at the exponential time with some probability that depends on x and waiting for the next chance, thereby effectively reducing the jump rate to different levels at different points. See the exercise below.

EXERCISE 4.2. Let τ be an exponential random variable with parameter λ , i.e. $P[\tau \geq t] = e^{-\lambda t}$. If N is a random variable with geometric distribution, i.e. for $k \geq 1$, $P[N = k] = p(1-p)^{k-1}$, then show that the distribution of $S = \tau_1 + \cdots + \tau_N$, where τ_i are independent copies of τ and are independent of N , is again exponential with parameter λp .

The strong Markov property is defined as follows. Let τ be any stopping time. Then

$$(4.10) \quad P_x[x(\tau+t) \in A \mid \mathcal{F}_\tau] = p(t, x(\tau), A)$$

i.e. the Markov property holds for stopping times.

THEOREM 4.1 *The strong Markov property holds for jump Markov processes that are given by generators of the form (4.9) with bounded $a(\cdot)$.*

PROOF: Just as in the proof of the optional stopping theorem for martingales, we note that the property holds for stopping times that take only a countable set of values. If $\{t_j\}$ are the set of possible values of τ and $E_j = \{\omega : \tau(\omega) = t_j\}$, it is enough to prove that for $B \in \mathcal{F}_\tau$ and $j = 1, 2, \dots$,

$$(4.11) \quad P_x[\{x(\tau + t) \in A\} \cap B \cap E_j] = \int_{B \cap E_j} p(t, x(\tau), A) dP_x$$

But $B_j = B \cap E_j \in \mathcal{F}_{t_j}$ and (4.11) reduces to

$$P_x[\{x(t_j + t) \in A\} \cap B_j] = \int_{B_j} p(t, x(t_j), A) dP_x$$

which follows from the Markov property. We then approximate τ by a sequence τ_n such that $\tau_n \downarrow \tau$. Since $x(\tau + t) = x(\tau)$ for sufficiently small t there is no problem in passing to the limit as $n \rightarrow \infty$ in

$$P_x[\{x(\tau_n + t) \in A\} \cap B] = \int_B p(t, x(\tau_n), A) dP_x$$

to obtain

$$P_x[\{x(\tau + t) \in A\} \cap B] = \int_B p(t, x(\tau), A) dP_x$$

which is the strong Markov property. □

If $\mathcal{X}$ is a finite set $\{1, \dots, k\}$ then L has a matrix representation

$$(Lf)(i) = \sum_j a_{i,j} f(j)$$

and if $p_{i,j}(t)$ are the transition probabilities at time t , one can check that for $i \neq j$

$$(4.12) \quad a_{i,j} = \lim_{t \downarrow 0} \frac{p_{i,j}(t)}{t} \geq 0$$

and

$$(4.13) \quad a_{i,i} = \lim_{t \downarrow 0} \frac{p_{i,i}(t) - 1}{t} \leq 0$$

Also $\sum_j p_{i,j}(t) \equiv 1$ leads to

$$(4.14) \quad \sum_j a_{i,j} = 0 \quad \text{for } i = 1, \dots, k.$$

The last property is simply the fact that $(L1)(x) \equiv 0$. Such matrices have a characterization that is important. If f is a function on the state space and $f(x_0)$ is the maximum of the function then $(T_t f)(x_0)$ being an average of f is at most $f(x_0)$. The inequality $(T_t f)(x_0) \leq f(x_0)$ leads, upon differentiation at $t = 0$, to $(Lf)(x_0) \leq 0$. This is called the maximum principle. One can verify that for

matrices $L = \{a_{i,j}\}$ the maximum principle is valid if and only if properties (4.12), (4.13), and (4.14) are valid.

The converse is valid as well. If L is a matrix satisfying properties (4.12), (4.13), and (4.14) then we can define $P(t) = \{p_{i,j}(t)\}$, by

$$P(t) = \exp[tL] = \sum_{r=0}^{\infty} e^{-t} \frac{(tL)^r}{r!}$$

and verify that the entries of the matrix $P(t)$ are nonnegative and the entries on every row sums to 1. Such matrices, called *stochastic matrices* are precisely the class of matrices that can be one step transition probabilities for a discrete time Markov process on the finite state space $\mathcal{X} = \{1, \dots, k\}$. If every off diagonal entry $a_{i,j}$ with $i \neq j$, satisfies $a_{i,j} > 0$, then it is easy to see that $P(t)$ is a transition matrix provided t is small. But $P(t) = [P(\frac{t}{n})]^n$ and it follows that $P(t)$ is a stochastic matrix for all $t > 0$. The case when $a_{i,j} \geq 0, \forall i \neq j$ is handled by approximating it by $\{a_{i,j}^{(n)}\}$ with $a_{i,j}^{(n)} > 0, \forall i \neq j$.

EXERCISE 4.3. If $P(t)$ are stochastic matrices satisfying $P(t+s) = P(t)P(s) = P(s)P(t)$ for all $t, s > 0$ and $P(t) \rightarrow I$ as $t \rightarrow 0$, then show that

$$L = \lim_{t \downarrow 0} \frac{P(t) - I}{t}$$

exists and satisfies (4.12), (4.13), and (4.14).

Hint: Show that if $u = \int_a^b P(s)v ds$ for some $v \in \mathbb{R}^k$ and $0 < a < b$, then $P(t)u$ is differentiable at $t = 0$. Then show that such vectors u span $\mathbb{R}^k$.

EXERCISE 4.4. With $\mathcal{X} = \{1, 2\}$ and

$$A = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$$

calculate the transition matrix $P(t)$ at time t .

4.3. Example: Birth and Death Processes

Imagine a population with a size that can be any nonnegative integer. The state space $\mathcal{X} = \mathbb{Z}^+ = \{n : n \geq 0\}$. If the current size of the population is n , either a birth or a death can take place and change the population from n to $n \pm 1$. Of course if the current population size is 0 only a birth can occur. Let us denote the birth rate by λ_n and the death rate by μ_n . This is interpreted as

$$\begin{aligned} a_{n,n+1} &= \lambda_n & \text{for } n \geq 0 \\ a_{n,n-1} &= \mu_n & \text{for } n \geq 1 \end{aligned}$$

and of course $\mu_0 = 0$ and

$$a_{n,n} = -(\lambda_n + \mu_n) \quad \text{for } n \geq 0.$$

If each individual gives birth at rate λ independently of others, then $\lambda_n = n\lambda$. Similarly if the death rate is μ for each individual we have $\mu_n = n\mu$. For a queue with a single server where the arrival time and the service time are exponential $\lambda_n = \lambda$ for $n \geq 0$ and $\mu_n = \mu$ for $n \geq 1$.

EXERCISE 4.5. One can think of a branching process in continuous time as well. Assume every individual in a population renews itself at rate λ , i.e. at a random time τ with $P[\tau \geq t] = e^{-\lambda t}$. At the renewal time the individual is replaced by a random number $Z \geq 0$ of offsprings. $P[Z = k] = p_k$. Write down the matrix $L = \{a_{i,j}\}$ for this model.

4.4. Markov Processes and Martingales

Let us consider a Markov process of the type considered earlier in Section 4.1. Suppose L is the infinitesimal generator given as

$$(4.15) \quad (Lf)(x) = a(x) \int_X [f(y) - f(x)] \Pi(x, dy)$$

where $a(x)$ is a bounded measurable function and Π is a transition probability on X . Let $T(t) = \exp[tL]$ be the associated semigroup operators

$$(4.16) \quad (T_t f)(x) = \int_X f(y) p(t, x, dy)$$

where $p(t, x, dy)$ are the transition probabilities corresponding to L . We saw earlier by differentiating term by term in the expansion

$$T_t = \sum_{k=0}^{\infty} \frac{(tL)^k}{k!}$$

of T_t in powers of L that

$$\begin{aligned} \frac{d T_t}{dt} &= \sum_{k=1}^{\infty} \frac{t^{k-1} L^k}{(k-1)!} \\ &= T_t L = L T_t \end{aligned}$$

In particular, (by integrating)

$$(4.17) \quad T_t f - f = \int_0^t T_s L f ds$$

In terms of the Markov family of measures $(\Omega, \mathcal{F}_t, \{P_x\})$, the relation (4.17) can be rewritten as

$$(4.18) \quad E^{P_x} \left[f(x(t)) - f(x(0)) - \int_0^t (Lf)(x(s)) ds \right] = 0$$

If we use the homogeneity in time of the transition probabilities of Markov processes $\{P_x\}$

$$\begin{aligned} &E^{P_x} \left[f(x(t)) - f(x(s)) - \int_s^t (Lf)(x(\sigma)) d\sigma \mid \mathcal{F}_s \right] \\ &= (T_{t-s} f)(x(s)) - f(x(s)) - \int_s^t (T_{\sigma-s} Lf)(x(s)) d\sigma \\ &= 0 \end{aligned}$$

Thus we have proved the following theorem.

THEOREM 4.2 *Let $\{P_x\}$ be the family of Markov processes with transition probabilities $\{p(t, x, dy)\}$ corresponding to a semigroup of operators $\{T_t\}$ with generator L of the form (4.15). Then for any x ,*

$$(4.19) \quad M_f(t) = f(x(t)) - f(x(0)) - \int_0^t (Lf)(x(\sigma))d\sigma$$

is a martingale with respect to any $(\Omega, \mathcal{F}_t, P_x)$.

A similar result was proved for processes with independent increments in Theorem 2.3. We can generate other martingales. We can use Lemma 1.5, with a damping factor $A(t) = e^{-\lambda t}$.

THEOREM 4.3 *For any λ*

$$M_f(t)e^{-\lambda t} - M_f(0) + \lambda \int_0^t M_f(s)e^{-\lambda s} ds$$

which reduces to

$$(4.20) \quad e^{-\lambda t} f(x(t)) - f(x(0)) - \int_0^t e^{-\lambda \sigma} [(Lf) - \lambda f](x(\sigma))d\sigma$$

is again a martingale.

PROOF: The proof is essentially a computation

$$\begin{aligned} e^{-\lambda t} \left[f(x(t)) - f(x(0)) - \int_0^t (Lf)(x(s))ds \right] + \lambda \int_0^t M_f(s)e^{-\lambda s} ds = \\ e^{-\lambda t} f(x(t)) - f(x(0)) + \int_0^t B'(s)ds \end{aligned}$$

with

$$\begin{aligned} B'(t) &= \lambda e^{-\lambda t} f(x(0)) + \lambda e^{-\lambda t} \int_0^t (Lf)(x(s))ds - e^{-\lambda t} (Lf)(x(t)) \\ &\quad + \lambda e^{-\lambda t} \left[f(x(t)) - f(x(0)) - \int_0^t (Lf)(x(s))ds \right] \\ &= e^{-\lambda t} [\lambda f(x(t)) - (Lf)(x(t))] \end{aligned}$$

□

One is often interested in the first exit time

$$\tau_A = \{\inf t : x(t) \notin A\}$$

of the process from sets $A \in \mathcal{B}$. Because the process stays at any site for a positive duration

$$\{\tau_A \leq t\} = \bigcup_{\substack{s \leq t \\ s \text{ rational}}} \{x(s) \notin A\} \cup \{x(t) \notin A\}$$

τ_A is seen to be a stopping time and if $\tau_A(\omega)$ is finite then $x(\tau_A) \in A^c$. The following theorem gives a way to compute the joint distribution of τ_A and $x(\tau_A)$.

THEOREM 4.4 For $\lambda > 0$ the function

$$U(x) = E_x[e^{-\lambda \tau_A} f(x(\tau_A))]$$

is the unique solution of

$$(LU)(x) = \lambda U(x) \quad \text{for } x \in A$$

with $U(x) = f(x)$ on A^c .

PROOF: Clearly for $x \in A^c$, $\tau_A = 0$ a.e. P_x and $U(x) = f(x)$. For $x \in A$, we wait for the first jump τ from x . By our construction of the jump Markov processes corresponding to $(a(\cdot), \Pi(x, dy))$ the time of the first jump τ and the landing point y are independent. τ is an exponential random variable with parameter $a(x)$ and y has the distribution $\Pi(x, dy)$. Therefore starting from

$$U(x) = E_x\{e^{-\lambda \tau_A} f(x(\tau))\}$$

and conditioning with respect to $\mathcal{F}_\tau$ where

$$\tau = \{\inf t : x(t) \neq x(0)\}$$

we get

$$\begin{aligned} U(x) &= E_x\{e^{-\lambda \tau} E_{x(\tau)}\{e^{-\lambda \tau_A} f(x(\tau_A))\}\} \\ &= E_x\{e^{-\lambda \tau} U(x(\tau))\} \\ &= E_x\{e^{-\lambda \tau} E_x\{U(x(\tau))\}\} \\ &= \frac{a(x)}{\lambda + a(x)} \int_X U(y) \Pi(x, dy) \end{aligned}$$

We can rewrite this as

$$(\lambda + a(x))U(x) = a(x) \int_X U(y) \Pi(x, dy)$$

or

$$(LU)(x) = a(x) \int_X [U(y) - U(x)] \Pi(x, dy) = \lambda U(x).$$

Conversely if U solves $LU = \lambda U$ on A^c with $U = f$ on A , we know from (4.20) that

$$e^{-\lambda t} U(x(t)) - U(x(0)) - \int_0^t e^{-\lambda s} [LU - \lambda U](x(s)) ds$$

is a P_x martingale. We apply Doob's stopping theorem with the stopping time τ_A . Because $(LU)(x(s)) - \lambda U(x(s)) = 0$ for $0 \leq s < \tau_A$ and $U(x(\tau_A)) = f(x(\tau_A))$, we conclude that

$$(4.21) \quad U(x) = E_x\{e^{-\lambda \tau_A} f(x(\tau_A))\}$$

proving uniqueness. However we need to be sure that the stopping time τ_A which need not be bounded or even almost surely finite, will work for Doob's optional stopping theorem. The expectation in (4.21) is clearly over the set where $\tau_A < \infty$.

The integrand is naturally 0 otherwise. In any case $\tau_A \wedge t$ is a bounded stopping time and we surely have

$$E^{P_x} [e^{-\lambda(\tau_A \wedge t)} U(x(\tau_A \wedge t))] = U(x)$$

Since $\lambda > 0$, we can let $t \rightarrow \infty$ and

$$\begin{aligned} U(x) &= \lim_{t \rightarrow \infty} E^{P_x} [e^{-\lambda(\tau_A \wedge t)} U(x(\tau_A \wedge t))] \\ &= E^{P_x} [e^{-\lambda\tau_A} U(x(\tau_A))] = E^{P_x} [e^{-\lambda\tau_A} f(x(\tau_A))] \end{aligned}$$

□

REMARK 4.3. There are some other related equations which have unique solutions expressible in terms of the process. For instance for $\lambda > 0$,

$$(4.22) \quad \lambda U - LU = g \quad \text{for } x \in A$$

with $U = f$ for $x \in A^c$ can be solved uniquely and the solution is given by

$$(4.23) \quad U(x) = E^{P_x} \left[e^{-\lambda\tau_A} f(x(\tau_A)) + \int_0^{\tau_A} e^{-\lambda s} g(x(s)) ds \right]$$

REMARK 4.4. To solve the equation

$$LU = 0 \quad \text{for } x \in A$$

uniquely with $U = f$ for $x \in A^c$ we require the assumption

$$P_x \{\tau_A < \infty\} = 1$$

for $x \in A$ and the solution is given by

$$(4.24) \quad U(x) = E^{P_x} [f(x(\tau_A))]$$

REMARK 4.5. To solve for a bounded g ,

$$LU = g \quad \text{for } x \in A$$

uniquely with $U = f$ for $x \in A^c$ we need the assumption

$$E_x \{\tau_A\} < \infty$$

for $x \in A$ and the solution is given by

$$(4.25) \quad U(x) = E^{P_x} \left[f(x(\tau_A)) - \int_0^{\tau_A} g(x(s)) ds \right]$$

EXERCISE 4.6. Prove the claim in Remark 4.3 and show that under suitable conditions λ can be taken as 0, thereby proving the claims made in Remarks 4.4 and 4.5.

REMARK 4.6. It is not hard to construct counterexamples with $P_x[\tau_A = \infty] > 0$ where the equation $LU = 0$ on A and $U = f$ on A^c has multiple solutions. The trivial example is $L \equiv 0$ where nothing ever moves. Any extension of f from A to $\mathcal{X}$ is a solution.

REMARK 4.7. It is possible that the process when it exits from A and enters A^c for the first time, it will only enter in a certain subset $B \subset A^c$. For instance when a birth and death process exits from the set $[0, 1, \dots, n-1]$ it can only exit at n . The function f of Theorem 4.4, formulae (4.24) and (4.25) need only be specified at points in B , which in this case consists of the single point n .

4.5. Explosion

When we have a state space $\mathcal{X}$ that is infinite, it can easily happen that the process, through successive jumps, moves to states that have shorter waiting times, i.e. have larger jump rates. This can result in an infinite number of jumps occurring in a finite time. Consider the example of a birth and death process with $\lambda_n = (n+1)^2$ and $\mu_n = 0$. The state space is $\mathbb{Z}^+$ the set of integers $\{0, 1, 2, \dots\}$. The process waits for an exponentially distributed random time at every site before moving one step. The waiting time τ'_n at site n has expected value $(n+1)^{-2}$. If the process starts at 0, it moves successively through $1, 2, 3, \dots$ waiting for an exponential time τ'_n at n with $E\{\tau'_n\} = \frac{1}{(n+1)^2}$ for $n \geq 0$. A simple calculation shows that

$$\sum_{n=0}^{\infty} E\{\tau'_n\} = \sum_{n=0}^{\infty} \frac{1}{(n+1)^2} < \infty$$

and by Fubini's theorem

$$\sum_{n=0}^{\infty} \tau'_n = \zeta < \infty \text{ a.e.}$$

If we define $\tau_n = \tau'_0 + \dots + \tau'_{n-1}$ and $\zeta = \lim_{n \rightarrow \infty} \tau_n$, the process will reach ∞ , i.e. exit from all finite sets by time ζ . We call ζ the explosion time and in our case it is finite with probability 1.

This can happen with any Markov process that we want to construct from its infinitesimal generator

$$(Lf)(x) = a(x) \int_{\mathcal{X}} [f(y) - f(x)] \Pi(x, dy)$$

if the function $a(\cdot)$ is unbounded. If ζ is the explosion time i.e. the first time when we have an infinite number of jumps, the process is well defined up to that time by the canonical construction.

THEOREM 4.5 *The Markov process corresponding to*

$$(4.26) \quad (Lf)(x) = a(x) \int_{\mathcal{X}} [f(y) - f(x)] \Pi(x, dy)$$

starting from x_0 explodes in finite time with positive probability if and only if there is a nonnegative bounded function U with $U(x_0) > 0$ that satisfies

$$(4.27) \quad (LU)(x) \geq \sigma U(x) \quad \text{for } x \in \mathcal{X}$$

for some $\sigma > 0$.

PROOF: Given $\sigma > 0$, we define

$$U_\sigma(x) = E^{P_x}[e^{-\sigma\zeta}]$$

Then $0 \leq U_\sigma(x) \leq 1$ and for any $x_0 \in \mathcal{X}$ and there is explosion with positive probability under P_{x_0} if and only if $U_\sigma(x_0) > 0$.

Let τ be the time of the first jump and $y = x(\tau)$ the new site after the jump. In our construction of the jump Markov process, we saw that τ and y were independent, τ having an exponential distribution with parameter $a(x)$ and y being distributed according to $\Pi(x, dy)$. Therefore

$$\begin{aligned} U_\sigma(x) &= E^{P_x}[E^{P_x}[e^{-\sigma\zeta} \mid \mathcal{F}_\tau]] \\ &= E^{P_x}[e^{-\sigma\tau} E^{P_x}[e^{-\sigma(\zeta-\tau)} \mid \mathcal{F}_\tau]] \\ &= E^{P_x}[e^{-\sigma\tau} U_\sigma(x(\tau))] \\ &= E^{P_x}[e^{-\sigma\tau}] E^{P_x}[U_\sigma(x(\tau))] \\ &= \frac{a(x)}{a(x) + \sigma} \int_{\mathcal{X}} U_\sigma(y) \Pi(x, dy) \end{aligned}$$

This can be rewritten as

$$(4.28) \quad (LU_\sigma)(x) = \sigma U_\sigma(x)$$

which is valid for every $x \in \mathcal{X}$. Conversely let $U(\cdot)$ satisfy (4.27) with $0 < U(x) \leq C$. Consider the martingale

$$\begin{aligned} M(t) &= e^{-\sigma t} U(x(t)) - \int_0^t e^{-\sigma s} [LU - \sigma U](x(s)) ds \\ &= e^{-\sigma t} U(x(t)) - A(t) \end{aligned}$$

It follows from (4.27) that $A(t)$ is an increasing function. This makes the process $e^{-\sigma t} U(x(t))$ a submartingale which in addition is uniformly bounded and therefore Doob's optional theorem can be used with unbounded stopping times. If τ_n is the time of the n -th jump and $x(\tau_n)$ the location of the process after the n -th jump, by Doob's stopping theorem, $e^{-\sigma\tau_n} U(x(\tau_n))$ is a submartingale. In particular

$$U(x_0) \leq E^{P_{x_0}}[e^{-\sigma\tau_n} U(x(\tau_n))] \leq C E^{P_{x_0}}[e^{-\sigma\tau_n}]$$

Letting $n \rightarrow \infty$,

$$U(x_0) \leq C E^{P_{x_0}}[e^{-\sigma\zeta}]$$

This proves that $P_{x_0}[\zeta < \infty] \geq \frac{U(x_0)}{C} > 0$. □

To prove nonexplosion, we need to show the nonexistence of a bounded solution, which is hard. The following theorem however provides a positive criterion that can be used to prove nonexplosion.

THEOREM 4.6 *If, for some $\sigma > 0$, there exists a nonnegative function $U(\cdot)$ on $\mathcal{X}$, that satisfies*

$$(LU)(x) \leq \sigma U(x) \quad \text{for all } x$$

and $U(x) \rightarrow \infty$ whenever $a(x) \rightarrow \infty$ then we can not have explosion.

PROOF: To see this we note that $e^{-\sigma t}U(x(t))$ is a supermartingale. Therefore for any finite T ,

$$U(x) \geq E_x[e^{-\sigma(\tau_n \wedge T)}U(x(\tau_n \wedge T))]$$

Letting $T \rightarrow \infty$ and using Fatou's lemma, it is not hard to see that

$$(4.29) \quad U(x) \geq E_x[e^{-\sigma\tau_n}U(x(\tau_n))]$$

Let Ω_0 be the set on which τ_n tends to a finite limit as $n \rightarrow \infty$. Since the waiting time at $x(\tau_n)$ is exponential with expected value $\frac{1}{a(x(\tau_n))}$, we must have the almost sure convergence of the series

$$\sum_n \frac{1}{a(x(\tau_n))}$$

on Ω_0 and as a consequence $\lim_{n \rightarrow \infty} a(x(\tau_n)) = \infty$. By our assumption $U(x(\tau_n)) \rightarrow \infty$ whenever $a(x(\tau_n)) \rightarrow \infty$. But the bound (4.29) does not allow $U(x(\tau_n)) \rightarrow \infty$ while τ_n remains bounded with nonzero probability. $\square$

If $a(x)$ is unbounded, then L is an unbounded operator. We cannot use the expansion

$$T_t = \sum_{j=0}^{\infty} \frac{(tL)^j}{j!}$$

and if there is explosion at a finite time with positive probability, the semigroup T_t is not determined by L i.e. by $a(\cdot)$ and $\Pi(x, dy)$. We will illustrate this in the context of birth and death processes.

We start with a birth and death process on $\mathbb{Z}^+$ the set of nonnegative integers $i \geq 0$. The transition rates from $i \rightarrow i+1$ and $i \rightarrow i-1$ are given, respectively, by λ_i and μ_i with μ_0 being 0. Given $\{\lambda_i\}, \{\mu_i\}$, we solve the equation

$$(4.30) \quad (LU)(i) = \sigma U(i)$$

which reduces in this case to

$$(4.31) \quad (LU)(i) = \lambda_i U(i+1) + \mu_i U(i-1) - (\lambda_i + \mu_i)U(i) = \sigma U(i)$$

for $i = 0, 1, 2, \dots, n-1$ with boundary condition $U(n) = 1$. Notice that because $\mu_0 = 0$,

$$\lambda_0 U(1) = (\lambda_0 + \sigma)U(0)$$

or

$$(4.32) \quad U(1) = \frac{\lambda_0 + \sigma}{\lambda_0} U(0)$$

The recurrence relation (4.31) can be rewritten as

$$(4.33) \quad U(i+1) = \frac{(\lambda_i + \mu_i + \sigma)U(i) - \mu_i U(i-1)}{\lambda_i}$$

Together with (4.32), (4.33) determines $U(i)$ successively for all $i \geq 2$. It follows from Theorem 4.4 and Remark 4.7 that if U is the solution of (4.30) with $U(n) = 1$ and $\tau_n = \inf\{t : x(t) = n\}$ then

$$U(i) = E_i[e^{-\sigma \tau_n}]$$

For the validity of this formula we need only the values of $\{\lambda_j, \mu_j\}$ for $0 \leq j \leq n-1$. We can define them arbitrarily for $j \geq n$ and so it does not matter if the sequence $\{\lambda_j, \mu_j\}$ is unbounded. On the other hand if $U(\cdot)$ is any solution of (4.30), so is

$$U_n(i) = \frac{U(i)}{U(n)}$$

and $U_n(n) = 1$. Hence

$$E_i[e^{-\sigma \tau_n}] = \frac{U(i)}{U(n)} \quad \text{for } 0 \leq i \leq n-1$$

Clearly $U(n)$ is $\uparrow$ in n and nonexplosion is equivalent to

$$\lim_{n \rightarrow \infty} U(n) = \infty$$

For determiner's $U(n)$ we can assume without loss of generality that $U(0) = 1$. We can rewrite equation (4.33) as

$$(4.34) \quad U(i+1) - U(i) = \frac{\mu_i}{\lambda_i}[U(i) - U(i-1)] + \frac{\sigma}{\lambda_i}U(i)$$

for $i \geq 1$ with

$$U(0) = 1, \quad U(1) = (1 + \frac{\sigma}{\lambda_0})$$

EXERCISE 4.7. Show that the solution of equation (4.34) is bounded if and only if the solution of

$$(4.35) \quad U(i+1) - U(i) = \frac{\mu_i}{\lambda_i}[U(i) - U(i-1)] + \frac{1}{\lambda_i}$$

for $i \geq 2$ with $U(0) = 0$ and $U(1) = 1$ is bounded.

EXERCISE 4.8. Show that the necessary and sufficient conditions for the solution of (4.35) with $U(0) = 0$ and $U(1) = 1$ to be bounded are

$$\sum_{n=1}^{\infty} \frac{\mu_1 \mu_2 \cdots \mu_n}{\lambda_1 \lambda_2 \cdots \lambda_n} \sum_{j=1}^n \frac{\lambda_1 \lambda_2 \cdots \lambda_{j-1}}{\mu_1 \mu_2 \cdots \mu_{j-1}} \frac{1}{\mu_j} < \infty$$

and

$$\sum_{n=1}^{\infty} \frac{\mu_1 \mu_2 \cdots \mu_n}{\lambda_1 \lambda_2 \cdots \lambda_n} < \infty$$

REMARK 4.8. Although we know that boundedness of the solution $U(\cdot)$ of (4.33) for some value of $\sigma > 0$ implies the same for all other positive values of σ , we can see it directly by the following simple inequality, which is a consequence of Jensen's inequality. For any convex function ϕ

$$(4.36) \quad (L\phi(u))(x) \geq \phi'(u(x))(Lu)(x)$$

In particular, if we denote the solution of

$$Lu = \sigma u \text{ for } x \in A; \quad u(x) = 1 \text{ on } A^c$$

by u_σ then for $\alpha > 1$, $v = u^\alpha$ satisfies, by (4.36)

$$Lv = Lu^\alpha \geq \alpha u^{\alpha-1} Lu = \sigma \alpha v$$

establishing for $\alpha > 1$, and $\sigma > 0$

$$u_\sigma \geq u_{\alpha\sigma} \geq u_\sigma^\alpha$$

EXERCISE 4.9. Prove the inequality (4.36).

EXERCISE 4.10. Prove the comparison property. If L satisfies

$$Lu \geq \sigma u \text{ for } x \in A; \quad u(x) = 1 \text{ on } A^c$$

and v solves

$$Lv = \sigma v \text{ for } x \in A; \quad v(x) = 1 \text{ on } A^c$$

then

$$v \geq u$$

REMARK 4.9. In fact, the probability of explosion in a finite time is either identically 1 for all initial states or identically equal to 0 for all states. To see this

$$P_i\{\zeta < \infty\} \geq E_i\{e^{-\zeta}\} = \lim_{n \rightarrow \infty} E_i\{e^{-\tau_n}\} = \lim_{n \rightarrow \infty} \frac{U(i)}{U(n)} = \frac{U(i)}{U(\infty)}$$

since $\lim_{n \rightarrow \infty} U(n) = U(\infty)$ exists. Now we see that

$$\lim_{i \rightarrow \infty} V(i) = 1$$

where

$$V(i) = P_i\{\zeta < \infty\}$$

On the other hand one can verify easily that

$$(LV)(i) = 0$$

for all i (including 0), and the only solution to this are the constants. Therefore $V(i) \equiv c$ and of course $c = 1$, if $U(\infty) < \infty$. Otherwise $V(i) \equiv 0$.

EXAMPLE 4.1. For a birth and death process with $\lambda_n - \mu_n \leq C(n+1)$ for all n , there is no explosion. Let us try $U(n) = (n+1)^\alpha$ with $\alpha < 1$. Since $U(\cdot)$ is concave

$$\begin{aligned} (LU)(n) &= \lambda_n(U(n+1) - U(n)) + \mu_n(U(n-1) - U(n)) \\ &\leq (\lambda_n - \mu_n)(U(n) - U(n-1)) \end{aligned}$$

$$\begin{aligned} &\leq C\alpha(n+1)(n+1)^{\alpha-1} \\ &= C\alpha U(n) \end{aligned}$$

4.6. Recurrence and Transience

For a birth and death process it is easy to investigate the solutions of the equation

$$\mu_n U(n-1) - (\lambda_n + \mu_n)U(n) + \lambda_n U(n+1) = 0$$

for $n \geq 1$. The equation can be rewritten as

$$U(n+1) - U(n) = \frac{\mu_n}{\lambda_n} [U(n) - U(n-1)]$$

If we normalize $U(1) - U(0) = 1$, we obtain

$$U(n+1) - U(n) = \prod_{j=1}^n \frac{\mu_j}{\lambda_j}$$

and with $U(0) = 0$

$$U(n+1) = 1 + \sum_{k=1}^n \prod_{j=1}^k \frac{\mu_j}{\lambda_j}$$

It is easy to see that $U(n)$ becomes unbounded as $n \rightarrow \infty$ if and only if

$$\sum_{k=1}^{\infty} \prod_{j=1}^k \frac{\mu_j}{\lambda_j} = \infty$$

Now we return to the question of recurrence. Given three states $0 < i < n$ we try to calculate the probability

$$P_i\{\tau_n < \tau_0\}$$

of visiting n before visiting 0 when the process starts from i . Here τ_j is the first visiting time of the state j . This probability is defined well before explosion and we can alter λ_k, μ_k for $k \geq n$ without changing the probability. In fact we can assume without any loss of generality that L is a bounded operator. If $LU(i) = 0$ for $0 \leq i \leq n$, then $U(x(\tau_n \wedge t))$ is seen to be a martingale because we can always modify L, U and g beyond n . In particular, if $U(0) = 0$, by Doob's theorem,

$$P_i\{\tau_n < \tau_0\} = \frac{U(i)}{U(n)}$$

and there is recurrence if and only if $U(n) \rightarrow \infty$. Actually we do not have to solve the equation $LU = 0$. If we can find a supersolution $LU \leq 0$ which goes to ∞ as $n \rightarrow \infty$ that works just as well to prove recurrence and the existence of a positive subsolution that remains bounded as $n \rightarrow \infty$ will establish transience. In particular, the necessary and sufficient condition for recurrence, with nonexplosion, is

$$\sum_{n=1}^{\infty} \frac{\mu_1 \cdots \mu_n}{\lambda_1 \cdots \lambda_n} = \infty$$

EXERCISE 4.11. For a birth and death process, if $\mu_n - \lambda_n \leq 0$ for $n \geq n_0$, then the process is recurrent.

4.7. Invariant Distributions

An invariant distribution $q(dy)$ on $(\mathcal{X}, \mathcal{B})$ is a probability distribution such that

$$q(A) = \int_{\mathcal{X}} p(t, x, A) q(dx)$$

for all $A \in \mathcal{B}$ and $t > 0$. If L is bounded this can be verified by verifying the infinitesimal relation

$$\int_{\mathcal{X}} (Lf)(x) q(dx) = 0$$

for all $f \in \mathbf{B}$. To see this we only have to observe that

$$\frac{d}{dt} \langle T_t f, q \rangle = \langle L T_t f, q \rangle = 0.$$

Or equivalently it is enough to verify that

$$(4.37) \quad (L^* q)(A) = \int_{\mathcal{X}} [\Pi(y, A) - \chi_A(y)] a(y) q(dy) = 0$$

The following theorem is therefore easy to prove.

THEOREM 4.7 *Let $p(t, x, dy)$ be the transition probability corresponding to a bounded L given by (4.9). Given a probability distribution $q(dy)$ on $\mathcal{X}$ it is an invariant distribution, i.e. it satisfies for all $x \in \mathcal{X}$, $t > 0$ and $A \in \mathcal{B}$,*

$$\int_{\mathcal{X}} p(t, x, A) q(dx) = q(A)$$

if and only if

$$L^* q = 0$$

or (4.37).

In general the invariant distribution need not be unique.

EXAMPLE 4.2. In the case of a birth and death process $\{q(i)\}$ is invariant if

$$q(i-1)\lambda_{i-1} + q(i+1)\mu_{i+1} = q(i)(\lambda_i + \mu_i)$$

for $i \geq 1$, with

$$q(1)\mu_1 = q(0)\lambda_0 \quad \text{or} \quad \frac{q(1)}{q(0)} = \frac{\lambda_0}{\mu_1}$$

We can assume for the moment that $q(0) = 1$. Then

$$q(i) = \frac{\lambda_0 \lambda_1 \cdots \lambda_{i-1}}{\mu_1 \mu_2 \cdots \mu_i}$$

satisfies

$$\begin{aligned} q(i-1)\lambda_{i-1} + q(i+1)\mu_{i+1} &= q(i-1) \left[\lambda_{i-1} + \frac{\lambda_{i-1}\lambda_i}{\mu_i} \right] \\ &= q(i-1) \frac{\lambda_{i-1}}{\mu_i} (\lambda_i + \mu_i) = q(i)(\lambda_i + \mu_i) \end{aligned}$$

or

$$q(i) = \frac{\lambda_{i-1}}{\mu_i} q(i-1)$$

The convergence of the series

$$(4.38) \quad Z = \sum_{i=0}^{\infty} q(i) = 1 + \sum_{i=1}^{\infty} \frac{\lambda_0 \lambda_1 \cdots \lambda_{i-1}}{\mu_1 \mu_2 \cdots \mu_i} < \infty$$

is necessary and sufficient for the measure $q(\cdot)$ to have finite total mass Z . We can then normalize it so that $\frac{1}{Z}q(i)$ is the invariant probability distribution. While it is clear that when L is bounded i.e. when $\{\lambda_n + \mu_n\}$ is bounded $q(\cdot)$, constructed under the assumption (4.38), is really an invariant probability for the process. This is not so obvious in the unbounded case. It is however true in the absence of explosion.

The proof is not very hard. We can truncate the system at size n by redefining $\lambda_n = 0$ and then the finite system has an invariant measure

$$q_n(i) = \frac{1}{Z_n} \frac{\lambda_0 \lambda_1 \cdots \lambda_{i-1}}{\mu_1 \mu_2 \cdots \mu_i}$$

with

$$Z_n = 1 + \sum_{j=1}^n \frac{\lambda_0 \lambda_1 \cdots \lambda_{j-1}}{\mu_1 \mu_2 \cdots \mu_j}$$

If we let $n \rightarrow \infty$, the absence of explosion yields

$$p_{i,j}^{(n)}(t) \rightarrow p_{i,j}(t)$$

as $n \rightarrow \infty$ for all i, j and $t > 0$. The invariance of $q_n(\cdot)$ for the truncated process implies that the limit $q(\cdot)$ of $\{q_n(\cdot)\}$ is an invariant distribution for L .

EXERCISE 4.12. Invariant measures and renewal times. Let τ be a stopping time such that for every $x \in \mathcal{X}$, $P_x[x(\tau) = x_0] = 1$ where $x_0 \in \mathcal{X}$. Assume that $E^{P_{x_0}}[\tau] = m_0 < \infty$. Then

$$q(A) = \frac{1}{m_0} E^{P_{x_0}} \left[\int_0^\tau \mathbf{1}_A(x(s)) ds \right]$$

is an invariant probability measure for the process.

EXERCISE 4.13. Consider a single server queue. It is a birth and death process with a constant birthrate λ and a constant death rate μ . Show that there is a finite invariant probability distribution if and only if $\mu > \lambda$. Find the invariant distribution. What is the expected proportion of times the server is idle?

4.8. Beyond Explosion

The choice of how to define the process beyond explosion is up to us. Anything reasonable will give us a process. For instance at the explosion time we can reinitiate at the state 0. Start afresh with a clean slate. Now the process is well defined for all times. One may see several explosions and each time we restart from 0. By the time we run out of luck again, i.e. when we see an infinite number of explosions, real time would have become infinite. In terms of analysis this corresponds to “boundary conditions” that U has to satisfy for the formal relation

$$(LU)(i) = f(i)$$

to be really valid. In the case we just discussed, it is

$$\lim_{i \rightarrow \infty} U(i) = U(0)$$

It is as if we are really on a circle and as $i \rightarrow \infty$, it approaches 0 from the other side. For each rule about how to continue after explosion there will be a different boundary condition. Another rule could be to create a new state called ∞ or “cemetery” and at the explosion time move the process to that state and bury it there, i.e. let it stay there for ever. Now for $LU = f$ to be well defined both U and f should have limits as $i \rightarrow \infty$ with $\lim_{i \rightarrow \infty} f(i) = 0$.

The proofs are left as exercises.

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CHAPTER 5

Brownian Motion**5.1. Definition of Brownian Motion**

We have constructed processes with independent stationary increments that have been built from jumps. The distributions involved have been variants of the Poisson distribution. We now turn our attention to the other basic process, one that is the building block for processes with continuous trajectories. The underlying distribution here is the Gaussian or normal distribution. We will consider a process with independent increments such that the distribution μ_t of the increment over an interval of length t has the Gaussian or normal distribution with mean mt and variance $\sigma^2 t$. We can assume without loss of generality that $m = 0$ and $\sigma^2 = 1$. If we want to construct such a process in some arbitrary but fixed time interval $[0, 1]$, we need to define the joint distribution $\mu_{t_1, \dots, t_n}$ of $x(t_1), \dots, x(t_n)$ at n time points $0 \leq t_1 < t_2 < \dots < t_n \leq 1$. This will be an n -dimensional joint normal distribution on R^n with mean 0 and covariance

$$\rho(t_i, t_j) = E\{x(t_i)x(t_j)\} = E\{x(t_i)[x(t_i) + (x(t_j) - x(t_i))]\} = t_i$$

if $t_i < t_j$. We conclude that for any s and t

$$E\{x(s)x(t)\} = s \wedge t = \min(s, t).$$

Since $x(0) = 0$ with probability 1 we need to define the measure $\mu_{t_1, \dots, t_n}$ for $0 < t_1 < t_2 < \dots < t_n \leq 1$. It has a multivariate density with respect to the Lebesgue measure on R^n given by

$$f_{t_1, \dots, t_n}(x_1, \dots, x_n) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi(t_i - t_{i-1})}} \exp\left\{-\sum_{i=1}^n \frac{(x_i - x_{i-1})^2}{2(t_i - t_{i-1})}\right\}$$

with the convention that $x_0 = 0$. The family $\{\mu_{t_1, \dots, t_n}\}$ is a consistent family of finite-dimensional distributions on the space Ω of real valued functions on $[0, 1]$.

The moments of the normal distribution with mean 0 and variance σ^2 are given by

$$E[X^{2n}] = c_n [E[X^2]]^n = c_n \sigma^{2n}$$

where $c_n = 1 \cdot 3 \cdot \dots \cdot (2n - 1)$. In particular

$$E[|x(t) - x(s)|^4] = 3[E[|x(t) - x(s)|^2]]^2 = 3|t - s|^2$$

By Theorem A.1 of Kolmogorov's proved in the Appendix, we can define a measure P on $C[0, 1]$ that corresponds to our process with Gaussian independent increments. This process is called the Brownian motion and the measure P is called the Wiener measure. It was constructed originally by Norbert Wiener in [8].

REMARK 5.1. Actually the Wiener measure is supported on trajectories that satisfy a Hölder condition with a positive exponent γ . Let us note that according to Remark A.5, γ can be any exponent satisfying $\gamma < \frac{1}{4}$. We can go to higher moments

$$E[|x(t) - x(s)|^{2n}] = c_n[E[|x(t) - x(s)|^2]]^n = c_n|t - s|^n$$

and improve it to $\gamma < \frac{n-1}{2n}$. Taking n large we can get any exponent $\gamma < \frac{1}{2}$.

EXERCISE 5.1. Show that for any constant A

$$P[x(\cdot) : |x(t) - x(s)| \leq A|t - s|^{\frac{1}{2}} \text{ for all } s, t] = 0$$

In particular, Brownian paths do not satisfy a Hölder condition with exponent $\frac{1}{2}$. [Hint: Observe that

$$A \geq \sup_{0 \leq s < t \leq T} \frac{|x(t) - x(s)|}{\sqrt{|t - s|}} \geq \sup_j \sqrt{n} \left| x\left(\frac{j+1}{n}\right) - x\left(\frac{j}{n}\right) \right|$$

A is larger than the maximum of the absolute values of n independent standard Gaussian random variables. Since n can be arbitrarily large and a Gaussian random variable is unbounded A has to be infinite.]

One can easily define the Wiener measure on $C[0, \infty)$. The natural σ -field $\mathcal{F}$ on $C[0, \infty)$ is generated by $\{x(s) : 0 \leq s < \infty\}$ and it has sub σ -fields $\mathcal{F}_t$ generated by $\{x(s) : 0 \leq s \leq t\}$. Since we know we can construct Brownian motion on $C[0, 1]$, we can take an infinite number of independent copies $\{y_j(\cdot)\}$ and just concatenate them. We define by induction for $n \leq t \leq n+1$

$$x(t) = x(n) + y_{n+1}(t - n)$$

This defines the Wiener measure P_0 on $C[0, \infty)$. We can start the process at some point x other than 0 by considering $x + x(t)$ instead of x . This defines a family $\{P_x\}$ of measures on $C[0, \infty)$.

REMARK 5.2. With this we have constructed *all* processes with homogeneous independent increments. Such a process has to consist of infinitely divisible distributions and in fact

$$E[e^{iyx(t)}] = \exp \left[t \left[i\gamma y - \frac{\sigma^2 y^2}{2} + \int \left[e^{izy} - 1 - \frac{izy}{1+z^2} \right] M(dz) \right] \right]$$

Earlier we had constructed all such processes with $\sigma^2 = 0$. We can represent $x(t)$ as an independent sum $x(t) = x_1(t) + x_2(t)$ where $x_1(t)$ corresponds to the Poisson component

$$E[e^{iyx_1(t)}] = \exp \left[t \left[\int \left[e^{izy} - 1 - \frac{izy}{1+z^2} \right] M(dz) \right] \right]$$

and $x_2(t)$ the Brownian component with mean γt and variance $\sigma^2 t$.

$$x_2(t) = \sigma y(t) + \gamma t$$

$y(t)$ being the standard Brownian motion.

5.2. Markov and Strong Markov Property

The family $\{P_x\}$ of probability measures corresponding to Brownian motion starting from $x \in \mathbb{R}$ has the following properties.

Markov Property. For any $t > s$,

$$(5.1) \quad P_x[x(t) \in A \mid \mathcal{F}_s] = p(t-s, x(s), A) = \frac{1}{\sqrt{2\pi(t-s)}} \int_A e^{-\frac{(y-x(s))^2}{2(t-s)}} dy$$

The Markov property is an immediate consequence of the independent increments property. The independence of the increments implies that $x(t) - x(s)$ is independent of $\mathcal{F}_s$ for $t > s$. Therefore,

$$(5.2) \quad P_x[x(t) - x(s) \in A \mid \mathcal{F}_s] = p(t-s, 0, A) = \frac{1}{\sqrt{2\pi(t-s)}} \int_A e^{-\frac{y^2}{2(t-s)}} dy$$

Since $x(s)$ is $\mathcal{F}_s$ measurable (5.1) is an immediate consequence of (5.2).

$$\begin{aligned} P_x[x(t) \in A \mid \mathcal{F}_s] &= P_x[x(t) - x(s) \in A - x(s) \mid \mathcal{F}_s] \\ &= p(t-s, 0, A - x(s)) = p(t-s, x(s), A) \end{aligned}$$

Strong Markov Property. Let τ be a stopping time. The increments $x(t + \tau) - x(\tau)$ are again distributed like a Brownian motion independent of $\mathcal{F}_\tau$. The proof goes along familiar lines. If τ is a stopping time that takes only a countable number of values $\{t_j\}$, by the definition of stopping times, $\{\omega : \tau(\omega) = t_j\} \in \mathcal{F}_{t_j}$ and $x(t) - x(t_j)$ is independent of $\mathcal{F}_{t_j}$. Hence for any t and $B \in \mathcal{F}_\tau$

$$\begin{aligned} &P_x[B \cap \{x(\tau + t) - x(\tau) \in A\}] \\ &= \sum_j P_x[B \cap \{\tau = t_j\} \cap \{x(t_j + t) - x(t_j) \in A\}] \\ &= \sum_j P_x[B \cap \{\tau = t_j\}] P_x[\{x(t_j + t) - x(t_j) \in A\}] \\ &= \sum_j P_x[B \cap \{\tau = t_j\}] p(t, 0, A) \\ &= P_x[B] p(t, 0, A) \end{aligned}$$

If τ is any stopping time it is approximated from above by a sequence $\{\tau_n\}$ and it is easy to pass to the limit. It is however more convenient to work with

$$\int_B f(x(\tau + t) - x(\tau)) dP_x = P_x(B) \int f(y) p(t, 0, y) dy$$

It is enough to verify the identity for continuous f and now it is easy to take the limit in n .

There are various equivalent versions of the strong Markov property. For example

$$(5.3) \quad P_x[x(t) \in A \mid \mathcal{F}_\tau] = p(t - \tau, x(\tau), A) \text{ a.e. on } \tau < t.$$

REMARK 5.3. There is slightly different formulation of the Markov property that is useful. For each $s \geq 0$ let us define a map θ_s of $\Omega = C[0, \infty) \rightarrow C[0, \infty)$ by

$$x(t, \theta_s \omega) = x(s + t, \omega)$$

or

$$\theta_s \omega(t) = \omega(s + t)$$

Then

$$P_x[\theta_s \omega \in A \mid \mathcal{F}_s] = P_{x(s)}[A] \text{ a.e. } P_x$$

For A of the form $A = \{\omega : x(t, \omega) \in E\}$ this is the old Markov property. For A of the form $A = \{\omega : x(t_1, \omega) \in E_1, \dots, x(t_n, \omega) \in E_n\}$ with $0 \leq t_1 < t_2 < \dots < t_n < \infty$, this is proved by induction on n . The rest is standard measure theory.

REMARK 5.4. A similar formulation exists for stopping times as well. Let τ be a stopping time that is finite. Then θ_τ is a map of $\Omega = C[0, \infty) \rightarrow \Omega$ defined by

$$\theta_\tau \omega(t) = \omega(\tau + t)$$

The strong Markov property can now be stated as

$$P_x[\theta_\tau \omega \in A \mid \mathcal{F}_\tau] = P_{x(\tau)}[A] \text{ a.e. } P_x$$

REMARK 5.5. The strong Markov property applies to stopping times that can some times be infinite. Let τ be such a stopping time. Then θ_τ is a map of $\Omega = C[0, \infty) \rightarrow \Omega$ defined by

$$\theta_\tau \omega(t) = \omega(\tau + t)$$

only on the set where $\tau(\omega) < \infty$. The strong Markov property can now be stated as

$$P_x[\theta_\tau \omega \in A \mid \mathcal{F}_\tau] = P_{x(\tau)}[A] \text{ a.e. } P_x \text{ on } \{\omega : \tau(\omega) < \infty\}$$

It is proved by replacing τ by $\tau \wedge T$ and letting $T \rightarrow \infty$.

Reflection Principle. One of the consequences of the strong Markov property is the ability to calculate for $\ell > 0$,

$$(5.4) \quad P_0[\sup_{0 \leq s \leq t} x(s) \geq \ell] = 2P_0[x(t) \geq \ell] = \frac{\sqrt{2}}{\sqrt{\pi t}} \int_\ell^\infty e^{-\frac{y^2}{2t}} dy$$

We write the measure $P_0[x(t) \in A] = p(t, 0, A)$ as the sum

$$P_0[\{x(t) \in A\} \cap \{\tau_\ell \leq t\}] + P_0[\{x(t) \in A\} \cap \{\tau_\ell > t\}] = p_1(A) + p_2(A)$$

where $\tau_\ell = \inf\{t : x(t) \geq \ell\}$. By the strong Markov property

$$p_1(A) = \int_{\tau \leq \ell} p(t - \tau, \ell, A) dP_0 = \int_A q_1(y)$$

and $q_1(y)$ is symmetric about $y = \ell$. On the other hand the measure $p_2(A)$ with density $q_2(y)$ has to be concentrated on $(-\infty, \ell)$ or $q_2(y) = 0$ for $y \geq \ell$. But

$$q_1 + q_2 = p(t, 0, y) = \frac{1}{\sqrt{2\pi t}} e^{-\frac{y^2}{2t}}$$

This uniquely determines

$$q_1(y) = \begin{cases} \frac{1}{\sqrt{2\pi t}} e^{-\frac{y^2}{2t}} & \text{if } y \geq \ell \\ \frac{1}{\sqrt{2\pi t}} e^{-\frac{(2\ell-y)^2}{2t}} & \text{if } y \leq \ell \end{cases}$$

$$q_2(y) = \begin{cases} \frac{1}{\sqrt{2\pi t}} e^{-\frac{y^2}{2t}} - \frac{1}{\sqrt{2\pi t}} e^{-\frac{(2\ell-y)^2}{2t}} & \text{if } y \leq \ell \\ 0 & \text{if } y \geq \ell \end{cases}$$

Finally

$$\begin{aligned} P_0\left[\sup_{0 \leq s \leq t} x(s) \geq \ell\right] &= P_0[\tau \leq \ell] = p_1(\mathbb{R}) \\ &= \int q_1(y) dy = 2 \int_{y \geq \ell} q_1(y) dy = 2P_0[x(t) \geq \ell] \end{aligned}$$

and

$$(5.5) \quad P_0[\tau > \ell, x(t) \in A] = \int_A q_2(y) dy$$

This proves the reflection principle (5.4). A consequence is the estimate

$$(5.6) \quad P_0\left[\sup_{0 \leq s \leq t} |x(s)| \geq \ell\right] \leq 2P_0\left[\sup_{0 \leq s \leq t} x(s) \geq \ell\right] = 4P_0[x(t) \geq \ell]$$

5.3. Heat Equation

The semigroup T_t corresponding to Brownian motion is defined by

$$\begin{aligned} (T_t f)(x) &= \int_R f(y) P(t, x, dy) = \int_R f(y) p(t, x, y) dy \\ &= \int_R f(y) \sqrt{\frac{1}{2\pi t}} \exp\left\{-\frac{(y-x)^2}{2t}\right\} dy \\ &= \int_R f(x+z) \sqrt{\frac{1}{2\pi t}} \exp\left\{-\frac{z^2}{2t}\right\} dz \\ &= \int_R f(x+z\sqrt{t}) \sqrt{\frac{1}{2\pi}} \exp\left\{-\frac{z^2}{2}\right\} dz \end{aligned}$$

The infinitesimal generator is calculated as

$$\begin{aligned} (Lf)(x) &= \lim_{t \rightarrow 0} \frac{(T_t f)(x) - f(x)}{t} \\ &= \lim_{t \rightarrow 0} \int_R \frac{[f(x+z\sqrt{t}) - f(x)]}{t} \sqrt{\frac{1}{2\pi}} \exp\left\{-\frac{z^2}{2}\right\} dz \end{aligned}$$

If we assume that f is bounded and has three bounded derivatives, we can expand by Taylor's formula

$$f(x + \sqrt{t}z) - f(x) = z\sqrt{t}f'(x) + \frac{tz^2}{2}f''(x) + t^{\frac{3}{2}}e(t, z)$$

with a bound of

$$|e(t, z)| \leq C|z|^3$$

A simple integration with respect to the standard Gaussian establishes

$$\lim_{t \rightarrow 0} \sup_{x \in \mathbb{R}} \left| \frac{(T_t f)(x) - f(x)}{t} - \frac{1}{2} f''(x) \right| = 0$$

and therefore the infinitesimal generator L for Brownian motion is given by

$$Lf = \frac{1}{2} f'' = \frac{1}{2} \frac{d^2 f}{dx^2}$$

One can relax the conditions somewhat and we only need two derivatives on f with a uniformly continuous second derivative. It is not difficult to see that, if we define,

$$u(t, x) = (T_t f)(x) = \int_{\mathbb{R}} f(y) \sqrt{\frac{1}{2\pi t}} \exp\left\{-\frac{(y-x)^2}{2t}\right\} dy$$

then u satisfies the “heat equation”

$$\frac{\partial u}{\partial t} = \frac{1}{2} \frac{\partial^2 u}{\partial x^2}$$

with the initial condition $u(t, x) \rightarrow f(x)$ as $t \rightarrow 0$.

EXERCISE 5.2. We define Brownian motion with variance σ^2 and drift m , starting from x at time 0 as

$$y(t) = x + \sigma x(t) + mt$$

where $x(t)$ is the standard Brownian motion. The measures $P_x^{\sigma, m}$ are again Markov, but the transition probability density is now given by

$$p^{\sigma, m}(t, x, y) = \sqrt{\frac{1}{2\pi\sigma^2 t}} \exp\left\{-\frac{(y-x-mt)^2}{2\sigma^2 t}\right\}$$

Show that the infinitesimal generator is

$$(Lf)(x) = \frac{\sigma^2}{2} f''(x) + m f'(x)$$

With the above exercise we have defined Markov processes, that are processes with independent increments, corresponding to every infinitely divisible distribution and the Lévy-Khintchine representation is closely linked to the infinitesimal generator. In fact

$$(L f_\xi)(x) = \psi(\xi) f_\xi(x)$$

where $f_\xi(x)$ is the exponential $e^{i\xi x}$ and $\psi(\xi)$ is given by the Levy-Khintchine formula

$$\psi(\xi) = -\frac{\sigma^2 \xi^2}{2} + im\xi + \int \left(e^{i\xi y} - 1 - \frac{i\xi y}{1+y^2} \right) M(dy)$$

Formally, if we think of $D = \frac{d}{dx}$ and $S_y = e^{yD}$ as the shift $f(\cdot) \rightarrow f(\cdot + y)$, then

$$L = \frac{1}{2}D^2 + mD + \int \left(e^{yD} - I - \frac{y}{1+y^2}D \right) M(dy)$$

Remember that in this language the generator of a Poisson process with jump size y and intensity λ is

$$L = \lambda(e^{yD} - I)$$

and the generator of a purely deterministic process with $x(t) \equiv mt$ is mD

REMARK 5.6. Already, in the deterministic case, we see that the expansion

$$(T_t f)(x) = \sum_k \frac{t^k (L^k f)(x)}{k!}$$

is the same as the Taylor series and does not in general converge nor even if it does, necessarily represent the original function.

5.4. Recurrence

With the one-dimensional Brownian motion certain exact calculation are possible. We already saw that the distribution of $M(t) = \sup_{0 \leq s \leq t} x(s)$ could be calculated exactly as

$$P_0[M(t) \geq \ell] = \sqrt{\frac{2}{\pi t}} \int_{\ell}^{\infty} \exp\left\{-\frac{y^2}{2t}\right\} dy$$

This is of course the same as

$$P_0[\tau_{\ell} \leq t] = \sqrt{\frac{2}{\pi t}} \int_{\ell}^{\infty} \exp\left\{-\frac{y^2}{2t}\right\} dy = \sqrt{\frac{2}{\pi}} \int_{\frac{\ell}{\sqrt{t}}}^{\infty} \exp\left\{-\frac{y^2}{2}\right\} dy$$

Differentiating with respect to t , produces the distribution of τ_{ℓ} as one with probability density

$$(5.7) \quad f_{\ell}(t) = \frac{1}{\sqrt{2\pi}} \frac{\ell}{t^{\frac{3}{2}}} \exp\left\{-\frac{\ell^2}{2t}\right\}$$

and because

$$\lim_{t \rightarrow \infty} P_0[\tau_{\ell} \leq t] = 1$$

we conclude that τ_{ℓ} is finite almost surely. In particular, the one-dimensional Brownian motion starting from 0 will get up to any level ℓ at some time. By independence of the increments (translation symmetry) and the symmetry with respect to $x \rightarrow -x$ we conclude that the one-dimensional Brownian motion starting from any arbitrary point will eventually visit any other point with probability 1. So it is point recurrent. Notice however that the tail of the distribution decays like $t^{-\frac{3}{2}}$

so that $E\{\tau_\ell\} = \infty$. This is to be expected, because the process is not positive recurrent. In fact, for any finite interval I

$$p(t, x, I) = \int_I \frac{1}{\sqrt{2\pi t}} e^{-\frac{(y-x)^2}{2t}} dy \leq \frac{|I|}{\sqrt{2\pi t}} \rightarrow 0$$

as $t \rightarrow \infty$.

5.5. Feynman-Kac Formula

Certain integrals with respect to Brownian motion satisfy partial differential equations involving the operator $L = \frac{1}{2} \frac{d^2}{dx^2}$. We saw for example that

$$u(t, x) = E^{P_x}[f(x(t))]$$

satisfies

$$\frac{\partial u}{\partial t} = \frac{1}{2} \frac{\partial^2 u}{\partial x^2}$$

with the initial condition $u(0, x) = f(x)$. We can take Laplace transforms in t and define

$$g_\lambda(x) = \int_0^\infty e^{-\lambda t} u(t, x) dt$$

and it is not difficult to check that

$$\lambda g_\lambda(x) - \frac{1}{2} g_\lambda''(x) = f(x)$$

From the theory of partial differential equation these equations are known to have unique solutions in the class of bounded solutions and one way then of calculating integrals of a certain type with respect to Brownian motion is to solve an associated differential equation. We will develop this further later, when we consider martingales associated with Brownian motion.

THEOREM 5.1 Suppose $V(x)$ is a bounded continuous function on R and we define

$$u(t, x) = E^{P_x} \left[\exp \left[\int_0^t V(x(s)) ds \right] f(x(t)) \right]$$

then u solves the equation

$$\frac{\partial u}{\partial t} = \frac{1}{2} \frac{\partial^2 u}{\partial x^2} + V(x)u(t, x)$$

with the initial condition $u(0, x) = f(x)$.

PROOF: The easiest proof is to do some elementary but tedious computation. Expand the exponential inside the expectation in a power series and compute term by term. Then the typical term looks like

$$\begin{aligned} & \frac{1}{n!} E^{P_x} \left[\int_0^t \cdots \int_0^t V(x(t_1)) \cdots V(x(t_n)) f(x(t)) dt_1 \cdots dt_n \right] \\ &= E^{P_x} \left[\int \cdots \int_{0 \leq t_1 \leq \cdots \leq t_n \leq t} V(x(t_1)) \cdots V(x(t_n)) f(x(t)) dt_1 \cdots dt_n \right] \end{aligned}$$

$$\begin{aligned}
&= \left[\int \cdots \int_{0 \leq t_n \leq \cdots \leq t_1 \leq t} V(x(t-t_1)) \cdots V(x(t-t_n)) f(x(t)) dt_1 \cdots dt_n \right] \\
&= \left[\int \cdots \int_{0 \leq t_n \leq \cdots \leq t_1 \leq t} T_{t-t_1} V T_{t_1-t_2} V \cdots T_{t_{n-1}-t_n} V T_{t_n} f dt_1 \cdots dt_n \right] (x)
\end{aligned}$$

where we have denoted by V the operator of multiplication by the function $V(\cdot)$. The last line is an easy computation using the Markov property and is arrived at by successive conditioning at times $t_1 < \cdots < t_n$. The term $n!$ was produced by ordering the times and the integrand is symmetric in $t_1, \dots, t_n$. We can now differentiate with respect to t term by term. The first term will produce

$$\frac{d}{dt}(T_t f)(x) = \frac{1}{2} \frac{d^2}{dx^2}(T_t f)(x)$$

All the other terms will produce two terms. Differentiating inside the integral is the same as applying $\frac{1}{2} \frac{d^2}{dx^2}$. Differentiating the t in the limit of the integral gets rid of the integral with respect to t_1 and replaces t_1 by t . This is the same as the preceding term multiplied by $V(x)$. Therefore

$$\frac{\partial u}{\partial t} = \frac{1}{2} \frac{d^2 u}{dx^2} + V u$$

proving the theorem.

5.6. Arcsine Law

We now calculate the distribution of $\xi_t = \frac{1}{t} \int_0^t \chi_{[0,\infty]}(x(s)) ds$, the proportion of time spent by Brownian motion starting from 0, on the positive side during the time interval $[0, t]$. The symmetry of the process tells us that

$$E[\xi_t] = \frac{1}{2}$$

for every $t > 0$. One could expect a law of large numbers of the form

$$\lim_{t \rightarrow \infty} \xi_t = \frac{1}{2}$$

It is easy to see that this is not the case. There is another symmetry of Brownian motion that starts from 0. For any $\lambda > 0$ the process $x_\lambda(t) = \frac{1}{\sqrt{\lambda}} x(\lambda t)$ is again a Brownian motion starting from 0.

$$\xi_t = \int_0^1 \chi_{[0,\infty]}(x(ts)) ds = \int_0^1 \chi_{[0,\infty]}(\sqrt{t} x_t(s)) ds = \int_0^1 \chi_{[0,\infty]}(x_t(s)) ds$$

has the same distribution as ξ_1 for all t . We will now show that this distribution is given by

$$P_0[\xi_1 \leq x] = \int_0^x \frac{1}{\pi \sqrt{y(1-y)}} dy = \frac{2}{\pi} \sin^{-1} \sqrt{x}$$

which is the arcsine law. We let $V(x) = \chi_{[0,\infty)}(x)$ and compute by Feynman-Kac formula

$$u(t, \sigma, x) = E^{P_x} \left\{ \exp \left[-\sigma \int_0^t V(x(s)) ds \right] \right\}$$

Although we are interested in $x = 0$ and $t = 1$ we need t and x to have an equation. The equation in question is

$$\frac{\partial u}{\partial t} = \frac{1}{2} \frac{\partial^2 u}{\partial x^2} - \sigma V u$$

with $u(0, x) \equiv 1$. We take a Laplace transform in t to get

$$g(\lambda, \sigma, x) = \int_0^\infty u(t, \sigma, x) e^{-\lambda t} dt$$

that will satisfy the equation

$$\lambda g(\lambda, \sigma, x) + \sigma V(x) g(\lambda, \sigma, x) - \frac{1}{2} \frac{d^2 g(\lambda, \sigma, x)}{dx^2} = 1$$

which can be rewritten as

$$\begin{cases} (\lambda + \sigma) g(\lambda, \sigma, x) - \frac{1}{2} \frac{d^2}{dx^2} g(\lambda, \sigma, x) = 1 & \text{if } x > 0 \\ \lambda g(\lambda, \sigma, x) - \frac{1}{2} \frac{d^2}{dx^2} g(\lambda, \sigma, x) = 1 & \text{if } x < 0 \end{cases}$$

□

There are conditions at $\pm\infty$ that require g to be bounded and matching conditions at 0 to ensure that g is continuously differentiable at 0. The general solution to the above equations are

$$\begin{cases} g(\lambda, \sigma, x) = \frac{1}{\lambda + \sigma} + A e^{\sqrt{2\lambda}x} + B e^{-\sqrt{2\lambda}x} & \text{if } x > 0 \\ g(\lambda, \sigma, x) = \frac{1}{\lambda} + C e^{\sqrt{2\lambda}x} + D e^{-\sqrt{2\lambda}x} & \text{if } x < 0 \end{cases}$$

The growth conditions at $\pm\infty$ tells us that $A = D = 0$. The matching conditions at 0 allow us to solve explicitly for B and C . When we do the calculation we get

$$g(\lambda, \sigma, 0) = \frac{1}{\lambda + \sigma} + B = \frac{1}{\lambda} + C = \frac{1}{\sqrt{\lambda}} \frac{1}{\sqrt{\lambda + \sigma}}$$

On the other hand

$$\begin{aligned} g(\lambda, \sigma, 0) &= E^{P_0} \left[\int_0^\infty \exp \left[-\lambda t - \sigma \int_0^t V(x(s)) ds \right] dt \right] \\ &= E^{P_0} \left[\int_0^\infty \exp[-\lambda t - \sigma t \xi_t] dt \right] \\ &= E^{P_0} \left[\int_0^\infty \exp[-\lambda t - \sigma t \xi_1] dt \right] \\ &= E^{P_0} \left[\frac{1}{\lambda + \sigma \xi_1} \right] \\ &= \frac{1}{\sqrt{\lambda}} \frac{1}{\sqrt{\lambda + \sigma}} \end{aligned}$$

We now set $\lambda = 1$ and for the distribution of the random variable $\xi = \xi_1$ under P_0 we have

$$E\left[\frac{1}{1 + \sigma\xi}\right] = \frac{1}{\sqrt{1 + \sigma}}$$

Expanding both sides in powers of σ and identifying coefficients we get

$$\begin{aligned} E[\xi^n] &= \frac{\frac{1}{2} \frac{3}{2} \dots \frac{2n-1}{2}}{n!} \\ &= \frac{1}{\pi} \frac{\Gamma(\frac{2n+1}{2})\Gamma(\frac{1}{2})}{\Gamma(n+1)} \\ &= \frac{1}{\pi} \int_0^1 x^{n-\frac{1}{2}} (1-x)^{-\frac{1}{2}} dx \\ &= \int_0^1 x^n p(x) dx \end{aligned}$$

with

$$p(x) = \frac{1}{\pi\sqrt{x(1-x)}}$$

which proves the arcsine law.

5.7. Harmonic Oscillator

We now take $V(x) = x^2$ and try to calculate

$$E^{P_x}\left[\exp\left[-\int_0^t [x(s)]^2 ds\right]\right]$$

This requires solving

$$\frac{\partial u}{\partial t} = \frac{1}{2} \frac{\partial^2 u}{\partial x^2} - x^2 u$$

with $u(0, x) = 1$. We will try for a solution of the form $u(t, x) = \exp[v(t, x)]$ with

$$v(t, x) = a(t) + b(t)x^2$$

mainly because Gaussians favor quadratic expression as exponents and we are dealing with even functions. The equation reduces to

$$\frac{\partial v}{\partial t} = \frac{1}{2} \frac{\partial^2 v}{\partial x^2} + \frac{1}{2} \left[\frac{\partial v}{\partial x}\right]^2 - x^2$$

Equating coefficients of 1 and x^2 we get two ordinary differential equations in t for $a(\cdot)$ and $b(\cdot)$.

$$\begin{aligned} \frac{db}{dt} &= 2b^2 - 1 \\ \frac{da}{dt} &= b \end{aligned}$$

with initial conditions $a(0) = c(0) = 0$. We can solve them explicitly to get

$$c(t) = -\frac{1}{\sqrt{2}} \tanh \sqrt{2} t$$

$$a(t) = -\frac{1}{2} \log(\cosh(\sqrt{2} t))$$

In particular

$$E^{P_0} \left[\exp \left[- \int_0^t [x(s)]^2 ds \right] \right] = \exp[a(t)] = \frac{1}{\sqrt{\cosh(\sqrt{2} t)}}$$

Notice that $a(t) \simeq -\frac{1}{\sqrt{2}}t$, giving an exponential decay rate with constant $\frac{1}{\sqrt{2}}$ for the expectation. On the other hand the typical value of

$$\int_0^t [x(s)]^2 ds = t \int_0^1 [x(ts)]^2 ds \simeq t^2 \int_0^1 x_t(s)^2 ds$$

is of order t^2 . The major contribution comes from paths that do not wander too far.

5.8. Exit Times from Bounded Intervals

We now take $V(x) = 0$ if $|x| \leq 1$ and $-\infty$ otherwise. The Feynman-Kac formula has to be carefully interpreted in this case. If $\tau(\omega) = \inf\{t : |x(t)| \geq 1\}$ then

$$u(t, x) = P_x[\tau > t]$$

is the expectation we are after. Paths that go out of $[-1, 1]$ do not contribute because the exponential term with V kills them. The function u will satisfy

$$\frac{\partial u}{\partial t} = \frac{1}{2} \frac{\partial^2 u}{\partial x^2}$$

on $[0, \infty) \times (-1, 1)$ with $u(t, \pm 1) = 0$. The initial condition is still $u(0, x) = 1$ for $x \in (-1, 1)$. We will find that the eigenfunctions of $\frac{1}{2} \frac{d^2}{dx^2}$ on $(-1, 1)$, with 0 boundary conditions, are given by $\{\sin \frac{k}{2} \pi x\}$ for even k and $\{\cos \frac{k}{2} \pi x\}$ for odd k , with eigenvalues $\{-\frac{k^2 \pi^2}{8}\}$, for $k = 1, 2, \dots$. By separation of variables

$$u(t, x) = \sum_{k \text{ even}} e^{-\frac{k^2 \pi^2}{8} t} a_k \sin \frac{k}{2} \pi x + \sum_{k \text{ odd}} e^{-\frac{k^2 \pi^2}{8} t} b_k \cos \frac{k}{2} \pi x$$

will solve the equation provided we match the initial conditions so that

$$1 = \sum_{k \text{ even}} a_k \sin \frac{k}{2} \pi x + \sum_{k \text{ odd}} b_k \cos \frac{k}{2} \pi x$$

Because we are dealing with even functions only the cosine series will be present.

$$b_k = \frac{\int_{-1}^1 \cos \frac{k}{2} \pi x dx}{\int_{-1}^1 (\cos \frac{k}{2} \pi x)^2 dx} = \frac{4 \sin \frac{k\pi}{2}}{k} = (-1)^{\frac{k-1}{2}} \frac{4}{k}$$

In particular

$$u(t, 0) = P_0[\tau \geq t] = 4 \sum_{k=0}^{\infty} \frac{(-1)^k}{k} e^{-\frac{(2k+1)^2 \pi^2}{8} t}$$

While this expansion is good for large t there is another expansion that is good for small t .

We try to construct the transition probability density $\hat{p}(t, x, y)$ for the measure

$$\hat{P}(t, x, E) = P_x[[\tau > t] \cap [x(t) \in E]]$$

as the solution of

$$\frac{\partial \hat{p}}{\partial t} = \frac{1}{2} \frac{\partial^2 \hat{p}}{\partial x^2}$$

with $\hat{p}(t, \pm 1, y) = 0$. Consider

$$\begin{aligned} \hat{p}(t, x, y) = & \sum_{j=-\infty}^{\infty} \frac{1}{\sqrt{2\pi t}} \exp\left[-\frac{(4j + x - y)^2}{2t}\right] \\ & - \sum_{j=-\infty}^{\infty} \frac{1}{\sqrt{2\pi t}} \exp\left[-\frac{(4j + 2 - x - y)^2}{2t}\right] \end{aligned}$$

It is clear that $\hat{p}$ satisfies the initial condition as well as the equation. Let us check the boundary conditions. Take $x = 1$. The two series are identical. Take $x = -1$. The two series become identical if we change $j \rightarrow j + 1$. In either case $\hat{p}$ is zero at both boundaries. Integrating with respect to y on $[-1, 1]$ after setting $x = 0$ evaluates $P_0[\tau > t]$. This series is good for small t .

5.9. Stochastic Integrals

We first show that almost surely the Brownian paths are not of bounded variation. For this purpose we actually prove a stronger statement

THEOREM 5.2 *In any finite interval $[0, T]$ almost all Brownian paths have a quadratic variation equal to T .*

PROOF: Let us divide the interval $[0, T]$ into 2^n equal subintervals of length $T2^{-n}$ each. Define

$$Q_n(\omega) = \sum_{j=1}^{2^n} \left[x\left(\frac{jT}{2^n}\right) - x\left(\frac{(j-1)T}{2^n}\right) \right]^2$$

Q_n is the sum of 2^n independent random variables each with mean $T2^{-n}$ and variance $2T^22^{-2n}$. Therefore $E[Q_n] = T$ and variance of Q_n equals $2T^22^{-n}$. Clearly $Q_n \rightarrow T$ almost surely as $n \rightarrow \infty$. $\square$

Since the paths are continuous, if they were of bounded variation, then the quadratic variation would have to be 0. But we can still define some sort of integrals with respect to $x(\cdot)$.

Wiener's Stochastic Integral. If $f \in L_2(R)$ we will define $\int_0^\infty f(t)dx(t)$. If $f(t)$ is equal to some constants $\{a_j\}$ on a finite number of disjoint bounded intervals $\{I_j = (s_j, t_j)\}$ and 0 everywhere else we can define

$$X_f = \int_0^\infty f(t)dx(t) = \sum_j a_j [x(t_j) - x(s_j)]$$

and an easy computation yields $E[X_f] = 0$ and

$$E[X_f^2] = \sum_j a_j^2 (t_j - s_j) = \int_0^\infty [f(t)]^2 dt$$

Given any function $f \in L_2(R)$ we can find a sequence of $\{f_n\}$ of functions of the type discussed above such that $\|f_n - f\|_{L_2} \rightarrow 0$ as $n \rightarrow \infty$. It follows from the linearity of the map $f \rightarrow X_f$ that

$$\lim_{m,n \rightarrow \infty} E[|X_{f_n} - X_{f_m}|^2] = 0$$

and we see that X_{f_n} has a limit in $L_2(P)$.

Itô's Stochastic Integrals. In Wiener's stochastic integrals the function f was deterministic and did not depend on ω . In Itô's theory certain type of dependence on ω is allowed. For a properly chosen class $\mathbf{F}$, the stochastic integrals are well defined in Itô's theory. Each $f \in \mathbf{F}$ is a function $f(s, \omega)$ mapping $[0, \infty) \times \Omega \rightarrow R$ that satisfies

- (i) For each $t > 0$, the restriction of f to $[0, t] \times \Omega$ is jointly measurable in (s, ω) as a map from the measurable space $([0, t] \times \Omega, \mathcal{B}[0, t] \times \mathcal{F}_t)$ to the real line. This property is referred to as progressive measurability.
- (ii) For each $t > 0$

$$E^{P_0} \left[\int_0^t |f(s, \omega)|^2 ds \right] < \infty$$

THEOREM 5.3 For each $f \in \mathbf{F}$ we can define the stochastic integral

$$X_f(t) = \int_0^t f(s, \omega) dx(s)$$

that has the following properties.

- (i) The map $f \rightarrow X_f$ is linear.
- (ii) $X_f(t)$ is progressively measurable and is a martingale with respect to $(\Omega, \mathcal{F}_t, P_0)$ and is almost surely continuous as a function of t .
- (iii) $X_f^2(t) - \int_0^t |f(s, \omega)|^2 ds$ is again a martingale with respect to $(\Omega, \mathcal{F}_t, P_0)$.

PROOF: Again we start with simple functions. Let $0 = t_0 < t_1 < t_2 < \dots < t_n < \infty$ be n time points. Let us suppose that for each $j = 1, 2, \dots, n$, on the interval $[t_{j-1}, t_j)$, $f(s, \omega)$ is identically equal to a function $f_{j-1}(\omega)$ which

is bounded and measurable with respect to $\mathcal{F}_{t_{j-1}}$, and that it is identically zero for $t \geq t_n$. We can define the stochastic integral in the natural way

$$X_f(t) = \sum_{i=1}^{k-1} f_{i-1}(\omega)[x(t_i) - x(t_{i-1})] + f_{k-1}(\omega)[x(t) - x(t_{k-1})]$$

if $t_{k-1} \leq t \leq t_k$ and $k = 1, 2, \dots, n$. For $t > t_n$ we define

$$X_f(t) = X_f(t_n)$$

It is easy to check that the properties (i), (ii) and (iii) are satisfied. Property (i) is obvious. Property (ii) for X_f follows from the fact that $x(t)$ itself is progressively measurable and is a martingale with respect to $\mathcal{F}_t$. To check the martingale property we note that $X_f(t)$ is a martingale in each of the intervals $[t_{j-1}, t_j]$, as well as in $[t_n, \infty)$ where it is a constant. If a function $Z(t)$ is a martingale in two intervals $[a, b]$ and $[b, c]$ with $a < b < c$ then it is a martingale in $[a, c]$. Finally we verify (iii). Because of the above comment we need only show that for $t_{j-1} \leq s < t \leq t_j$,

$$E^{P_0} \left[X_f^2(t) - X_f^2(s) - \int_s^t f^2(\sigma, \omega) d\sigma \mid \mathcal{F}_s \right] = 0$$

We can calculate

$$\begin{aligned} & E^{P_0} [X_f^2(t) - X_f^2(s) \mid \mathcal{F}_s] \\ &= E^{P_0} [(X_f(s) + X_f(t) - X_f(s))^2 - X_f^2(s) \mid \mathcal{F}_s] \\ &= E^{P_0} [(X_f(t) - X_f(s))^2 + 2X_f(s)(X_f(t) - X_f(s)) \mid \mathcal{F}_s] \\ &= E^{P_0} [(X_f(t) - X_f(s))^2 \mid \mathcal{F}_s] \\ &= E^{P_0} [f_{j-1}^2(\omega)(x(t) - x(s))^2 \mid \mathcal{F}_s] \\ &= E^{P_0} [f_{j-1}^2(\omega)(t - s) \mid \mathcal{F}_s] \\ &= E^{P_0} \left[\int_s^t f^2(s, \omega) ds \mid \mathcal{F}_s \right] \end{aligned}$$

and that verifies (iii). Note that the boundedness of $f_j(\cdot)$ provides all the integrability that we need.

A consequence of Doob's inequality for martingales is the following estimate:

$$E^{P_0} \left\{ \sup_{0 \leq s \leq t} |X_f(s)|^2 \right\} \leq 4E^{P_0} \left\{ \int_0^t f^2(s, \omega) ds \right\}$$

This estimate allows us to extend the definition of $X_f(\cdot)$ from simple functions to larger classes. If we approximate $f(s, \omega)$ by a sequence of functions $f_n(s, \omega)$ (as above) such that

$$E^{P_0} \left\{ \int_0^t |f_n(s, \omega) - f(s, \omega)|^2 ds \right\} \rightarrow 0$$

then

$$\lim_{m,n \rightarrow \infty} E^{P_0} \left\{ \sup_{0 \leq s \leq t} |X_{f_n}(s) - X_{f_m}(s)|^2 \right\} = 0$$

and then the limit $X_f(\cdot)$ of $X_{f_n}(\cdot)$ exists in the sense that

$$\lim_{n \rightarrow \infty} E^{P_0} \left\{ \sup_{0 \leq s \leq t} |X_{f_n}(s) - X_f(s)|^2 \right\} = 0$$

It would then follow that X_f satisfies the properties (i), (ii) and (iii).

We now have to construct the approximations. First, given $f(s, \omega)$ we can define $f_T(s, \omega)$ to be f cutoff for $s \geq T$. Then clearly f_T is supported on a finite interval and f_T approximates f in the required sense. Any progressively measurable $f(s, \omega)$ can be approximated by bounded progressively measurable functions $f_n(s, \omega)$ by taking $f_n(s, \omega) = f(s, \omega)$ if $|f(s, \omega)| \leq n$ and 0 otherwise. A bounded, almost surely continuous $f(s, \omega)$ can be approximated by simple functions, if we define $f_n(s, \omega)$ to equal $f(\frac{j}{n}, \omega)$ for $\frac{j}{n} \leq s < \frac{j+1}{n}$ and it is easy to verify that in all these approximations

$$\lim_{n \rightarrow \infty} E^{P_0} \left\{ \int_0^t |f_n(s, \omega) - f(s, \omega)|^2 ds \right\} = 0$$

for every $t > 0$. There remains only one approximation to carry out. To show that a bounded progressively measurable function can be approximated by bounded almost surely continuous progressively measurable functions. We do this as follows. Let $h = \frac{1}{n} \rightarrow 0$. We define $f_n(s, \omega)$ to be 0 for $s \leq h$, and for $s \geq h$ define

$$f_n(s, \omega) = \frac{1}{h} \int_{s-h}^s f(s, \omega) ds$$

Because we look back, f_n will be progressively measurable. It is clearly almost surely continuous and it is a standard result in the theory of Lebesgue integrals that for any function $f(s)$ that is square integrable on every finite interval,

$$\lim_{h \rightarrow 0} \int_0^t \left| \frac{1}{h} \int_{s-h}^s f(\sigma) d\sigma - f(s) \right|^2 ds = 0$$

for each $t > 0$. Since we have a uniform bound on f we can take expectations and by the bounded convergence theorem

$$\lim_{n \rightarrow \infty} E^{P_0} \left\{ \int_0^t |f_n(s, \omega) - f(s, \omega)|^2 ds \right\} = 0$$

□

REMARK 5.7. It is possible to define the stochastic integral

$$X_f(t) = \int_0^t f(s, \omega) dx(s)$$

for the larger class of progressively measurable functions that satisfy for almost ω

$$A(t) = \int_0^t |f(s, \omega)|^2 ds < \infty$$

without the assumption that the expectation $E[A(t)] < \infty$. We can use the stopping time $\tau_\ell = \inf\{t : A(t) \geq \ell\}$ and define

$$f_\ell(s, \omega) = f(s, \omega)\mathbf{1}_{\tau_\ell \geq t}$$

It is easy to see that

$$X_\ell(t) = \int_0^t f_\ell(s, \omega) dx(s)$$

does not change after $\ell > \int_0^t |f(s, \omega)|^2 ds$. The limit $X_f(t) = \lim X_\ell(t)$ exists. But we can not control the integrals and $E[X_f(t)]$ may fail to exist.

REMARK 5.8. One can attempt a similar definition for all square integrable martingales. Let $M(t)$ be a martingale. Since $[M(t)]^2$ is a submartingale there exists a nondecreasing process $A(t) = A(t, \omega)$ such that $[M(t)]^2 - A(t)$ is a martingale. We assume that $A(t)$ is almost surely continuous and attempt to define the stochastic integral

$$\int_0^t f(s, \omega) dM(s)$$

as a square integrable martingale provided f is progressively measurable and

$$E\left[\int_0^t |f(s, \omega)|^2 dA(s)\right] < \infty$$

The only step that offers some difficulty is, the approximation of a bounded progressively measurable f by a continuous one. Simple averaging will not work if we have no control over A . One has to be more careful. But if

$$A(t) = \int_0^t a(s, \omega) ds$$

with a bounded a , then there is no difficulty.

REMARK 5.9. But there are some surprises. Let $M(t) = N(t) - t$, the compensated Poisson process which is a martingale. $M^2(t) - t$ is a martingale. We can define

$$X(t) = \int_0^t N(s) dM(s) = \int_0^t N(s) dN(s) - \int_0^t N(s) ds$$

If $\{\tau_j\}$ are the times of successive jumps of $N(t)$, then we expect

$$X(t) = \sum_{j: \tau_j \leq t} N(\tau_j) - \int_0^t N(s) ds$$

What is the value of $N(\tau_j)$? It is $j - 1$ and not j . The integrand prefers to be left continuous. This is a consequence of looking back in the approximation.

EXAMPLE 5.1. $\int_0^t x(s) dx(s) \equiv \frac{1}{2}[x^2(t) - t]$ a.e. P_0 .

Since both sides are almost surely continuous it is sufficient to prove that for each $t > 0$, $\int_0^t x(s) dx(s) = \frac{1}{2}[x^2(t) - t]$ a.e. P_0 . If we divide the interval $[0, t]$

into 2^n equal parts we can evaluate the left-hand side by

$$\begin{aligned}
 & \int_0^t x(s) dx(s) \\
 &= \lim_{n \rightarrow \infty} \sum_{j=1}^{2^n} x\left(\frac{(j-1)t}{2^n}\right) \left[x\left(\frac{jt}{2^n}\right) - x\left(\frac{(j-1)t}{2^n}\right) \right] \\
 &= \lim_{n \rightarrow \infty} \frac{1}{2} \left[\sum_{j=1}^{2^n} \left[x\left(\frac{(j-1)t}{2^n}\right) + x\left(\frac{jt}{2^n}\right) \right] \left[x\left(\frac{jt}{2^n}\right) - x\left(\frac{(j-1)t}{2^n}\right) \right] \right] \\
 &\quad - \lim_{n \rightarrow \infty} \frac{1}{2} \left[\sum_{j=1}^{2^n} \left[x\left(\frac{jt}{2^n}\right) - x\left(\frac{(j-1)t}{2^n}\right) \right]^2 \right] \\
 &= \frac{1}{2} [x^2(t) - x^2(0)] - \frac{1}{2} t
 \end{aligned}$$

The following lemma is a useful variant.

LEMMA 5.4 *Suppose $f(t, x)$ is a bounded continuous function of t and x . Then we have*

$$\lim_{h \rightarrow 0} \left[\sum_{\substack{j \\ 0 \leq jh \leq t}} f(jh, x(jh)) [x((j+1)h) - x(jh)]^2 \right] = \int_0^t f(s, x(s)) ds$$

in probability with respect to P_0 .

PROOF: The idea of the proof is quite simple. If f is a constant we already know that

$$\lim_{h \rightarrow 0} \left[\sum_{\substack{j \\ 0 \leq jh \leq t}} [x((j+1)h) - x(jh)]^2 \right] = t$$

and if we approximate the continuous function $z(s) = f(s, x(s))$ by $z_n(s)$ which is piecewise constant being equal to $f(\frac{j}{n}, x(\frac{j}{n}))$ in the interval $\frac{j}{n} \leq s < \frac{j+1}{n}$, then for every $n \geq 1$ it is easily seen that

$$\lim_{h \rightarrow 0} \left[\sum_{\substack{j \\ 0 \leq jh \leq t}} z_n(jh) [x((j+1)h) - x(jh)]^2 \right] = \int_0^t z_n(s) ds$$

Since $z_n \rightarrow z$ uniformly on $[0, t]$, it is trivial to justify the interchanging of the limits involving h and n . $\square$

An important result in the theory of stochastic integrals is Itô's formula.

THEOREM 5.5 *Let $f(t, x)$ be a bounded continuous function of t and x with one bounded continuous t derivative and two bounded continuous x derivatives. Then*

$$\begin{aligned}
 f(t, x(t)) - f(0, x(0)) = & \int_0^t f_s(s, x(s)) ds + \frac{1}{2} \int_0^t f_{xx}(s, x(s)) ds + \int_0^t f_x(s, x(s)) dx(s)
 \end{aligned}$$

PROOF: First we note that if we can prove it for functions f that are C^∞ and have compact support, we can then prove it for the class in the statement of the theorem by approximation. For a nice f , we write $f(t, x(t)) - f(0, x(0))$ as the limit of a telescoping sum

$$\begin{aligned} & \sum_{0 \leq jh < t}^j [f((j+1)h, x((j+1)h)) - f(jh, x(jh))] \\ &= \sum_{0 \leq jh < t}^j \left[hf_s(jh, x(jh)) + f_x(jh, x(jh))[x((j+1)h) - x(jh)] \right. \\ & \quad \left. + \frac{1}{2} f_{xx}(jh, x(jh))[x((j+1)h) - x(jh)]^2 \right] + \text{Error} \end{aligned}$$

The error comes from the Taylor expansion and is clearly dominated by

$$C \Delta_h(\omega) \sum_{0 \leq jh < t}^j [h + [x((j+1)h) - x(jh)]^2]$$

where

$$\Delta_h(\omega) = \sup_{0 \leq jh < t} |x((j+1)h) - x(jh)|$$

and by the almost sure continuity of paths, $\Delta_h(\omega) \rightarrow 0$ as $h \rightarrow 0$. $\square$

COROLLARY 5.6 *Let $f(t, x)$ be a bounded continuous function with derivatives f_s and f_{xx} that are again continuous functions of t and x . Suppose*

$$f_s(s, x) + \frac{1}{2} f_{xx}(s, x) = g(s, x)$$

is also a bounded continuous function, then

$$f(t, x(t)) - f(0, x(0)) - \int_0^t g(s, x(s)) ds$$

is a martingale with respect to $(\Omega, \mathcal{F}_t, P_x)$.

PROOF: Although we proved Itô's Formula for P_0 it holds just as well for any P_x . It involves only the increments of $x(\cdot)$ and they do not depend on x . By Itô's Formula the quantity above is a stochastic integral and is therefore a martingale. The only problem is bounds on f_x . We can change the definition of f outside a large interval $\{|x| \leq \ell\}$ and clearly for the modified function

$$f_\ell(t, x(t)) - f_\ell(0, x(0)) - \int_0^t g_\ell(s, x(s)) ds$$

is a martingale. If we denote by τ_ℓ the exit time from the interval $\{|x| \leq \ell\}$ because $f = f_\ell$ in the interval

$$f(\tau_\ell \wedge t, x(\tau_\ell \wedge t)) - f(0, x(0)) - \int_0^{\tau_\ell \wedge t} g(s, x(s)) ds$$

is a martingale and there is no problem in letting $\ell \rightarrow \infty$ and, since $\tau_\ell \rightarrow \infty$ as $\ell \rightarrow \infty$, we are done. $\square$

There are some important exponential martingale that are associated with stochastic integrals. These are analogues of the exponential martingales $e^{\theta\beta x(t) - \theta^2 t/2}$ in the case of Brownian motion.

THEOREM 5.7 *Let $c(s, \omega)$ be progressively measurable and bounded. Then*

$$(5.8) \quad Z(t) = \exp \left[\int_0^t c(s, \omega) dx(s) - \frac{1}{2} \int_0^t c^2(s, \omega) ds \right]$$

is a martingale with respect to $(\Omega, \mathcal{F}_s, P_0)$.

PROOF: There is no loss of generality in restricting ourselves to an arbitrary but finite interval $[0, T]$. The theorem clearly holds for constants c . In fact, $c(t)$ can be piecewise constant equal to c_i on a finite number of intervals $[t_i, t_{i+1}]$ with $0 < t_1 < \dots < t_i < t_{i+1} < \dots < t_n = T$. In fact, $c_i = c_i(\omega)$ can be a function of ω so long as c_i is $\mathcal{F}_i$ measurable. It follows that if $c(s, \omega)$ is simple and bounded $Z(t) = Z_{c(\cdot)}(t)$ given by (5.8) is a martingale. We note that

$$Z_c(t)^2 \leq Z_{2c} e^{C^2 t}$$

where C is a bound on $c(s, \omega)$. A bounded progressively measurable c can be approximated by simple c_n with c_n remaining uniformly bounded. Therefore if we let $c_n \rightarrow c$ the approximating martingales $Z_{c_n(\cdot)}(t)$ are now uniformly bounded in $L_2[P_0]$ and the limit $Z(t) = Z_{c(\cdot)}(t)$ is again a martingale. $\square$

REMARK 5.10. Sometimes it is necessary to estimate higher moments of stochastic integrals of the form

$$\xi(t) = \int_0^t c(s, \omega) dx(s)$$

in terms of the second term

$$A(t) = \int_0^t c^2(s, \omega) ds$$

that controls the expected mean square. We replace c by λc in (5.8) and develop $Z_\lambda(t)$ as a power series in λ . Each term will be a martingale. In particular, the term $\frac{\lambda^4}{4!}$ is given by

$$M_4(t) = \xi(t)^4 + 3A(t)^2 - 6A(t)\xi(t)^2$$

and since $M_4(t)$ is a martingale

$$E[\xi(t)^4] = 6E[A(t)\xi(t)^2] - 3E[A(t)^2] \leq \frac{1}{2}E[\xi(t)^4] + 15E[A(t)^2]$$

Therefore

$$E[\xi(t)^4] \leq 30E[A(t)^2]$$

EXERCISE 5.3. Let V be a bounded continuous function and let $U(t, x)$ be a reasonable solution of

$$U_t(t, x) + \frac{1}{2}U_{xx}(t, x) + V(x)U(t, x) = 0$$

Then

$$Z(t) = U(t, x(t)) \exp \left\{ \int_0^t V(x(s)) ds \right\}$$

is a martingale with respect to any P_x . (Part of the exercise is to determine reasonable conditions on V and U). What is the connection to Feynman-Kac formula?

5.10. Brownian Motion with a Drift, Girsanov Formula

We saw earlier that Brownian motion with a drift, i.e. $x(t) = \beta(t) + bt$ correspond to the operator

$$\frac{1}{2} \frac{d^2}{dx^2} + b \frac{d}{dx}$$

it is possible that the drift b depends on the current location $x(t)$ of the path. $x(t)$ would then be a solution of the equation

$$(5.9) \quad x(t) = x + \beta(t) + \int_0^t b(x(s)) ds$$

Here $\beta(\cdot)$ is the standard Brownian motion. We will assume that $b(\cdot)$ satisfies the Lipschitz condition

$$(5.10) \quad |b(x) - b(y)| \leq C|x - y|$$

We then have the following theorem.

THEOREM 5.8 *Given x and the continuous function $\beta(\cdot)$, equation (5.9) has a unique solution and defines a map $\Phi_x : \Omega \rightarrow \Omega$. This induces a distribution Q_x on the space of paths starting from 0. If we denote by P_x the measure corresponding to the Brownian motion starting from x at time 0, then $Q_x \ll P_x$ and the Radon-Nikodym derivative on the σ -field $\mathcal{F}_t$ is given by*

$$R_t(\omega) = \exp \left[\int_0^t b(x(s)) dx(s) - \frac{1}{2} \int_0^t b^2(s, x(s)) ds \right]$$

which is of course a martingale with respect to $(\Omega, \mathcal{F}_t, P_x)$

PROOF: We do the Picard iteration scheme, with a forcing term $\beta(\cdot)$ starting from $x_0(t) \equiv x + \beta(t)$ and defining successively

$$x_n(t) = x + \beta(t) + \int_0^t b(x_{n-1}(s)) ds$$

It is clear that each $x_n(t)$ is well defined and using the Lipschitz condition (5.10) and restricting t to an arbitrary but fixed interval $[0, T]$,

$$z_n(t) = x_n(t) - x_{n-1}(t)$$

satisfies for $n \geq 2$

$$(5.11) \quad |z_n(t)| \leq C \int_0^t |z_n(s)| ds$$

and

$$z_1(t) = \int_0^t b(x + \beta(s)) ds \leq c(x, T, \beta(\cdot))t$$

where c is a constant that depends on x , T and $\beta(\cdot)$. Successive iteration, using (5.11) produces by induction the estimate

$$|z_n(t)| \leq c(x, T, \beta(\cdot)) \frac{C^n t^n}{n!}$$

valid for $0 \leq t \leq T$. It is now clear that $x_n(t) = x_0(t) + \sum_{j=1}^n z_j(t)$ converges uniformly in any finite interval $[0, T]$ and the limit $x(t) = \lim_{n \rightarrow \infty} x_n(t)$ is a solution of (5.12).

If $x(t)$ and $y(t)$ are two different solutions to (5.9), then using the Lipschitz condition (5.10)

$$|x(t) - y(t)| \leq \int_0^t |b(x(s)) - b(y(s))| ds \leq C \int_0^t |x(s) - y(s)| ds$$

Since $x(t)$, $y(t)$ are both continuous solutions $|x(t) - y(t)|$ is bounded by some constant $c(T)$ on the finite interval $[0, T]$. Then, on that interval, $\delta(t) = |x(t) - y(t)|$ satisfies $\delta(t) \leq c(T)$ and

$$\delta(t) \leq C \int_0^t \delta(s) ds$$

By induction

$$\delta(t) \leq c(T) \frac{C^n t^n}{n!}$$

Since this holds for every n , letting $n \rightarrow \infty$ we have $\delta(t) \equiv 0$ and therefore we have existence and uniqueness. We now have a map $\Phi_x : C[0, \infty) \rightarrow C[0, \infty)$ that sends $\beta(\cdot)$ to $x(\cdot)$. The image of the Wiener measure on the Brownian paths $\beta(\cdot)$ is transformed into a new measure Q_x that is the distribution of $x(\cdot)$. The claim is that Q_x is a Markov process with transition probability given by

$$q(t, x, A) = P_0[\Phi_x(\beta(\cdot))(t) \in A]$$

Because of uniqueness we have the identity

$$x(t) = \Phi_x(\beta(\cdot))(t) = \Phi_{x(s)}(\beta'(\cdot))(t - s)$$

where $\beta'(t) = \beta(s + t) - \beta(s)$. From the independence of the increments of the Brownian motion $\beta(\cdot)$ and the measurability of $x(s)$ with respect to Brownian σ -field up to time s ,

$$Q_x[x(t) \in A \mid \mathcal{F}_s] = P_0[\Phi_{x(s)}(\beta'(\cdot))(t - s) \in A] = q(t - s, x(s), A)$$

This proves the Markov property. The strong Markov property is totally similar and depends on the similar property for the Brownian motion.

Finally, we turn to proving the Girsanov formula that provides the Radon-Nikodym derivative of Q_x with respect to P_x . Let us first assume that b is bounded. We define a new measure $\widehat{Q}_x$ with Radon-Nikodym derivative

$$(5.12) \quad \left. \frac{d\widehat{Q}_x}{dP_x} \right|_{\mathcal{F}_t} = R_t(\omega) = \exp \left[\int_0^t b(x(s)) dx(s) - \frac{1}{2} \int_0^t b^2(x(s)) ds \right]$$

From Theorem 5.7 it follows that R_t is a martingale and the above definition of $\widehat{Q}_x$ being consistent for different values of t , is well defined. In addition

$$R_t(\theta, \omega) = \exp \left[\int_0^t (\theta + b(x(s))) dx(s) - \frac{1}{2} \int_0^t (b(x(s)) + \theta)^2 ds \right]$$

is a martingale for every θ . This is the same as

$$S_t(\theta) = \exp \left[\int_0^t \theta dx(s) - \frac{\theta^2 t}{2} - \frac{1}{2} \int_0^t \theta b(x(s)) ds \right]$$

being a martingale with respect to $\widehat{Q}_x$. This can be restated as

$$\exp \left[\theta \left[x(t) - x - \int_0^t b(x(s)) ds \right] - \frac{\theta^2 t}{2} \right]$$

are martingales with respect to $\widehat{Q}_x$. But this means $y(t) = x(t) - x - \int_0^t b(x(s)) ds$ is distributed as the Brownian motion. Since $\Phi_x(y(\cdot)) = x(\cdot)$ this identifies $\widehat{Q}_x$ as the same as Q_x .

To go from bounded to unbounded b is not very difficult. We can do the approximation and since the martingales are nonnegative by Fatou's lemma $Z(t)$ will be a supermartingale. Because equality in Fatou's lemma implies uniform integrability, $Z(t)$ will be a martingale if and only if $E^{P_0}[Z(t)] \equiv 1$. We can always replace $b(x)$ by a bounded function beyond a ball of radius ℓ and if τ_ℓ is the first exit time from this ball $Z(\tau_\ell \wedge t)$ will be a martingale. It is not difficult to see that

$$(5.13) \quad Q_x \left[\sup_{0 \leq t \leq T} |x(t)| < \ell \right] = Q_x[\tau_\ell > T] = \int_{\tau_\ell > T} Z(\tau_\ell \wedge T) dP_x$$

In general if $b(x)$ grows too fast the solution might explode. Uniform Lipschitz condition guarantees at most linear growth and that prevents explosion. In any case, letting $\ell \rightarrow \infty$ in (5.13)

$$Q_x[\tau_\infty > T] = \int Z(T) dP_x$$

In particular, Girsanov formula is always valid. □

EXERCISE 5.4. Let $b(x) = x^2$, and $x(0) = x = 1$. Show that the solution of

$$x(t) = 1 + \beta(t) + \int_0^t |x(s)|^2 ds$$

blows up with positive probability before $t = 2$ by comparing it to the solution of

$$x(t) = \int_0^t |x(s)|^2 ds$$

If $\beta(t) \geq -1$ for $0 \leq t \leq T$, then $x(t) \geq \int_0^t |x(s)|^2 ds$. Use this to show blow up. How do you show that for any $t > 0$ there is positive probability of blow up before time t ?

The operator

$$L = \frac{1}{2} \frac{d^2}{dx^2} + b(x) \frac{d}{dx}$$

is the infinitesimal generator of the process corresponding to Q_x . This relationship can be exhibited in many ways. We saw that

$$\exp \left[\theta(x(t) - x(0)) - \theta \int_0^t b(x(s)) ds - \frac{\theta^2 t}{2} \right]$$

is a martingale with respect to Q_x . It is not hard to carry out an analytic continuation and switch from θ to $i\theta$.

$$\exp \left[i\theta(x(t) - x(0)) - i\theta \int_0^t b(x(s)) ds + \frac{\theta^2 t}{2} \right]$$

are martingales with respect to Q_x . Now we can appeal to Lemma 1.7 with

$$A(t) = \exp \left[i\theta \int_0^t b(x(s)) ds - \frac{\theta^2 t}{2} \right]$$

to conclude that with $f(x) = e^{i\theta x}$,

$$f(x(t)) - f(x(0)) - \int_0^t (Lf)(x(s)) ds$$

is a martingale with respect to Q_x .

5.11. Ornstein-Uhlenbeck Process

The Ornstein-Uhlenbeck or the O-U process, as it is often called, is Brownian motion with the drift $b(x) = -cx$ where $c > 0$ is positive constant. This corresponds to the equation

$$x(t) = x + \beta(t) - c \int_0^t x(s) ds$$

which can be explicitly solved to give

$$x(t) = e^{-ct} x + \int_0^t e^{c(s-t)} d\beta(s)$$

In particular, the distribution of $x(t)$ is Gaussian with mean $e^{-ct}x$ and variance $\sigma^2(t) = \int_0^t e^{2c(s-t)} ds = \frac{1}{2c}(1 - e^{-2t})$. The O-U process therefore has the transition probability

$$(5.14) \quad p(t, x, A) = \int_A \frac{1}{\sqrt{2\pi\sigma^2(t)}} \exp \left[-\frac{(y - e^{-ct}x)^2}{2\sigma^2(t)} \right] dy$$

The O-U process has a stationary invariant distribution which is a Gaussian distribution with mean 0 and variance $\sigma^2 = \frac{1}{2c}$. That this Gaussian is invariant can be seen easily from the identity

$$e^{-2ct}\sigma^2 + \sigma^2(t) = \sigma^2$$

which shows that, if $x(0)$ is Gaussian with mean 0 and variance $\frac{1}{2c}$, then so is $x(t)$ for any $t > 0$. One can guess at this distribution by observing that it is the limit of the distributions $p(t, x, \cdot)$ as $t \rightarrow \infty$. The dependence on x disappears in the limit and we get the invariant distribution.

Once we have an invariant density we can construct a stationary stochastic process on $C(-\infty, \infty)$. The measure can be defined on $C[-T, \infty)$ by starting with the stationary distribution at time $-T$ and proceeding forward according to the Markovian evolution. In other words the measure Q^{-T} on $C[-T, \infty)$ is defined as

$$Q^{-T} = \int Q_x^{-T} \phi(x) dx$$

where Q_x^{-T} is the O-U process starting from $x(-T) = x$ and defined for $t \geq -T$. $\phi(x)$ is the invariant Gaussian density

$$\phi(x) = \frac{\sqrt{c}}{\sqrt{\pi}} e^{-cx^2}$$

The family $\{Q^{-T}\}$ is consistent as $T \rightarrow -\infty$ and defines a unique probability measure Q on $C(-\infty, \infty)$.

THEOREM 5.9 *The measure Q on $C(-\infty, \infty)$ is invariant under translations and is ergodic. It is a Gaussian process with*

$$\begin{aligned} E[x(t)] &= 0 \\ E[x(s)x(t)] &= \frac{1}{2c} e^{-c|t-s|} \end{aligned}$$

PROOF: Stationarity is just a consequence of the invariance of the distribution $\phi(x)dx$. Our construction provides the same limit if we replace $-T$ by $-T + k$. This establishes invariance of Q under translations. Proving ergodicity requires some work. We need to show that if E is a translation invariant set in $C(-\infty, \infty)$ then $Q(E) = 0$ or 1. By standard measure theory any measurable set can be approximated by sets from the σ -field $\mathcal{F}_t^s = \sigma\{x(u) : s \leq u \leq t\}$. In other words, given any $\epsilon > 0$ there exists $\tilde{E} \in \mathcal{F}_t^s$ such that $Q[E \Delta \tilde{E}] < \epsilon$ since E is invariant it can just as well be approximated by the translates of $\tilde{E}$. These translates can be pushed in either direction. This allows $\tilde{E}$ to be chosen from $\mathcal{F}_\ell^{-\infty}$ or $\mathcal{F}_\ell^\infty$ with a large fixed ℓ .

Because ϵ is arbitrary and ℓ can be arbitrarily chosen and fixed independently of ϵ , we can now conclude that $E \in \mathcal{F}_\ell^{-\infty} \cap \mathcal{F}_\ell^\infty$ for every ℓ . Recalling the definitions $\mathcal{F}_{-\infty}^{-\infty} = \bigcap_\ell \mathcal{F}_\ell^{-\infty}$ and $\mathcal{F}_\infty^\infty = \bigcap_\ell \mathcal{F}_\ell^\infty$ we conclude that $E \in \mathcal{F}_{-\infty}^{-\infty} \cap \mathcal{F}_\infty^\infty$. An invariant set belongs to the remote past as well as the remote future.

But for a Markov process, given the present, the past and the future are independent. This is not hard to check and is left as an exercise. But given this fact, given the present, any invariant set E , because it is both in the past and the future, has to be independent of itself. A set that is independent of itself has probability 0 or 1. In other words $P[E \mid \mathcal{F}_0^0] = \mathbf{1}_A(x(0))$. But invariant sets of the form $x(0) \in A$ exist only if

$$p(t, x, A) = \mathbf{1}_A(x)$$

Since $p(t, x, A)$ is either positive for all x or 0 for all x , there are no such nontrivial sets. In other words Q is ergodic. $\square$

REMARK 5.11. This is a fairly general argument which only depends on establishing the irreducibility of the state space. If $p(t, x, A)$ all have densities $p(t, x, y)$ with respect to a reference measure μ and $p(t, x, y) > 0$ a.e. μ for all x , that will do it!

EXERCISE 5.5. Let $Z = X_1 \times X_2 \times X_3$ be a product of three spaces with associated σ -fields. We have a Markov chain on Z defined by

$$Q_{1,2,3}(dx_1, dx_2, dx_3) = \mu(dx_1)P_{1,2}(x_1, dx_2)P_{2,3}(x_2, dx_3)$$

where $P_{1,2}$ and $P_{2,3}$ are transition probabilities. Write the joint distribution $Q_{1,2}$ given by $\mu_1(dx_1)P_{1,2}(dx_1, dx_2)$ as $\mu_2(dx_2)P_{2,1}(x_2, dx_1)$. Then show that

$$E^Q[f_1(x_1)f_2(x_2)f_3(x_3)] = \int \left[\int f_1(x_1)P_{2,1}(x_2, dx_1) \int f_3(x_3)P_{2,3}(x_2, dx_3) \right] f_2(x_2)\mu_2(dx_2)$$

O-U Semigroup. The semigroup corresponding to the O-U process is explicit and from the transition probability (5.14) we see that

$$\begin{aligned} (T_t f)(x) &= \int f(y)p(t, x, y)dy \\ &= \int f\left(e^{-ct}x + \sqrt{\frac{1-e^{-2ct}}{2c}}y\right) \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy \end{aligned}$$

as is the infinitesimal generator

$$(Lf)(x) = \frac{1}{2} \frac{d^2}{dx^2} - cx \frac{d}{dx}$$

The transition probability density, which is a Gaussian with mean $e^{-ct}x$ and variance $\sigma^2(t) = \frac{1-e^{-2ct}}{2c}$, can be written as

$$\begin{aligned} p(t, x, y) &= \frac{1}{\sqrt{2\pi\sigma^2(t)}} \exp\left[-\frac{(y - e^{-ct}x)^2}{2\sigma^2(t)}\right] \\ &= q(t, x, y) \sqrt{\frac{c}{\pi}} \exp[-cy^2] \end{aligned}$$

where

$$\begin{aligned} q(t, x, y) &= \frac{1}{\sqrt{2\pi\sigma^2(t)}} \sqrt{\frac{\pi}{c}} \exp\left[-\frac{(y - e^{-ct}x)^2}{2\sigma^2(t)} + cy^2\right] \\ &= \frac{e^{h(t,x,y)}}{\sqrt{1 - e^{-2ct}}} \end{aligned}$$

and

$$\begin{aligned} h(t, x, y) &= -\frac{(y - e^{-ct}x)^2}{2\sigma^2(t)} + c y^2 = -\frac{c(y - e^{-ct}x)^2}{1 - e^{-2ct}} + c y^2 \\ &= -\frac{ce^{-2ct}}{1 - e^{-2ct}}(x^2 + y^2) + 2cxy \frac{e^{-ct}}{1 - e^{-2ct}} \end{aligned}$$

The point of this calculation is that $q(t, x, y)$ is symmetric, so that the semigroup T_t is symmetric in $L_2[\phi]$, where ϕ is the Gaussian weight

$$\phi(y) = \sqrt{\frac{c}{\pi}} \exp[-c y^2]$$

which is also the invariant density. The generator L is self adjoint as well and will have in fact a discrete spectrum of real eigenvalues and eigenfunctions.

EXERCISE 5.6. Take $c = \frac{1}{2}$. Show that the eigenvalues of L are the non positive integers and half-integers i.e. $\{0, -\frac{1}{2}, -1, -\frac{3}{2}, -2, \dots\}$ and the corresponding eigenfunctions are the Hermite polynomials.

EXERCISE 5.7. Calculate

$$\int q(t, x, x) \phi(x) dx = \frac{1}{1 - e^{-\frac{t}{2}}}$$

Can you explain it?

5.12. Invariant Densities

If $b(x)$ is any reasonable drift and we consider the generator

$$L = \frac{1}{2} \frac{d^2}{dx^2} + b(x) \frac{d}{dx}$$

and the corresponding process Q_x with transition probabilities $p(t, x, A)$ given either by solving (5.9) or through Girsanov formula (5.12) the formal adjoint equation

$$L^* \phi = \frac{1}{2} \frac{d^2 \phi}{dx^2} - \frac{d(b\phi)}{dx} = \frac{d\psi}{dx} = 0$$

where

$$\psi(x) = \frac{1}{2} \phi'(x) - b(x) \phi(x)$$

If we are looking for a ϕ which is a probability density, then $\psi(x)$ can not be a nonzero constant. The choice therefore reduces to

$$(5.15) \quad \phi(x) = \exp \left[\int_0^x 2b(y) dy \right]$$

We shall sketch a proof that if $\phi(x)$ is integrable it is the invariant probability distribution for the process. In particular, if $p(t, x, y)$ is the transition probability density

$$\phi(y) = \int p(t, x, y) \phi(x) dx$$

we need to show that for any smooth function f

$$\frac{d}{dt} \langle T_t f, \phi \rangle = 0$$

Formally

$$\frac{d}{dt} \langle T_t f, \phi \rangle = \langle L T_t f, \phi \rangle = \langle T_t f, L^* \phi \rangle = 0$$

But this needs $u(t, x) = T_t f$ to be smooth enough to be differentiable twice in x and once in t . Then integration by parts can be done. The details are left as an exercise.

EXERCISE 5.8. Show that the map $\Phi_x : C[0, \infty) \rightarrow C[0, \infty)$ which is the solution of (5.9) is a smooth map as a function of x provided $b(\cdot)$ is smooth. Do a variation of parameters argument (i.e. differentiate implicitly both sides in the equation) to estimate the derivatives of Φ_x with respect to x as functions of $\beta(\cdot)$. Conclude that $(T_t f)(x)$ is smooth provided b, f are smooth. If b is not smooth approximate by smooth b and pass to the limit both in the semigroup as well as the invariant distribution.

It is of course quite possible that $\phi(x)$ defined in (5.15) may not be integrable and the process not be positive recurrent. For instance when $b(x) \equiv 0$!

REMARK 5.12. Given a $\phi(x) > 0$ we can define $b(x) = \frac{\phi'(x)}{2\phi(x)}$ and this drift will have ϕ as invariant density.

5.13. Local Times

Local time is one of the most important and useful objects in the theory of Markov processes. It is also strange. Let us consider the standard Brownian motion $x(t)$ with its measures $\{P_x\}$ on $C[0, \infty)$. If the path $x(\cdot)$ is given in an interval $[0, t]$ we can define a random measure $L(t, A, \omega)$ by

$$(5.16) \quad L(t, \omega, A) = \int_0^t \mathbf{1}_A(x(s)) ds$$

Note that for any function $V(x)$ which is bounded and continuous on $\mathbb{R}$

$$\int V(y) L(t, \omega, dy) = \int_0^t V(x(s)) ds$$

THEOREM 5.10 *For almost all ω with respect to P_x , the random measure $L(t, \omega, A)$ satisfies the following properties.*

- (i) $L(t, \omega, A)$ is absolutely continuous with respect to the Lebesgue measure on $\mathbb{R}$ with Radon-Nikodym derivative $l(t, \omega, y)$, i.e.

$$L(t, \omega, A) = \int_A l(t, \omega, y) dy$$

- (ii) *It is possible to choose a version $l(t, \omega, y)$ that is jointly continuous as a function of t and x .*

- (iii) For each x the function $l(t, \omega, x)$ is a nondecreasing function of t which increases only when $x(t)$ is at x .
- (iv) For any $t_1 < t_2$, if $x(t)$ visits x during (t_1, t_2) , then $l(t_2, \omega, x) > l(t_1, \omega, x)$

PROOF: The starting point is Tanaka's formula which is a special case of Itô's formula. If we take $f(x) = |x - y|$ and apply Itô's formula to it we get

$$|x(t) - y| - |x(0) - y| = \int_0^t \text{sign}(x(s) - y) dx(s) + \int_0^t \delta_y(x(s)) ds$$

Since formally $l(t, \omega, y)$ wants to be $\int_0^t \delta_y(x(s)) ds$ one can define it as

$$(5.17) \quad l(t, \omega, y) = |x(t) - y| - |x(0) - y| - \int_0^t \text{sign}(x(s) - y) dx(s)$$

and show that it works.

Step 1. Define $l(t, \omega, y)$ as above. The integrand in the stochastic integral is progressively measurable and bounded. For each y the stochastic integral exists as an almost surely continuous martingale in t . We would like to choose a version which for each t is almost surely a continuous function of y . For this purpose we consider for $a < b$ the difference

$$\begin{aligned} D(t, a, b) &= \int_0^t [\text{sign}(x(s) - a) - \text{sign}(x(s) - b)] dx(s) \\ &= \int_0^t \mathbf{1}_{[a, b]}(x(s)) dx(s) \end{aligned}$$

and estimate it by

$$(5.18) \quad E[[D(t, a, b)]^4] \leq C(t)|a - b|^2$$

We can then apply Theorem A.5 to conclude that such a choice is possible. According to Remark 5.10 we can estimate it by

$$E[[D(t, a, b)]^4] \leq 30E\left[\left[\int_0^t \mathbf{1}_{[a, b]}(x(s)) ds\right]^2\right]$$

We need the following

LEMMA 5.11 *Let $x(t)$ be a Markov process. $V(x) \geq 0$ and assume that*

$$\sup_x E_x\left[\int_0^T V(x(s)) ds\right] \leq C$$

Then

$$E_x\left[\left[\int_0^T V(x(s)) ds\right]^n\right] \leq n! C^n$$

PROOF: We write

$$\left[\int_0^T V(x(s)) ds\right]^n = n! \int \cdots \int_{0 \leq t_1 \cdots < t_n < T} V(x(t_1)) \cdots V(x(t_n)) dt_1 \cdots dt_n$$

We note that

$$E \left[\int_{t_{n-1}}^T V(x(t_n)) dt_n \mid \mathcal{F}_{t_{n-1}} \right] = E_{x(t_{n-1})} \left[\int_0^{T-t_{n-1}} V(x(s)) ds \right] \leq C$$

Successive application of the same step gives the result. $\square$

Continuing with the proof of the theorem

$$\begin{aligned} E_x \left[\int_0^t \mathbf{1}_{[a,b]}(x(s)) ds \right] &= \int_a^b \int_0^t \frac{1}{\sqrt{2\pi t}} e^{-\frac{(x-y)^2}{2t}} dy \\ &\leq \sqrt{\frac{2t}{\pi}} |a - b| \end{aligned}$$

We now have the estimate (5.18) which proves the continuity in y for fixed t . There is a version of Kolmogorov's theorem valid for processes $\eta(x, \omega)$ that depend on $x \in \mathbb{R}^k$ rather than $\mathbb{R}$. The estimate needed is

$$E[|\eta(x) - \eta(y)|^\beta] \leq C|x - y|^{k+\alpha}$$

for some $\alpha > 0$. We can look at

$$\eta(t, a) = \int_0^t \text{sign}(x(s) - a) dx(s)$$

and get an estimate of the form

$$E[|\eta(t, a) - \eta(s, b)|^6] \leq C(|t - s| + |a - b|)^3$$

which is enough. The derivation of the above estimate is quite similar to the previous one involving fourth moments.

Step 2. We now show that

$$\int l(t, \omega, y) V(y) dy = \int_0^t V(x(s)) ds$$

We start with equation (5.17) and multiply both sides with a smooth function $V(y)$ with compact support and integrate. Denoting by

$$u(x) = \int |x - y| V(y) dy$$

clearly $u'(x) = \int \text{sign}(x - y) V(y) dy$ and $u''(y) = 2V(y)$. Moreover

$$\begin{aligned} u(x(t)) - u(x(0)) - \int_0^t u'(x(s)) dx(s) &= \int_0^t V(x(s)) ds \\ &= \int V(y) l(t, \omega, y) dy \end{aligned}$$

This shows that $l(t, \omega, y)$ is increasing and increases only when $x(t) = y$.

Step 3. Finally, to prove (iii), by the strong Markov property, it is enough to show that for $t > 0$

$$P_0[l(t, \omega, 0) > 0] = 1$$

To prove it, let us denote by $\tau = \{\inf t : l(t, \omega, 0) > 0\}$. τ is not a stopping time, but $\tau + \epsilon$ is for every $\epsilon > 0$. This is good enough to apply Doob's stopping theorem. We can apply it $\tau + \epsilon$ and let $\epsilon \rightarrow 0$. Therefore

$$E^{P_0}[|x(\tau \wedge t)|] = E^{P_0}[l(\tau, \omega, 0)] = 0$$

Since $P_0[|x(t)| > 0] = 1$ for every $t > 0$. Obviously this cannot be true unless $P_0[\tau = 0] = 1$.

□

5.14. Reflected Brownian Motion

If $x(t)$ is Brownian motion $y(t) = |x(t)|$ is a Markov process. Its transition probability density is

$$q(t, x, y) = p(t, x, y) + p(t, x, -y) = \frac{1}{\sqrt{2\pi t}} [e^{-\frac{(x-y)^2}{2t}} + e^{-\frac{(x+y)^2}{2t}}]$$

is symmetric under $x \rightarrow -x$. This makes $q(t, x, y)$ the transition probabilities of a Markov process that lives on $[0, \infty)$. This is called the reflected Brownian motion. Note that $y(t)$ is a submartingale and the decomposition $y(t) = M(t) + l(t)$ is the same as Tanaka's formula for the local time at 0.

There are other descriptions of reflected Brownian motion. Let $f(t)$ be any continuous function on $[0, \infty)$. Let $f(0) \geq 0$. $f(t)$ may become negative at some point. Let us define the class $\mathcal{J}$ of functions $a(\cdot) : [0, \infty) \rightarrow [0, \infty)$ as follows

$$\mathcal{J} = \{a(\cdot) : a(0) = 0; a(t) \text{ is } \uparrow; g(t) = f(t) + a(t) \geq 0 \text{ for all } t \geq 0\}$$

THEOREM 5.12 *Among all $a(\cdot)$ from $\mathcal{J}$ there is a unique $a(\cdot)$ which is minimal, i.e. for any $b(\cdot) \in \mathcal{J}$, $b(t) \geq a(t)$. It is characterized by the property that the measure $da(t)$ is supported on the set $\{t : f(t) + a(t) = 0\}$.*

REMARK 5.13. One can imagine $f(t)$ as the fortunes of a gambler and $a(t)$ the help provided by an indulgent but thrifty relative, who will pay exactly what it takes to keep the gambler out of debt when he or she is totally broke and never ask for the money back when the gambler is lucky enough to win.

PROOF: If $b(\cdot) \in \mathcal{J}$, then $g(t) = b(t) + f(t) \geq 0$. But $b(t)$ is nondecreasing. Therefore

$$b(t) = \sup_{0 \leq s \leq t} b(s) = \sup_{0 \leq s \leq t} [g(s) - f(s)] \geq \sup_{0 \leq s \leq t} [-f(s)] = - \inf_{0 \leq s \leq t} f(s)$$

On the other hand $a(t) = - \inf_{0 \leq s \leq t} f(s)$ is in $\mathcal{J}$ and is therefore the minimal element of $\mathcal{J}$. It is clear that $a(t)$ increases only when $f(t)$ is currently at its minimum in $[0, t]$, i.e. $g(t) = 0$. Suppose we have two solutions $\{a_i(t); i = 1, 2\}$ from $\mathcal{J}$ such that $a_i(t)$ increases only when $g_i(t) = f(t) + a_i(t) = 0$, then

$$\begin{aligned} [a_1(t) - a_2(t)]^2 &= \int_0^t 2[a_1(s) - a_2(s)][da_1(s) - da_2(s)] \\ &= \int_0^t 2[g_1(s) - g_2(s)][da_1(s) - da_2(s)] \end{aligned}$$

$$\begin{aligned}
&= -2 \int_0^t [g_2(s) da_1(s) + g_1(s) da_2(s)] \\
&\leq 0
\end{aligned}$$

Hence $a_1(t) \equiv a_2(t)$ and we have uniqueness. $\square$

If $y(t)$ is the reflected Brownian motion then

$$y(t) = l(t, \omega, 0) + \int_0^t \text{sign } x(s) dx(s) = l(t, \omega, 0) + \beta(t)$$

$l(t, \omega, 0)$ increases only when $y(t) = 0$ and $\beta(t)$ is another Brownian motion. From the uniqueness proved earlier in Theorem 5.12 the local time $l(t, \omega, 0)$ has the representation

$$l(t, \omega, 0) = - \inf_{0 \leq s \leq t} \beta(s)$$

in terms of a Brownian motion. As a stochastic process the distribution of $l(\cdot, \omega, 0)$ is the same as that of $-\inf_{0 \leq s \leq t} \beta(s)$ or by symmetry $\sup_{0 \leq s \leq t} \beta(s)$. In particular,

$$P_0[l(t, \omega, 0) \geq \ell] = 2P_0[x(t) > \ell] = \sqrt{\frac{2}{\pi t}} \int_{\ell}^{\infty} e^{-\frac{y^2}{2t}} dy$$

EXERCISE 5.9. Use exponential martingales to verify that $\int_0^t \text{sign } x(s) dx(s)$ is again a Brownian motion.

REMARK 5.14. We saw that the

$$p(t, x, y) = \frac{1}{\sqrt{2\pi t}} e^{-\frac{(x-y)^2}{2t}}$$

is the fundamental solution of the heat equation

$$\frac{\partial p}{\partial t} = \frac{1}{2} \frac{\partial^2 p}{\partial x^2}$$

in the sense that

$$\int f(y) p(t, x, y) dy = u(t, x)$$

solves

$$\frac{\partial u}{\partial t} = \frac{1}{2} \frac{\partial^2 u}{\partial x^2}$$

with $u(0, x) = f(x)$. Moreover $p(t, x, y)$ is the transition probability density of Brownian motion on $\mathbb{R}$. If we replace $\mathbb{R}$ by $[0, \infty)$ and kill the Brownian path if it reaches 0, then we have a transition probability that is sub-stochastic that can be calculated from the reflection principle.

$$\begin{aligned}
P_x[\{x(t) \in A\} \cap \{\tau_0 > t\}] &= \int_A p_0(t, x, y) dy \\
&= \int_A \left[\frac{1}{\sqrt{2\pi t}} e^{-\frac{(x-y)^2}{2t}} - \frac{1}{\sqrt{2\pi t}} e^{-\frac{(x+y)^2}{2t}} \right] dy
\end{aligned}$$

where τ_0 is the first time of reaching 0. It is also a fundamental solution of the heat equation on $[0, \infty)$ but with the additional Dirichlet boundary condition, $p(t, x, y) \rightarrow 0$ as $x \rightarrow 0$.

The transition density of the reflected Brownian motion

$$\widehat{p}(t, x, y) = \left[\frac{1}{\sqrt{2\pi t}} e^{-\frac{(x-y)^2}{2t}} + \frac{1}{\sqrt{2\pi t}} e^{-\frac{(x+y)^2}{2t}} \right] dy$$

is also a fundamental solution but satisfies instead the Neumann boundary condition $p_x(t, x, y) \rightarrow 0$ as $x \rightarrow 0$.

5.15. Excursion Theory

Reflected Brownian motion is a nonnegative process $x(t)$ which returns to zero often and its set of zeros is a closed uncountable set of Lebesgue measure zero. The local time $l(t)$ is a continuous nondecreasing function that increases only when $x(t) = 0$ and $x(t) - l(t) = \beta(t)$ is a Brownian motion. The set where $x(t) > 0$ is an open set and is therefore a countable union of disjoint intervals. In each interval $x(t) > 0$ and $x(t) \rightarrow 0$ at the end points. The path in each such interval is called an excursion. The excursions have duration or length which is the length of the interval. The excursion theory describes the statistics of these excursions and tells how to reassemble the reflected Brownian motion from its excursions. If we make each excursion be positive or negative with probability $\frac{1}{2}$ and independently so for different excursions, we recover the Brownian motion.

Let $x > 0$ and $T > s > 0$ be given. We want to start a Brownian motion at time s from x and condition it to exit from $(0, \infty)$ at time T . By the reflection principle the distribution of $\tau = \{\inf t : x(t) = 0\}$ is given by

$$(5.19) \quad \widehat{p}(s, x, \tau) d\tau = \frac{1}{\sqrt{2\pi}} \frac{x}{(\tau - s)^{\frac{3}{2}}} \exp\left[-\frac{x^2}{2(\tau - s)}\right] d\tau$$

We can calculate the conditional probability density that such a conditioned process is at y at some intermediate time t .

$$p_T(s, x, t, y) = r(s, x, t, y) \frac{\widehat{p}(t, y, T)}{\widehat{p}(s, x, T)}$$

where

$$(5.20) \quad r(s, x, t, y) = \frac{1}{\sqrt{2\pi(t-s)}} \left[\exp\left[-\frac{(x-y)^2}{2(t-s)}\right] - \exp\left[-\frac{(x+y)^2}{2(t-s)}\right] \right]$$

$p_T(s, x, t, y)$ can serve as the transition probabilities of a Markov process. Although defined initially only for $x > 0$ we can take the limit as $x \rightarrow 0, s \rightarrow 0$ that yields

$$p_T(0, 0, t, y) = \frac{y^2}{\sqrt{2\pi t}} \frac{T^{\frac{3}{2}}}{(T-t)^{\frac{3}{2}} t^{\frac{3}{2}}} \exp\left[-\frac{y^2 T}{2t(T-t)} - \frac{x^2(T-t)}{2Tt}\right]$$

This describes an inhomogeneous Markov process Q_T on excursions of length T . Brownian scaling implies that the map $x(\cdot) \rightarrow y(\cdot)$ defined by $y(Tt) = \sqrt{T}x(t)$

that maps excursions of length 1 into excursions of length T takes Q_1 into Q_T . The distribution of the length τ of an excursion starting from $x > 0$ has the distribution

$$q(x, \tau) d\tau = \sqrt{\frac{1}{2\pi}} \frac{x}{\tau^{\frac{3}{2}}} \exp\left[-\frac{x^2}{2\tau}\right]$$

Suitably normalized this converges to

$$q(\tau) d\tau = \sqrt{\frac{1}{2\pi}} \frac{d\tau}{\tau^{\frac{3}{2}}}$$

as $x \rightarrow 0$.

We now construct a point process on $[0, \infty)$. This will be a marked point process. The points come with labels τ and an excursion of length τ . we call this space $\mathcal{E}$. We have a σ -finite measure on $\mathcal{E}$.

$$d\lambda = \sqrt{\frac{1}{2\pi}} \frac{d\tau}{\tau^{\frac{3}{2}}} dQ_\tau$$

We define a Poisson point process on $[0, \infty) \times \mathcal{E}$ has intensity $ds d\lambda$. Our points are $(s_j, \tau_j, x_j(\cdot))$. We try and recover the reflected Brownian motion by doing the following. The times s_j are used only to order the excursions. We define

$$A(s) = \sum_{i: s_i \leq s} \tau_i$$

Since there are an infinite number of small jumps in any interval of time $A(s)$ is strictly increasing. For each time t , we consider $A^{-1}(t)$ defined as

$$A^{-1}(t) = \{\inf s : A(s) \geq t\} = \{\sup s : A(s) \leq t\}$$

Either $A(s) = t$ in which case we define $x(t) = 0$. More likely $A(s)$ jumps over t at a specific time s_j with a specific τ_i extending from t^- to $t^+ = t^- + \tau_i$ with $t^- < t < t^+$. We fit the excursion in this interval so that $x(t) = x_j(t - t^-)$ in the interval (t^-, t^+) with $x(t^-) = x(t^+) = 0$.

Finally, we need to check that our construction really gives the reflected Brownian motion. Let us run a Brownian motion on $[0, \infty)$ and when it hits 0 we move it to $\delta > 0$ and start over again. This process continues indefinitely. We just have a concatenation of excursions starting from δ . It is clear that as $\delta \rightarrow 0$ this process denoted by $x_\delta(t)$ converges to the reflected Brownian motion. We can prove it by showing that the process can be obtained by erasing the part of the reflected Brownian motion when it leaves 0 until it hits δ and that the total duration of these erased periods up to any time t goes to 0 with δ .

If we denote the excursion measure on $\mathcal{E}$ by

$$d\lambda_\delta = \sqrt{\frac{1}{2\pi}} \frac{\delta}{\tau^{\frac{3}{2}}} \exp\left[-\frac{\delta^2}{2\tau}\right] d\tau dQ_{\delta, \tau}$$

we can repeat the way we constructed the reflected Brownian motion from the point process corresponding to $ds d\lambda$, but now with $ds d\lambda_\delta$. The process we get will be $x_\delta(t)$. Letting $\delta \rightarrow 0$ will do it.

The measure Q_T was defined as an inhomogeneous Markov process on $[0, T]$ with transition probability

$$p_T(s, x, t, y) = r(s, x, t, y) \frac{\widehat{p}(t, y, T)}{\widehat{p}(s, x, T)}$$

where r and $\widehat{p}$ are as in (5.19) and (5.20). One can ask what its infinitesimal generator is. Since

$$\widehat{p}(s, x, T) = \int r(s, x, t, y) \widehat{p}(t, y, T) dy$$

for fixed T , $\widehat{p}(s, x, T)$ is a solution of the heat equation.

$$\widehat{p}_t + \frac{1}{2} \widehat{p}_{xx} = 0$$

The time dependent generator L_t is obtained as

$$\begin{aligned} (L_t f)(x) &= \lim_{h \rightarrow 0} \frac{1}{h} \left[\int p_T(t-h, x, t, y) f(y) dy - f(x) \right] \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \left[\int r(t-h, x, t, y) \frac{\widehat{p}(t, y, T)}{\widehat{p}(t-h, x, T)} f(y) dy - f(x) \right] \\ &= \lim_{h \rightarrow 0} \frac{1}{\widehat{p}(t-h, x, T)} \frac{1}{h} \left[\int r(t-h, x, t, y) \widehat{p}(t, y, T) f(y) dy \right. \\ &\quad \left. - \widehat{p}(t-h, x, T) f(x) \right] \end{aligned}$$

Since $r(s, x, t, y)$ is the fundamental solution of the heat equation,

$$\begin{aligned} (L_t f)(x) &= \frac{1}{\widehat{p}(t, x, T)} \left[\frac{1}{2} \frac{d^2}{dx^2} \widehat{p}(t, x, T) f(x) + \frac{d}{dt} \widehat{p}(t, x, T) f(x) \right] \\ &= \frac{1}{2} f''(x) + \frac{\widehat{p}_x(t, x, T)}{\widehat{p}(t, x, T)} f'(x) \\ &= \frac{1}{2} f''(x) + \left[\frac{1}{x} - \frac{x}{T-t} \right] f'(x) \end{aligned}$$

The drift keeps the Brownian motion away from 0 during $0 \leq t < T$, but forces it to go to zero at time T .

5.16. Invariance Principle

In the volume on probability we had discussed central limit theorems under various conditions. The simplest among them is for sums of independent identically distributed random variables. If

$$S_n = X_1 + \cdots + X_n$$

is the sum of n independent and identically distributed random variables with mean $E[X_i] = 0$ and variance $E[X_i^2] = 1$, then $\frac{S_n}{\sqrt{n}}$ has asymptotically a Gaussian distribution with mean 0 and variance 1. This allows us to say for example that

$$\lim_{n \rightarrow \infty} P[S_n \geq a\sqrt{n}] = \int_a^\infty \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$$

The invariance principle or the functional central limit allows us to conclude that

$$\lim_{n \rightarrow \infty} P\left[\sup_{1 \leq j \leq n} S_j \geq a\sqrt{n}\right] = Q\left[\sup_{0 \leq s \leq 1} x(t) \geq a\right] = 2 \int_a^\infty \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$$

where Q is the Wiener measure. The method consists of considering the random walk $S_j : 1 \leq j \leq n$ and viewing it as path on $[0, 1]$ by defining

$$X_n\left(\frac{j}{n}\right) = \frac{S_j}{\sqrt{n}}$$

and interpolating linearly between $\frac{j}{n}$ and $\frac{j+1}{n}$. The resulting process will have a distribution on $C[0, 1]$ that we denote by P_n . If P_n has a weak limit on $C[0, 1]$ it can only be the Wiener measure Q . Usual central limit theorem, or actually the multidimensional version of it, will ensure that any limit will have the same finite-dimensional distribution as Wiener measure. It is therefore enough to prove that the sequence $\{P_n\}$ is compact in the space $\mathcal{M}$ of probability measures on $C[0, 1]$.

Once we establish weak convergence it will follow as a consequence that any continuous function from $f : C[0, 1] \rightarrow \mathbb{R}$ the distribution of f under P_n will converge to the distribution of f under Q . Taking $f(x(\cdot)) = \sup_{0 \leq t \leq 1} x(t)$ will prove our result.

To prove the compactness of P_n we will assume for simplicity that the fourth moment is finite. Then it is not too hard to obtain the following estimate.

LEMMA 5.13

$$E^{P_n}[|x(t) - x(s)|^4] = E[|X_n(t) - X_n(s)|^4] \leq C|t - s|^2$$

PROOF: Let $s < t$ belong to some $[\frac{i}{n}, \frac{i+1}{n}]$. Then

$$E[|X(t) - X(s)|^4] = |t - s|^4 n^2 E[X_{i+1}^4] = C|t - s|^2$$

since $|t - s| \leq \frac{1}{n}$. Let s, t be in different subintervals. Then we can estimate

$$\|X(t) - X(s)\|_4 \leq \|X(s) - X(t_1)\|_4 + \|X(t_1) - X(t_2)\|_4 + \|X(t_2) - X(t)\|_4$$

The two end pieces correspond to incomplete subintervals and $t_i = \frac{k_1}{n}$ and $t_2 = \frac{k_2}{n}$ with $k_2 - k_1 = \ell = n(t_2 - t_1)$ which can be 0. The end pieces are estimated by $C[|s - t_1|^{\frac{1}{2}} + |t_2 - t|^{\frac{1}{2}}]$ and the middle piece by

$$\frac{1}{n^2} E\left[\sum_{j=k_1+1}^{k_2} X_j\right]^4 = \frac{1}{n^2} [3\ell(\ell - 1) + C\ell] \leq C|t_1 - t_2|^2$$

We can add the three pieces to prove the lemma. □

5.17. Representation of Martingales

The aim of this section is to examine what all the square integrable martingales on Wiener space are. We have examples that are stochastic integrals of the form

$$M(t) = \int_0^t c(s, \omega) dx(s)$$

Is that all?

THEOREM 5.14 *Any square integrable martingale $M(t)$ with respect to the natural σ -field $\mathcal{F}_t$ on the Wiener space $(\Omega, \mathcal{F}_t, P)$ is in fact given by a stochastic integral*

$$M(t) = c_0 + \int_0^t c(s, \omega) dx(s)$$

with a suitable progressively measurable $c(\cdot, \cdot)$, satisfying

$$E \left[\int_0^t c^2(s, \omega) ds \right] < \infty$$

for every $t > 0$. The choice of c_0 must of course be $c_0 = E[M(t)]$.

PROOF: The proof is in two steps.

Step 1. We note that we can fix a time interval $[0, T]$ and work on square integrable martingales in the time interval $[0, T]$. Clearly the representing $c(\cdot, \cdot)$ is uniquely determined by $M(\cdot)$ and is consistently defined when we expand the interval. Let us assume $T = 1$ without any loss of generality. The real question is: can every $\mathcal{F}_1$ measurable square integrable function $X(\omega)$ with mean zero be represented as

$$X(\omega) = \int_0^1 e(s, \omega) dx(s)$$

for some $c(\cdot, \cdot)$ with

$$E \left[\int_0^t c^2(s, \omega) ds \right] < \infty?$$

Because the correspondence between $X \leftrightarrow c(\cdot, \cdot)$ is an isometry the set of X 's that admit a representation is a closed subspace. It is therefore sufficient to prove that for any n and any smooth function f on $\mathbb{R}^n$ and $0 \leq t_1 < t_2 < \dots < t_n \leq 1$, $X = f(x(t_1), x(t_2), \dots, x(t_n))$ admits a representation. We will prove it by induction on n .

Step 2. Given f on $\mathbb{R}^n$ and $0 \leq t_1 < t_2 < \dots < t_n \leq 1$ let us define

$$U(t_1, x_1, t_2, x_2, \dots, t_{n-1}, x_{n-1}, t, x) = \int_{\mathbb{R}} f(x_1, x_2, \dots, x_{n-1}, y) \frac{1}{\sqrt{2\pi(t_n - t)}} \exp \left[-\frac{(y - x)^2}{2(t_n - t)} \right] dy$$

Because U satisfies

$$\frac{\partial U}{\partial t} + \frac{1}{2} \frac{\partial^2 U}{\partial x^2} = 0$$

we obtain by Itô's formula

$$U(t_1, x_1, t_2, x_2, \dots, t_{n-1}, x_{n-1}, t_n, x(t_n)) - V(t_1, x_1, t_2, x_2, \dots, t_{n-1}, x_{n-1}) = \int_{t_{n-1}}^{t_n} U_x(t_1, x_1, t_2, x_2, \dots, t_{n-1}, x_{n-1}, s, x(s)) dx(s)$$

where

$$V(t_1, x_1, t_2, x_2, \dots, t_{n-1}, x_{n-1}) = U(t_1, x_1, t_2, x_2, \dots, t_{n-1}, x_{n-1}, t_{n-1}, x_{n-1})$$

The problem now reduces to representing V which is a function of $n - 1$ variables. $\square$

COROLLARY 5.15 *If $M(t)$ is a square integrable martingale with respect to $(\Omega, \mathcal{F}_t, P)$ then it has an almost surely continuous version.*

PROOF: Let us represent $M(1)$ as a stochastic integral

$$M(1) = c_0 + \int_0^1 c(s, \omega) dx(s)$$

Then the stochastic integral

$$\widehat{M}(t) = c_0 + \int_0^t c(s, \omega) dx(s)$$

is an almost surely continuous version of $M(t)$. $\square$

CHAPTER 6

One-Dimensional Diffusions

6.1. Stochastic Differential Equations

A diffusion process is a stochastic process which behaves locally like a Brownian motion with some mean b and variance a . However b and a are no longer constants, but depend on the position $x(t)$ and perhaps even explicitly on t . We are given two functions $a(t, x)$ and $b(t, x)$ and our goal is to construct a Markov process with continuous paths such that, roughly speaking,

$$E[x(t+h) - x(t) \mid \mathcal{F}_t] \simeq hb(t, x(t)) + o(h)$$

and

$$E[(x(t+h) - x(t))^2 \mid \mathcal{F}_t] \simeq ha(t, x(t)) + o(h)$$

Given the starting point x , $a(\cdot, \cdot)$ and $b(\cdot, \cdot)$ should determine the process uniquely. This problem is rather complex and one can use a variety of methods with their own individual advantages and shortcomings. We shall first consider the method of stochastic differential equations. The idea due to K. Itô, is to try to realize the increments $x(t+h) - x(t)$ as

$$x(t+h) - x(t) \simeq \sqrt{a(t, x(t))} [\beta(t+h) - \beta(t)] + b(t, x(t))h$$

or equivalently

$$(6.1) \quad dx(t) = \sqrt{a(t, x(t))} d\beta(t) + b(t, x(t))dt$$

where $\beta(\cdot)$ is the Brownian motion. This needs careful interpretation because $\beta(\cdot)$ is not of bounded variation in t . Taking $\sigma(t, x) = \sqrt{a(t, x)}$ the problem is formulated in the following terms.

Let $(\Omega, \mathcal{F}_t, \beta(t), P)$ be the Wiener space, i.e. the space $\Omega = C[0, \infty)$ of continuous functions $\beta(\cdot)$ with values in $\mathbb{R}$, with the Wiener measure P on it. Let $s \geq 0$ be given along with a random variable $\zeta(\omega)$ that is $\mathcal{F}_s$ measurable. We are looking for an almost surely continuous, progressively measurable function $x(t, \omega)$, defined for $t \geq s$, that satisfies

$$(6.2) \quad x(t, \omega) = \zeta(\omega) + \int_s^t \sigma(u, x(u)) d\beta(u) + \int_s^t b(u, x(u)) du$$

for all $t \geq s$. We had earlier defined stochastic integrals of the type that arise here in Section 5.9. We would like to make enough assumptions on $\zeta(\cdot)$, $a(\cdot, \cdot)$ and $b(\cdot, \cdot)$ to guarantee us both existence and uniqueness in some natural class. We remark that in the special case when $\sigma(\cdot, \cdot) \equiv 0$, the stochastic differential equation reduces to an ordinary differential equation, one that is usually solved by

Picard's iteration scheme. This method provides both existence and uniqueness if we assume that $b(\cdot, \cdot)$, satisfies a Lipschitz condition in x . In an analogous manner we will assume that

$$(6.3) \quad |\sigma(t, x) - \sigma(t, y)| \leq A|x - y|$$

$$(6.4) \quad |b(t, x) - b(t, y)| \leq A|x - y|$$

In addition we need some control on the size of $\sigma(\cdot, \cdot)$ and $b(\cdot, \cdot)$ and we shall assume

$$(6.5) \quad |\sigma(t, x)| \leq A(1 + |x|)$$

$$(6.6) \quad |b(t, x)| \leq A(1 + |x|)$$

Because of the Lipschitz assumption this is basically a hypothesis of the form $|\sigma(t, 0)| \leq A$ and $|b(t, 0)| \leq A$. Given $\zeta(\omega)$, which is $\mathcal{F}_s$ measurable and satisfies

$$(6.7) \quad \int_{\Omega} |\zeta(\omega)|^2 dP < \infty$$

we are looking for progressively measurable almost surely continuous solution $x(\cdot, \cdot)$ defined for $t \geq s$ that satisfies

$$(6.8) \quad E^P \left\{ \int_s^t |x(u, \omega)|^2 du \right\} < \infty$$

and the stochastic integral equation (6.2). We will prove existence as well as uniqueness in the class of such functions $x(\cdot, \cdot)$.

THEOREM 6.1 *There exists a unique solution to the stochastic integral equation (6.2) under the assumptions (6.3) through (6.7).*

PROOF: *Step 1.* We start with

$$x_0(t, \omega) \equiv \zeta(\omega)$$

and define successively

$$x_{n+1}(t, \omega) = \zeta(\omega) + \int_s^t \sigma(u, x_n(u, \omega)) d\beta(u) + \int_s^t b(u, x_n(u, \omega)) du$$

We note that, by induction on n , we can establish the following preliminary facts: for each $n \geq 0$, $x_{n+1}(\cdot, \cdot)$ is well defined, is progressively measurable and almost surely continuous. In addition we can assert that, for every $t > s$,

$$\sup_{s \leq u \leq t} E^P \{ |x_{n+1}(u, \omega)|^2 \} < \infty$$

implying

$$E^P \left\{ \int_s^t |\sigma(u, x_{n+1}(u, \omega))|^2 du \right\} < \infty$$

so that the next induction step can be carried out.

Step 2. We define, for $t \geq s$, the difference

$$\begin{aligned} y_n(t, \omega) &= x_{n+1}(t, \omega) - x_n(t, \omega) \\ &= \int_s^t [\sigma(u, x_n(u, \omega)) - \sigma(u, x_{n-1}(u, \omega))] d\beta(u) \\ &\quad + \int_s^t [b(u, x_n(u, \omega)) - b(u, x_{n-1}(u, \omega))] du \end{aligned}$$

If we denote by

$$\Delta_n(t) = E^P \{|y_n(t, \omega)|^2\}$$

we obtain by using the Lipschitz condition, the properties of stochastic integrals and Schwarz's inequality

$$\Delta_n(t) \leq 2A^2(1+t-s) \int_s^t \Delta_{n-1}(u) du$$

If we limit ourselves to an arbitrary but fixed interval $s \leq t \leq T$ we can rewrite the inequality as

$$(6.9) \quad \Delta_n(t) \leq 2A^2(1+T-s) \int_s^t \Delta_{n-1}(u) du = C_T \int_s^t \Delta_{n-1}(u) du$$

For $n = 0$ we estimate directly

$$y_0(t, \omega) = \int_0^t \sigma(u, \zeta(\omega)) d\beta(u) + \int_0^t b(u, \zeta(\omega)) du$$

and

$$(6.10) \quad \Delta_0(t) = E^P \{|y_0(t, \omega)|^2\} \leq C C_T t$$

where C_T is the same constant as above and $C = E^P \{1 + |\zeta(\omega)|^2\}$.

Step 3. Inequalities (6.9) and (6.10) produce upon successive iteration

$$(6.11) \quad \Delta_n(t) \leq C \frac{[C_T]^{n+1} t^{n+1}}{(n+1)!}$$

establishing rapid convergence of $\sup_{s \leq t \leq T} \Delta_n(t)$ to 0. In particular

$$\left\| \sup_{0 \leq t \leq T} |y_n(t)| \right\|_{L_2(P)} \leq \left\| \sup_{0 \leq t \leq T} |y_n^{(1)}(t)| \right\|_{L_2(P)} + \left\| \sup_{0 \leq t \leq T} |y_n^{(2)}(t)| \right\|_{L_2(P)}$$

where

$$y_n^{(1)}(t) = \int_s^t [\sigma(u, x_n(u, \omega)) - \sigma(u, x_{n-1}(u, \omega))] d\beta(u)$$

and

$$y_n^{(2)}(t) = \int_s^t [b(u, x_n(u, \omega)) - b(u, x_{n-1}(u, \omega))] du$$

Since $y_n^{(1)}(\cdot)$ is a martingale, by Doob's inequality, we obtain

$$\left\| \sup_{s \leq t \leq T} |y_n^{(1)}(t)| \right\|_{L_2(P)} \leq 2 \left\| |y_n^{(1)}(T)| \right\|_{L_2(P)} \leq A_T \left\| \int_s^T \Delta_{n-1}(u) du \right\|_{L_2(P)}$$

At the same time, by Schwartz's inequality, it is not difficult to see that

$$\left\| \sup_{s \leq t \leq T} |y_n^{(2)}(t)| \right\|_{L_2(P)} \leq A_T \left\{ \int_s^T \Delta_{n-1}(u) du \right\}^{\frac{1}{2}}$$

Step 4. We now conclude by our estimates (6.11) that

$$\sum_n \left\| \sup_{0 \leq t \leq T} |y_n(t)| \right\|_{L_2(P)} < \infty$$

proving the almost sure uniform convergence of $\sum_n y_n(\cdot)$ on every finite interval. This clearly establishes the almost sure uniform convergence in every finite interval $[0, T]$ of $x_n(t, \omega)$ to a limit $x(t, \omega)$, which is clearly a progressively measurable and almost surely continuous solution of (6.2).

Step 5. Uniqueness: The uniqueness is straightforward. If we have two solutions $x(\cdot, \cdot)$ and $x'(\cdot, \cdot)$ the difference

$$y(t, \omega) = x(t, \omega) - x'(t, \omega)$$

satisfies the estimate

$$E^P \{|y(t, \omega)|^2\} \leq C_T \int_s^t E^P \{|y(u, \omega)|^2\} du$$

with an initial estimate

$$E^P \{|y(t, \omega)|^2\} \leq C_T t$$

valid for $s \leq t \leq T$. Iteration produces the following estimate by induction:

$$E^P \{|y(t, \omega)|^2\} \leq \frac{[C_T]^n t^n}{n!}$$

If we let $n \rightarrow \infty$ we get uniqueness. □

6.2. Properties of the Solution

Markov and Strong Markov Property. We can represent the solution $x(t, \omega)$ as a function $x(t, \omega, s, \zeta(\omega))$ of the starting time and initial value. One of the consequences of uniqueness is the identity

$$x(t, \omega, s, \zeta(\omega)) = x(t, \omega, s', x(s', \omega, s, \zeta(\omega)))$$

which says nothing more than the obvious fact that if we solve first up to time $s' > s$ and then start from s' and solve for times $t > s'$, we do not get anything different from the solution obtained by solving in a single step. If we denote by $x' = x(s', \omega, s, \zeta(\omega))$, then

$$x(t, \omega, s, \zeta(\omega)) = x(t, \omega', s', x')$$

where ω' represents the increments of the Brownian motion after time s' and is therefore independent of the field $\mathcal{F}_{s'}$. The independence of x' and ω' implies that the conditional distribution of $x(t)$ given $\mathcal{F}_{s'}$ is given by

$$P[x(t) \in A \mid \mathcal{F}_{s'}] = P[x(t, \omega', s', x') \in A]$$

with $x' = x(s', \omega, s, \zeta(\omega))$ which is the Markov property. Since Brownian motion has the strong Markov property the equation (6.2) can be solved just as well, when we prescribe $\zeta(\omega)$ to be $\mathcal{F}_\tau$ measurable where τ is an arbitrary stopping time. The solution is defined uniquely for $t > \tau(\omega)$ on the set $\{\omega : \tau(\omega) < \infty\}$ and we have the relation

$$x(t, \omega, s, \zeta(\omega)) = x(t, \omega', \tau(\omega), x(\tau(\omega), \omega, s, \zeta(\omega)))$$

where ω' represents the increments $\beta(\tau(\omega) + \cdot) - \beta(\tau(\omega))$ that is independent of $\mathcal{F}_\tau$. This will yield the strong Markov property for the solution just as the earlier representation provided a verification of the Markov property.

Itô's Formula for Solutions of SDE. If $\sigma(\cdot, \cdot)$ and $b(\cdot, \cdot)$ are progressively measurable functions satisfying

$$E^P \left\{ \int_0^t |\sigma(s, \omega)|^2 ds \right\} < \infty$$

and

$$E^P \left\{ \int_0^t |b(s, \omega)| ds \right\} < \infty$$

then

$$\xi(t, \omega) = x_0 + \int_0^t \sigma(s, \omega) d\beta(s) + \int_0^t b(s, \omega) ds$$

is well defined as a stochastic process with continuous paths. Formally we write it as

$$d\xi(t) = \sigma(t) d\beta(t) + b(t) dt$$

Itô's formula allows us to calculate

$$df(t, \xi(t)) = f_t(t, \xi(t)) dt + f_\xi(t, \xi(t)) d\xi(t) + \frac{1}{2} f_{\xi\xi}(t, \xi(t)) \sigma^2(t) dt$$

or

$$\begin{aligned} f(t, \xi(t)) - f(0, \xi(0)) &= \int_0^t f_\xi(t, \xi(t)) \sigma(t, \omega) d\beta(t) \\ &\quad + \int_0^t \left[f_t(t, \xi(t)) + f_\xi(t, \xi(t)) b(t, \omega) \right. \\ &\quad \left. + \frac{1}{2} f_{\xi\xi}(t, \xi(t)) \sigma^2(t, \omega) \right] dt \end{aligned}$$

PROOF: The proof of the above identity can be deduced from Itô's formula for Brownian motion.

Step 1. If we replace Brownian motion by $\xi(t) = \sigma\beta(t) + bt$ where σ and b are constants

$$f(t, \xi(t)) = g(t, \beta(t))$$

with $g(t, \xi) = f(t, \sigma\xi + bt)$ and the previous formula applied to g yields the new formula.

Step 2. If $\sigma(t, \omega)$ and $b(t, \omega)$ are simple functions, we can obtain it by piecing together formulas in intervals where the functions σ and b are constants. That the constants are functions of ω causes no problem.

Step 3. Finally, we approximate arbitrary progressively measurable functions by simple functions and the passage to the limit is routine. In particular, if $x(\cdot)$ is a solution of (6.2) then

$$\begin{aligned} f(t, x(t)) - f(0, x(0)) &= \int_0^t f_x(s, x(s))\sigma(s, x(s))d\beta(s) \\ &\quad + \int_0^t \left[f_s(s, x(s)) + b(s, x(s))f_x(s, x(s)) \right. \\ &\quad \left. + \frac{1}{2}\sigma^2(s, x(s))f_{xx}(s, x(s)) \right] ds \end{aligned}$$

□

Girsanov Formula. If we denote by $P_{s,x}$ and $Q_{s,x}$ the diffusion processes that correspond to the distributions of $x(t)$ and $y(t)$ that are solutions of

$$dx(t) = \sigma(t, x(t))d\beta(t) + b(t, x(t))dt$$

and

$$dy(t) = \sigma(t, y(t))d\beta(t) + c(t, x(t))dt$$

that start at the same place and have the same diffusion term but differ in their drift terms. Let us suppose that σ, b and c satisfy the Lipschitz and growth conditions. In addition let $a(t, x) > 0$ for all t and x , and

$$c(t, x) - b(t, x) = \sigma(t, x)e(t, x)$$

for some $e(t, x)$ satisfying $|e(t, x)| \leq C$. Then

THEOREM 6.2 *For every s, x and $t > s$ the measures $P_{s,x}$ and $Q_{s,x}$ are mutually absolutely continuous on the σ -field $\mathcal{F}_t^s$ and the Radon-Nikodym derivative is given by*

$$\frac{dQ_{s,x}}{dP_{s,x}} \big| \mathcal{F}_t^s = R(t, \omega) = \exp \left[\int_s^t e(s, x(s))d\beta(s) - \frac{1}{2} \int_s^t e^2(s, x(s))ds \right]$$

Here

$$d\beta(t) = \frac{1}{\sigma(t, x(t))} [dx(t) - b(t, x(t))dt]$$

and the Radon-Nikodym derivative is measurable with respect to the σ -field $\mathcal{F}_t^s = \sigma\{x(u) : s \leq u \leq t\}$.

PROOF: The proof is not all that different from the one carried out in Section 5.10. where $a(t, x) \equiv 1$.

Step 1. Let $e(s, \omega) = e(s, x(s, \omega))$. We know from Theorem 5.7 that

$$R(t, \omega) = \exp \left[\int_s^t e(s, x(s))d\beta(s) - \frac{1}{2} \int_s^t e^2(s, x(s))ds \right]$$

is a martingale with respect to $(\Omega, \mathcal{F}_t^s, P_{s,x})$. It defines a measure $Q = Q_{s,x}$ on $(\Omega, \mathcal{F}_\infty^s)$ by the relation $dQ = R(t, \omega) dP_{s,x}$ on $\mathcal{F}_t^s$. The function $\gamma(\cdot)$ defined by

$$\gamma(t) = \beta(t) - \int_0^t e(s, \omega) ds$$

is a Brownian motion relative to $(\Omega, \mathcal{F}_t, Q)$. More precisely under Q , $\gamma(t) - \gamma(s)$ is distributed independently of $\mathcal{F}_s$, and has the same distribution as Brownian motion. The proof is based on verifying that

$$\xi_\lambda(t) = \exp \left[\lambda \gamma(t) - \frac{\lambda^2}{2} t \right]$$

is a martingale relative to $(\Omega, \mathcal{F}_t, Q)$. By the definition of Q , this is the same as checking that $\xi_\lambda(t) R(t, \omega)$ is a martingale with respect to $(\Omega, \mathcal{F}_t, P)$. But

$$\begin{aligned} \xi_\lambda(t) R(t, \omega) &= \exp \left[\lambda \gamma(t) - \frac{\lambda^2}{2} t + \int_0^t e(s, \omega) d\beta(s) - \frac{1}{2} \int_0^t e^2(s, \omega) ds \right] \\ &= \exp \left[\int_0^t [\lambda + e(s, \omega)] d\beta(s) - \frac{1}{2} \int_0^t [\lambda + e(s, \omega)]^2 ds \right] \end{aligned}$$

is clearly a $(\Omega, \mathcal{F}_t, P)$ martingale.

Step 2. Let us rewrite the solution of

$$dx(t) = \sigma(t, x(t)) d\beta(t) + b(t, x(t)) dt$$

as

$$dx(t) = \sigma(t, x(t)) d\gamma(t) + c(t, x(t)) dt$$

with

$$\gamma(t) = \beta(t) - \int_0^t \sigma(t, x(t)) e(t, x(t)) dt$$

and compare it with the solution

$$dy(t) = \sigma(t, y(t)) d\beta(t) + c(t, y(t)) dt$$

Since $\beta(\cdot)$ under P and $\gamma(\cdot)$ under Q are both Brownian motions, the distribution of $y(\cdot)$ under P is the same as the distribution of $x(\cdot)$ under Q . So the Radon-Nikodym derivative of $Q_{0,x}$ with respect to $P_{0,x}$ can be computed by computing the Radon-Nikodym derivative of the distribution of $x(\cdot)$ under Q with respect to the distribution of the same solution $x(\cdot)$ under P . If we can show that $R(t, \omega)$ is measurable with respect to $\widehat{\mathcal{F}}_t = \sigma\{x(s) : 0 \leq s \leq t\}$ we are done. To establish this all we need to note is that because $a(t, x) > 0$, $\sigma(t, x)$ is nonvanishing and we can invert the stochastic integral to represent

$$d\beta(t) = \frac{1}{\sigma(t, x(t))} [dx(t) - b(t, x(t)) dt].$$

Therefore $R(t, \omega)$ can be expressed as a progressively measurable function of $x(\cdot)$.

Some technical work is needed to show that stochastic integrals with respect to $x(\cdot)$ are well defined. But this is routine. $\square$

EXERCISE 6.1. Under what conditions on $a(t, x)$ can one be sure that $\sigma(t, x) = \sqrt{a(t, x)}$ is Lipschitz? Show that if $a(t, x)$ satisfies a Lipschitz condition in x and has a uniform positive lower bound then $\sigma(t, x)$ is also Lipschitz in x . Otherwise, if $a(t, x)$ is only nonnegative, if it has two bounded derivatives then show that $\sigma(t, x)$ is Lipschitz in x . Use calculus to show that if $f(x) \geq 0$ and $|f''(x)| \leq C$, then $|f'(x)| \leq \sqrt{2Cf(x)}$. This guarantees that $|\sigma'(x)| \leq \sqrt{2C}$.

Transition Probabilities. If we are given functions $a(t, x)$ and $b(t, x)$ such that $\sigma = \sqrt{a}$ and b are Lipschitz, then we can construct processes $x(t; s, x)$ defined for $t \geq s$ that satisfy $x(s) = x$. These Markov processes will be inhomogeneous in time and will have transition probabilities

$$p(s, x, t, A) = P[x(t; s, x) \in A]$$

Because we have uniqueness of the solutions and the consequent Markov property $p(s, x, t, A)$ will satisfy the Chapman-Kolmogorov equations.

$$p(s, x, t, dy)p(t, y, u, A) = p(s, x, u, A)$$

for all Borel sets A and $s \leq t \leq u$.

EXERCISE 6.2. Show that

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{1}{h} \int_{|y-x| \leq 1} (y-x) p(s, x, s+h, dy) &= b(s, x) \\ \lim_{h \rightarrow 0} \frac{1}{h} \int_{|y-x| \leq 1} (y-x)^2 p(s, x, s+h, dy) &= a(s, x) \\ \lim_{h \rightarrow 0} \frac{1}{h} \int_{|y-x| \geq \delta} p(s, x, s+h, dy) &= 0 \quad \text{for any } \delta > 0 \end{aligned}$$

Time Homogeneous Case. A special class of coefficients $a(t, x)$, $b(t, x)$ are those that do not depend explicitly on t . In that case one can always take $s = 0$ when solving the stochastic differential equations, and the transition probabilities are given by

$$p(s, x, t, A) = \tilde{p}(t-s, x, A) = P[x(t-s, 0, x) \in A]$$

6.3. Connections with Differential Equations

Let us begin with the time homogeneous case. We have a family of solutions $x(t) = x(t; x)$ that solve

$$x(t) = x + \int_0^t \sigma(x(s)) d\beta(s) + \int_0^t b(x(s)) ds$$

and we have the transition probabilities

$$p(t, x, A) = P[x(t, x) \in A]$$

as well as the family of Markov measures $\{P_x\}$ on $C[0, \infty)$ that satisfy $P_x[x(0) = x] = 1$ and P_x is Markov with transition probability $p(\cdot, \cdot, \cdot)$ given above. The

measures P_x are the distributions of the solutions $x(t, x)$ that map Brownian motion, i.e. $C[0, \infty)$ with Wiener measure into the trajectories of our Markov process.

$$\Phi_x : \beta(\cdot) \rightarrow x(\cdot)$$

P_x is the image of the Wiener measure under Φ_x . There is the semigroup of linear transformations T_t defined by

$$(T_t f)(x) = \int f(y) p(t, x, dy)$$

that maps the space of bounded measurable functions into itself.

EXERCISE 6.3. Obtain an estimate of the form $E[|x(t, x) - x(t, y)|^2] \leq C(t)|x - y|^2$

EXERCISE 6.4. Use Exercise 6.3 to verify that T_t is a semigroup which maps the space $C(\mathbb{R})$ of bounded continuous functions on $\mathbb{R}$ into itself.

We have the infinitesimal generator of the semigroup T_t defined by the relation

$$(L f)(x) = \lim_{h \rightarrow 0} \frac{(T_h f)(x) - f(x)}{h}$$

We can use the results of Exercise 6.2 to calculate

$$(6.12) \quad (L f)(x) = \frac{a(x)}{2} f''(x) + b(x) f'(x)$$

for any f that is bounded and twice continuously differentiable. Just like the heat equation and Brownian motion, the operator L has close links to the process and many differential equations involving L describe quantities associated with the process. Some are established directly and some require methods from the theory of partial differential equations. We will illustrate through some examples. The simplest one is

$$(6.13) \quad \begin{aligned} u(t, x) &= (T_t f)(x) = \int f(y) p(t, x, dy) \\ &= E^{P_x}[f(x(t))] = E[f(x(t)) \mid x(0) = x] \end{aligned}$$

The function $u(t, x)$ satisfies the equation

$$(6.14) \quad \frac{\partial u}{\partial t} = \frac{a(x)}{2} \frac{\partial^2 u}{\partial x^2} + b(x) \frac{\partial u}{\partial x}$$

with the initial condition $u(0, x) = f(x)$. But to show that $u(t, x)$ defined by (6.13) satisfies (6.14) we must first show that $u(t, x)$ is sufficiently smooth, i.e. it has two derivatives in x and one derivative in t . Using stochastic integral equations one can show this provided σ , b and f have two derivatives in x . But we will not pursue this here. We will however show that if u is any bounded solution of (6.14) then u satisfies (6.13).

This is simple application of Itô's formula. Let us fix t and consider $u(t-s, x(s)) = z(s)$ and calculate $dz(s)$ using Itô's formula.

$$\begin{aligned} dZ(s) &= -u_t(t-s, x(s))ds + u_x(t-s, x(s))dx(s) \\ &\quad + \frac{1}{2}a(s, x(s))u_{xx}(t-s, x(s)) \\ &= \left(-u_t + bu_x + \frac{1}{2}au_{xx} \right)(t-s, x(s))ds \\ &\quad + u_x(t-s, x(s))\sigma(x(s))d\beta(s) \\ &= u_x(t-s, x(s))\sigma(x(s))d\beta(s) \end{aligned}$$

Therefore

$$Z(s) = Z(0) + \int_0^s u_x(t-s', x(s'))\sigma(x(s'))d\beta(s')$$

is a martingale. We can equate expectations and conclude that

$$E^{P_x}[f(x(t))] = E[Z(t)] = E[Z(0)] = u(t, x)$$

which proves (6.13). Other relationships which can be proved in a similar way are

(i) If a smooth function $u(x)$ on $[a, b]$ solves the equation

$$(Lu)(x) \equiv 0; \text{ for } x \in (a, b);$$

then

$$u(x) = P_x[x(\tau) = a]u(a) + P_x[x(\tau) = b]u(b)$$

where $\tau = \inf\{t : x(t) \notin (a, b)\}$. $u(x(t))$ is a martingale and stop it at τ .

(ii) For $\lambda > 0$, the quantity

$$E^{P_x}[e^{-\lambda\tau}] = u(x)$$

where again $\tau = \inf\{t : x(t) \notin (a, b)\}$ can be obtained by solving

$$Lu = \lambda u \text{ in } (a, b); u(a) = u(b) = 1$$

The process $u(x(t))e^{-\lambda t}$ is a martingale.

(iii) For $\lambda > 0$ and bounded f , the quantity

$$E^{P_x} \left[\int_0^\infty e^{-\lambda t} f(x(t)) dt \right] = u(x)$$

can be obtained as the unique bounded solution of

$$\lambda u - Lu = f$$

Use the fact that

$$e^{-\lambda t} u(x(t)) + \int_0^t e^{-\lambda s} f(x(s)) ds$$

is a martingale. Equate expectations at $t = 0$ and $t = \infty$.

(iv) The Feynman-Kac formula is valid and one can evaluate

$$u(t, x) = E^{P_x} \left[\exp \left[\int_0^t V(x(s)) ds \right] f(x(t)) \right]$$

as the solution of

$$\frac{\partial u(t, x)}{\partial t} = \frac{1}{2} a(x) \frac{\partial^2 u(t, x)}{\partial x^2} + b(x) \frac{\partial u(t, x)}{\partial x} + V(x) u(t, x)$$

with $u(0, x) = f(x)$.

The time dependent case is not all that different. The function defined as $u(t, x) = \int f(y) p(s, x, t, dy)$ for fixed x is not very natural. x is the starting point at time s and t is the final time. It is more natural to consider the function $u(s, x) = \int f(y) p(s, x, t, dy)$ for fixed t . This will satisfy the equation

$$(6.15) \quad \frac{\partial u}{\partial s} + \frac{1}{2} a(s, x) \frac{\partial^2 u}{\partial x^2} + b(s, x) \frac{\partial u}{\partial x} = 0$$

with the terminal condition $u(t, x) = f(x)$. This is called Kolmogorov's backward equation and it is solved for $s < t$. In the time homogeneous case $u(s, x) = (T_{t-s} f)(x)$. If at time s one has a random initial position specified by a density $\phi(x)$ then at time $t > s$ the random variable $x(t)$ will have a density $\phi(t, x)$ and $\phi(t, x)$ will evolve as a solution of the "forward equation of Kolmogorov"

$$(6.16) \quad \frac{\partial \phi}{\partial t} = \frac{1}{2} \frac{\partial^2 [a(t, x) \phi(t, x)]}{\partial x^2} + \frac{\partial [b(t, x) \phi(t, x)]}{\partial x}$$

with $\phi(s, x) = \phi(x)$. If u is a solution of (6.15) and ϕ a solution of (6.16) then

$$\frac{d}{dt} \int u(t, x) \phi(t, x) dx = 0$$

after performing an integration by parts. This establishes the duality between solutions u and ϕ .

In nice situations, the transition probability $p(s, x, t, dy)$ will have a density $p(s, x, t, y)$ with respect to Lebesgue measure it will have the property that $u(s, x) = \int p(s, x, t, y) f(y) dy$ for fixed t and y will satisfy (6.15) and $\phi(t, x) = \int p(s, x, t, y) \phi(s, y) dy$ for fixed s, x will be a solution of (6.16). The initial and terminal conditions will be δ functions. When such an object exists it is called the fundamental solution. For Brownian motion $\frac{1}{\sqrt{2\pi t}} \exp[-\frac{(y-x)^2}{2t}]$ is it.

EXERCISE 6.5. Verify that for any O-U process with $a(x) = 1$ and $b(x) = -cx$, the fundamental solution given in (5.14) satisfies both the forward and backward equations.

6.4. Martingale Characterization

We have seen that the proofs of many results that relate partial differential equations with Markov processes use martingales to make the connection. This raises the question of whether Markov processes can be characterized in terms of their martingales.

The answer for Brownian motion is clearly in the affirmative. If we have a measure P , on $C[0, \infty)$ with $P[x(0) = 0] = 1$ and

$$(6.17) \quad X_\lambda(t) = \exp \left[i\lambda x(t) + \frac{t\lambda^2}{2} \right]$$

is a martingale with respect to $(\Omega, \mathcal{F}_t, P)$ for all $\lambda \in \mathbb{R}$ then P has to be the Wiener measure. Clearly

$$E^P \left[\exp[i\lambda(x(t) - x(s))] \mid \mathcal{F}_s \right] = e^{-\frac{\lambda^2(t-s)}{2}}$$

and this implies that the increment $x(t) - x(s)$ is distributed as a Gaussian with mean 0 and variance $t - s$ and is independent of $\mathcal{F}_s$. One can equivalently characterize the Wiener measure by the property $P[x(0) = 0] = 1$ and

$$(6.18) \quad f(x(t)) - f(x(0)) - \int_0^t \frac{1}{2} f''(x(s)) ds$$

are martingales with respect to $(\Omega, \mathcal{F}_t, P)$ for all smooth (bounded and with two bounded continuous derivatives) functions.

One can go from one class to another by a familiar route. In (6.18) one can restrict to exponentials, and for them the equivalence between (6.17) and (6.18) is a consequence of Lemma 1.7.

For diffusions processes in one dimension

$$(6.19) \quad f(x(t)) - f(x(0)) - \int_0^t \left[\frac{1}{2} a(x(s)) f''(x(s)) + b(x(s)) f'(x(s)) \right] ds$$

is a martingale for all f . This follows from of Itô's formula

$$df(x(t)) = \left[\frac{1}{2} a(x(t)) f''(x(t)) + b(x(t)) f'(x(t)) \right] dt + f'(x(t)) \sigma(x(t)) d\beta(t)$$

It is possible to characterize processes that are obtained by solving the stochastic differential equation

$$dx(t) = b(x(t))dt + \sigma(x(t))d\beta(t)$$

through (6.19). Let P be such that $P[x(0) = x] = 1$ and let (6.19) hold. One can pass from here to exponential martingales again using Lemma 1.7. Instead of (6.17) we now get

$$(6.20) \quad X_\lambda(t) = \exp \left[i\lambda x(t) - i\lambda \int_0^t b(x(s))ds + \frac{\lambda^2}{2} \int_0^t a(x(s))ds \right]$$

are martingales. One can then define $y(t) = x(t) - \int_0^t b(x(s))ds$ and it will have the property

$$X_\lambda(t) = \exp \left[i\lambda y(t) + \frac{\lambda^2}{2} \int_0^t a(x(s))ds \right]$$

are martingales. One can define stochastic integrals with respect to $y(t)$ and

$$Z_\lambda(t) = \exp \left[i\lambda \int_0^t c(s, \omega) dy(s) + \frac{\lambda^2}{2} \int_0^t a(x(s)) c^2(s, \omega) ds \right]$$

are now martingales. If we make the choice of $c(s, \omega) = \sigma(x(s))^{-1} = a(s, x)^{-\frac{1}{2}}$, and define $z(t) = \int_0^t \sigma^{-1}(s, \omega) dy(s)$, then

$$Z_\lambda(t) = \exp \left[i\lambda z(t) + \frac{\lambda^2 t}{2} \right]$$

are martingales. In other words $z(t)$ is Brownian motion. Retracing the steps we obtain

$$x(t) = x + \int_0^t b(x(s)) ds + \int_0^t \sigma(x(s)) d\beta(s)$$

which identifies P as the solution of the stochastic differential equation.

6.5. Random Time Change

If $x(t)$ is the diffusion process corresponding to the generator

$$L = \frac{1}{2} a(x) \frac{d^2}{dx^2} + b(x) \frac{d}{dx}$$

then for any $k > 0$, $y(t) = x(kt)$ will clearly have as its generator kL . This is seen from the limit

$$\lim_{h \rightarrow 0} \frac{T_{kh} - I}{h} = k \lim_{h \rightarrow 0} \frac{T_{kh} - I}{kh} = kL$$

However what is surprising is that one can do a time change that produces a generator $k(x)L$ where $0 < k_1 \leq k(x) \leq k_2 < \infty$ is a uniformly positive bounded measurable function. This requires a time change with a rate that is random and varies at each location. More precisely the particle carries its own clock with a speed that depends on location.

$$A(t) = \int_0^t \frac{ds}{k(x(s))}$$

We define τ_t by $A(\tau_t) = t$. Then τ_t is the original time when the new time is t . Notice that if $k(x) \equiv k$ then $\tau_t = kt$. $\{\tau_t\}$ is an increasing family of stopping times. We have a map $\Phi_{k(\cdot)} : C[0, \infty) \rightarrow C[0, \infty)$ defined by

$$x(t) \rightarrow y(t) = x(\tau_t)$$

One can check that $\Phi_{k(\cdot)} \circ \Phi_{k'(\cdot)} = \Phi_{k(\cdot)k'(\cdot)}$. This proves that $\Phi_k(\cdot)$ is one to one and invertible. The inverse is $\Phi_{k'(\cdot)}$ with $k'(x) = \frac{1}{k(x)}$.

EXERCISE 6.6. Verify the relation $\Phi_{k(\cdot)} \circ \Phi_{k'(\cdot)} = \Phi_{k(\cdot)k'(\cdot)}$

THEOREM 6.3 *Let $\{P_x\}$ be a Markov processes with generator L . Let $Q_x = P_x \Phi_{k(\cdot)}^{-1}$ be the image of P_x under $\Phi_{k(\cdot)}$. Then $\{Q_x\}$ is a Markov process that corresponds to $\tilde{L} = k(x)L$.*

PROOF: With respect to $(C[0, \infty), \mathcal{F}_t, P_x)$

$$f(x(t)) - f(x) - \int_0^t (Lf)(x(s)) ds$$

is a martingale. With stopping times τ_t defined by

$$t = \int_0^{\tau_t} \frac{ds}{k(x(s))}$$

$$f(x(\tau_t)) - f(x) - \int_0^{\tau_t} (Lf)(x(s))ds$$

are martingales with respect to $\mathcal{F}_{\tau_t}$. If we define $y(t) = x(\tau_t)$ then $y(t)$ is $\mathcal{F}_{\tau_t}$ measurable. We can change variables $t \rightarrow \tau_t$ in the integral we get

$$\int_0^{\tau_t} (Lf)(x(s))ds = \int_0^t k(x(\tau_s))(Lf)(x(\tau_s))ds$$

and

$$f(y(t)) - y(0) - \int_0^t k(y(s))(Lf)(y(s))ds$$

is a martingale with respect to $\mathcal{F}_{\tau_t}$. This determines $Q_x = P_x \Phi_{k(\cdot)}^{-1}$ as the Markov process corresponding to $k(x)L$. $\square$

REMARK 6.1. Actually, in the time homogeneous case we have constructed diffusion processes for much larger class of coefficients than those for which $\sigma(x)$ and $b(x)$ are Lipschitz. Starting from $L = \frac{1}{2} \frac{d^2}{dx^2}$ we can do a random time change and go to $a(x)L = \frac{1}{2} a(x) \frac{d^2}{dx^2}$ with $0 < a_1 \leq a(x) \leq a_2 < \infty$. Then we can do a Girsanov transformation and use the Radon-Nikodym derivative to go to

$$L = \frac{1}{2} a(x) \frac{d^2}{dx^2} + b(x) \frac{d}{dx}$$

with any bounded measurable b .

6.6. Some Examples

In this section we will work out some examples and sketch some additional results concerning one-dimensional diffusions. We will not give all the details of the proofs, but leave some of it to the reader.

Geometric Brownian Motion. It is defined as the solution of

$$dx(t) = \sigma x(t) d\beta(t) + cx(t)dt$$

where $\beta(t)$ is the standard Brownian motion. This can be explicitly solved as

$$x(t) = x(0) \exp \left[\sigma \beta(t) + \left(c - \frac{\sigma^2}{2} \right) t \right]$$

as can be verified by Itô's formula. Note that even if $c > 0$, if $c < \frac{\sigma^2}{2}$ then $x(t)$ decays to 0. The interpretation of the geometric Brownian motion is that it is the value of an investment that fluctuates. Note that

$$E[x(t)] = x(0)e^{ct}$$

so that if $c > 0$ the expected value grows while in reality it will not. The major contribution to the expectation comes from large possible values with small probability.

If one can trade in and out of the position without any fees, then it may be possible to make money even from this investment which is going down in the long run. If the total asset at time t is $A(t)$, a fixed portion p of it can be invested in the fluctuating instrument and the rest can be held risk free. Then the value of the asset at time t satisfies

$$dA(t) = pcA(t)dt + pA(t)\sigma d\beta(t)$$

The solution is

$$A(t) = A(0) \exp \left[p\sigma\beta(t) + \left(pc - \frac{p^2\sigma^2}{2} \right) t \right]$$

So long as $c > 0$ one can choose p so that $pc - \frac{p^2\sigma^2}{2}$ has the largest value possible which is $\frac{c^2}{2\sigma^2}$.

Option Pricing. Let $x(t)$ be a diffusion process in $\mathbb{R}$ corresponding to the generator

$$\frac{a(x)}{2} \frac{d^2}{dx^2} + b(x) \frac{d}{dx}$$

If one makes a contract to provide a payoff of $\phi(x(T))$ based on the value of $x(T)$ at a future time T , what is the value of the contract today at time $t < T$? This is a form of gambling. Let us pretend for a moment that the value should be the expectation. Then the current value of the contract, at time t when the process is at x , should be

$$u(t, x) = E[\phi(x(T)) \mid x(t) = x] = \int \phi(y) p(T - t, x, dy)$$

What if the process is actually a stock that is traded in public and $\phi(x) = x$? If $u(t, x) < x$, we can sell the stock (even if we do not own it), buy the contract now, hold our breath and with the payoff at time T buy back the stock, making an assured profit of $x - u(t, x)$. A similar strategy will work if $x < u(t, x)$. One that sells the contract and buys the stock now and holds it till time T when it is sold to pay off the obligation. This suggests the value has to be x and not $u(t, x)$.

In other words when computing the value we have to pretend that $x(t)$ is a martingale. We change the model to a diffusion with generator

$$\frac{a(x)}{2} \frac{d^2}{dx^2}$$

and take

$$u(t, x) = E[\phi(x(T)) \mid x(t) = x] = \int \phi(y) q(T - t, x, dy)$$

In other word $u(t, x)$ is a solution of

$$\frac{\partial u}{\partial t} + \frac{a(x)}{2} \frac{\partial^2 u}{\partial x^2} = 0; \quad u(T, x) = \phi(x)$$

We know it works for $\phi(x) = x$, because $u(t, x) \equiv x$. Why does it work for any ϕ ? In our position which is the contract we have an asset with fluctuating value. The current value being $u(t, x(t))$. By Itô's formula

$$du(t, x(t)) = \left[u_t + \frac{a(x)}{2} u_{xx} \right] + u_x dx(t) = u_x dx(t)$$

If we hold the position of $-u_x(t, x(t)) = c(t, x(t))$ in the stock trading in and out without cost and maintaining that position, we can hedge our risk completely and our net position will be one without any risk.

Explosion. One of the interesting questions regarding one-dimensional diffusions that can be answered fully is the question of explosion in a finite time. These tests go under the name of Feller's tests. Let

$$L = \frac{1}{2}a(x)\frac{d^2}{dx^2} + b(x)\frac{d}{dx}$$

be the generator of a diffusion on (a, b) where a and or b can be infinite (a can be $-\infty$ while b can be $+\infty$) The coefficients are assumed to be nice, with $a(x) > 0$ inside (a, b) , but can behave badly at the end points. We define stopping times $\tau_c = \{\inf t : x(t) = c\}$, $\tau_a = \lim_{c \downarrow a} \tau_c$ and $\tau_b = \lim_{c \uparrow b} \tau_c$. The question is when $P_x[\tau_a < \infty]$ or $P_x[\tau_b < \infty]$ is positive.

The first step is to solve the equation

$$Lu = 0$$

This has an obvious solution, constants. Since it is a second order linear equation in one variable the set of solutions has dimension two and there is a linearly independent solution. A calculation reveals

$$u(x) = \int_0^x e^{-2 \int_0^y \frac{b(z)}{a(z)} dz} dy$$

is such a choice. The choice of 0 is only for convenience. We assume that $a < 0 < b$.

LEMMA 6.4 *If $\lim_{x \uparrow b} u(x) = \infty$, i.e.*

$$\int_0^b e^{-2 \int_0^y \frac{b(z)}{a(z)} dz} dy = \infty$$

then $P_x[\tau_b < \infty] = 0$. There is a similar result at the end a . If

$$\int_a^0 e^{-2 \int_0^y \frac{b(z)}{a(z)} dz} dy = \infty$$

then $P_x[\tau_a < \infty] = 0$.

PROOF: For $a < a' < x < b' < b$,

$$P_x[\tau_{b'} > \tau_{a'}] = \frac{u(b') - u(x)}{u(b') - u(a')}$$

and it tends to 1 as $b' \rightarrow b$. Therefore $P_x[\tau_b > \tau_{a'}] = 1$. The path from x to b has to go to a' first. But if it is to go to b from a' it has to cross x , and then by strong

Markov property has to return to a' again. Clearly it is getting a run around and is never going to make it to b . $\square$

Let us suppose that $u(b) < \infty$. Then we can change coordinates by defining $y(t) = u(x(t))$. Because $Lu = 0$ the generator for the $y(t)$ process, to be denoted by $\{Q_y\}$, will have the form

$$\frac{A(y)}{2} \frac{d^2}{dy^2}$$

where $A(u(x)) = a(x)u_x^2$ for $y < u(b)$. The question is: does $y(t)$ ever reach $u(b)$? One can construct the $y(t)$ process by a random time change of Brownian motion. We have $y(t) = \beta(\tau_t)$ where

$$t = \int_0^{\tau_t} \frac{ds}{A(\beta(s))}$$

If τ is the hitting time of $u(b)$ then for $Q_y[\tau_{u(b)} < \infty] > 0$ we need

$$\int_0^{\tau_{u(b)}} \frac{ds}{A(\beta(s))} < \infty.$$

It is sufficient that

$$E \left[\int_0^{\tau_{u(b)}} \frac{ds}{A(\beta(s))} \right] < \infty$$

which can be seen to be equivalent to

$$(6.21) \quad \int_{u(0)}^{u(b)} \frac{u(b) - y}{A(y)} dy < \infty$$

We will conclude this section with the following theorem.

THEOREM 6.5 *In order that the process $x(t)$ to exit from b at a finite time with positive probability it is necessary and sufficient that $u(b) < \infty$ and condition (6.21) hold.*

All we need to prove is the necessity of (6.21) which is the following lemma.

LEMMA 6.6 *Let $\beta(s)$ be Brownian motion starting from $x \in (0, \infty)$. Let $\tau_{(a,b)}$ denote the exit time from (a, b) . In order that $\int_0^{\tau_{(a,b)}} V(\beta(s)) ds$ be finite with positive probability it is necessary that*

$$\int_0^1 y V(y) dy < \infty$$

The function $V(y)$ is assumed to be nonnegative, and continuous on $(0, \infty)$ but may be unbounded as $y \rightarrow 0$.

PROOF: Define

$$F_a(x) = \int_{\beta(\tau_{(a,1)})=0} \exp \left[\int_0^{\tau_{(a,1)}} V(\beta(s)) ds \right] dP_x$$

$F_a(x)$ is $\downarrow$ as $x \uparrow$ and solves

$$\frac{1}{2}F_a''(x) = V(x)F_a(x); \quad F_a(a) = 1, \quad F_a(1) = 0$$

and for each fixed x , $F_a(x)$ is $\downarrow$ as $a \downarrow$ to 0. Moreover it is bounded below. Taking the limit as $a \rightarrow 0$ we get a positive solution to

$$\frac{1}{2}F''(x) = V(x)F(x)$$

on $(0, 1)$ which is not possible unless $\int_0^\delta x V(x) dx < \infty$ for $\delta > 0$. $\square$

Bessel Processes. If $x_1(t), x_2(t), \dots, x_n(t)$ are independent Brownian motions the radial part $|x(t)| = \sqrt{\sum_i x_i(t)^2}$ is again a Markov process. If $n = 1$ it is the reflected Brownian motion. To see that $|x(t)|$ is a Markov process we need to verify that

$$\int \frac{1}{(2\pi t)^{\frac{n}{2}}} e^{-\frac{1}{2t} \sum_{j=1}^n (x_i - y_i)^2} f(|y|) dy_1 \cdots dy_n = u(t, x) = \tilde{u}(t, |x|)$$

is a function depending only on $|x|$. If S is any rotation on $\mathbb{R}^n$, then $\Pi p(t, x_i, y_i) = q(t, x, y)$ satisfies $q(t, Sx, Sy) = q(t, x, y)$ and $f(|y|)$ is clearly invariant under S . Therefore

$$u(t, x) = \int q(t, x, y) f(|y|) dy$$

is invariant under S and is a function $\tilde{u}(t, |x|)$ of $|x|$. It is easier to do the calculation with $|x|^2 = \sum_i x_i^2$. Note that $x_i(t) = x_i + \beta_i(t)$.

$$E[|x(t)|^2 - |x|^2] = nt$$

and

$$E[(|x(t)|^2 - |x|^2)^2] = 4|x|^2 t + o(t)$$

The generator for the Markov process $z(t) = |x(t)|^2$ is therefore

$$2z \frac{d^2}{dz^2} + n \frac{d}{dz}$$

Now if we change variables to $r = \sqrt{z}$ we get the Bessel operator

$$\frac{1}{2} \frac{d^2}{dr^2} + \frac{n-1}{2r} \frac{d}{dr}$$

More generally we can consider the class of generators

$$\frac{1}{2} \frac{d^2}{dx^2} + \frac{a}{x} \frac{d}{dx}$$

We can ask the question of when the process will reach 0 in a finite time with positive probability. The first step is to solve for

$$\frac{1}{2} \frac{d^2 u}{dx^2} + \frac{a}{x} \frac{du}{dx} = 0$$

We try $u(x) = x^p$ that works if $\frac{p(p-1)}{2} + ap = 0$ or $p = 1 - 2a$. If $a = \frac{1}{2}$, $\log x$ works. It is clear that $u(0) > -\infty$ only if $p > 0$. With $u(x) = x^p$, $A(x^p) = cx^{2p-2}$ or $A(y) = y^{2-\frac{2}{p}}$.

$$\frac{y}{A(y)} = y^{\frac{2}{p}-1}$$

which is integrable if $p > 0$. Therefore $a < \frac{1}{2}$ is necessary and sufficient for the process to be able to reach 0 in finite time.

Doob's h-Transform. Let us start a Brownian motion at $x = 1$ and insist that it stays positive for ever. We cannot do it because it has probability 0. But we can condition it to stay positive up to time T and then let $T \rightarrow \infty$. Does the limit of the conditioned processes exist and if it does what is it?

It is better to consider the family $\{P_{s,x}^T\}$ of processes starting from $x > 0$ at time s conditioned to stay positive up to time T . They will be Markov and we will try to describe them. From the reflection principle

$$u_T(s, x) = P_{s,x}[x(t) > 0; \text{ for all } s \leq t \leq T] = \sqrt{\frac{2}{\pi(T-s)}} \int_0^x e^{-\frac{y^2}{2(T-s)}} dy$$

The transition probability for $\{P_{s,x}^T\}$ can be calculated by Bayes' rule and

$$p^T(\sigma, y, t, z) = q(t - \sigma, y, z) \frac{u_T(t, z)}{u_T(\sigma, y)}$$

where

$$q(t, y, z) = p(t, y - z) - p(t, y + z)$$

with

$$p(t, y) = \frac{1}{\sqrt{2\pi t}} e^{-\frac{y^2}{2t}}$$

We can now take the limit as $T \rightarrow \infty$ and obtain

$$p^\infty(\sigma, y, t, z) = q(t - \sigma, y, z) \frac{z}{y}$$

which is homogeneous in time and has generator

$$(Lf) = \frac{1}{2} \frac{(xf)''}{x} = \frac{1}{2} f'' + \frac{1}{x} f'$$

the Bessel process with $a = 1$.

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CHAPTER 7

General Theory of Markov Processes

7.1. Introduction

We have limited our discussion up to this point mostly to two classes of Markov processes. One class of processes, modeled after the Poisson process, have only a finite number of jumps in a finite time interval. These processes stay at a site for a positive time and then jump. The exceptions have been processes with independent increments with infinite Lévy measure. The generator L of these processes take the form

$$(Lf)(x) = \int_{\mathbb{R}} \left[f(x+y) - f(x) - \frac{y}{1+y^2} f'(x) \right] M(dy)$$

The transition probability $p(t, x, A)$ is translation invariant and is of the form

$$p(t, x, A) = p(t, 0, A - x) = p_0(t, A - x)$$

with the corresponding semigroup

$$(T_t f)(x) = \int_{\mathbb{R}} f(y) p(t, x, dy) = \int_{\mathbb{R}} f(x+y) p_0(t, dy)$$

The independent increments property is reflected in the invariance of the transition probability $p(t, x, A) = p_0(t, A - x)$ under translations. This is equivalent to the commutativity of T_t and the translations S_a acting on functions as $(S_a f)(x) = f(x+a)$.

The second class of processes modeled after Brownian motion, have continuous paths and are described either through a stochastic differential equation

$$dx(t) = b(x(t))dt + \sigma(x(t))d\beta(t)$$

or the associated differential operator

$$\frac{a(x)}{2} \frac{d^2}{dx^2} + b(x) \frac{d}{dx}$$

which is the infinitesimal generator. The general Markov processes on $\mathbb{R}$ will have as its generator a combination of the form

$$\begin{aligned} (Lf)(x) &= \frac{a(x)}{2} \frac{d^2}{dx^2} + b(x) \frac{d}{dx} \\ &+ \int_{\mathbb{R}} \left[f(x+y) - f(x) - \frac{y}{1+y^2} f'(x) \right] M(x, dy) \end{aligned}$$

But the construction of such processes is by no means easy and depends very much on the nature of the dependence of $a(x)$, $b(x)$ and $M(x, dy)$ on x .

On the other hand given the transition probabilities $p(t, x, A)$ on $(\mathcal{X}, \mathcal{B})$ that satisfy the Chapman-Kolmogorov equations (4.2), one can define the semigroup $\{T_t\}$ on $\mathbf{B}$ the space of bounded measurable functions on $(\mathcal{X}, \mathcal{B})$ as in (4.5). One can construct a family $\{P_x\}$ of probability measures on $(\Omega, \mathcal{K})$ where $\Omega = \prod_{t \in [0, \infty)} \mathcal{X}$ and $\mathcal{K}$ is the Kolmogorov σ -field. But to be able to proceed further it is necessary to have a topology on $\mathcal{X}$ and some regularity on $\{p(t, x, dy)\}$ to ensure that the measure $\{P_x\}$ can live on decent subsets $\Omega_0 \subset \Omega$. The appendix contains some results that provides conditions under which the processes can be constructed on the space of continuous paths or paths with only jump discontinuities.

We shall limit our discussion to the time homogeneous case and assume for simplicity that the state space $\mathcal{X}$ is the real line $\mathbb{R}$.

7.2. Semigroups, Generators and Resolvents

Given transition probabilities $p(t, x, dy)$ for a Markov process on $\mathbb{R}$, we can define a semigroup of linear transformations on $\mathbf{B}(\mathbb{R})$ the space of bounded measurable functions by

$$(T_t f)(x) = \int f(y) p(t, x, dy)$$

DEFINITION 7.1 The Markov process with transition probability $p(t, x, dy)$ is called a *Feller process* if the corresponding semigroup leaves the space $C(\mathbb{R})$ of bounded continuous functions on $\mathbb{R}$ invariant.

REMARK 7.1. All processes with independent increments are Feller processes because the semigroup consists of convolution operators. Any operator of the form

$$(Tf)(x) = \int f(x + y) \mu(dy)$$

maps the space of bounded continuous functions into itself.

THEOREM 7.1 Any Markov process that has almost surely right continuous paths and is a Feller process has the strong Markov property.

PROOF: For a stopping time with a finite or countable set of values the strong Markov property is no different from the Markov property. For a general stopping time we approximate from above by stopping times that take at most a countable set of values. We have

$$P_x[x(\tau_n + t) \in A \mid \mathcal{F}_{\tau_n}] = P_{x(\tau_n)}[x(t) \in A] = p(t, x(\tau_n), A)$$

or equivalently for all continuous functions

$$E^{P_x}[f(x(\tau_n + t)) \mid \mathcal{F}_{\tau_n}] = E[f(x(t))] = \int f(y) p(t, x(\tau_n), A)$$

We take a set $E \in \mathcal{F}_\tau \subset \mathcal{F}_{\tau_n}$. Then

$$\int_E [E^{P_x}[f(x(\tau_n + t))]] dP = \int_E (T_t f)(x(\tau_n)) dP$$

We can let $n \rightarrow \infty$. Use the right continuity of $x(\cdot)$ and the continuity of f to conclude

$$\int_E [E^{P_x}[f(x(\tau + t))]] dP = \int_E (T_t f)(x(\tau)) dP$$

This is true for all bounded continuous functions. If two measures give the same integral for all continuous functions they agree on all Borel sets. We can now replace f by the indicator of a set A and we have the strong Markov property. $\square$

The strongly continuous center of a Feller semigroup T_t is defined as the subspace of C_0 of $C(\mathbb{R})$ consisting of

$$C_0 = \{f : \lim_{t \rightarrow 0} \|T_t f - f\| = 0\}$$

where $\|f\| = \sup_x |f(x)|$ is the supremum norm. Since $\|T_t f\| \leq \|f\|$

$$\|T_t(T_s f) - (T_s f)\| = \|T_s(T_t f - f)\| \leq \|T_t f - f\|$$

and it follows that $T_s C_0 \subset C_0$. The infinitesimal generator L of the semigroup is defined as

$$Lf = \lim_{h \rightarrow 0} \frac{T_h f - f}{h}$$

if it exists in the supremum norm i.e. $\|Lf - \frac{T_h f - f}{h}\| \rightarrow 0$ as $h \rightarrow 0$. It may not exist for all f . We define the domain $\mathcal{D}(L)$ as $\{f : \lim_{h \rightarrow 0} \frac{T_h f - f}{h} = Lf \text{ exists}\}$. The following facts are available from the theory of semigroups of operators and can be found in [3].

- (i) $T_t \mathcal{D}(L) \subset \mathcal{D}(L)$ and for all $f \in \mathcal{D}(L)$, $LT_t f = T_t Lf$.
- (ii) For $f \in \mathcal{D}(L)$, $T_t f$ is differentiable in t and

$$\frac{d}{dt} T_t f = LT_t f = T_t Lf$$

In particular

$$T_t f - f = \int_0^t T_s Lf ds = \int_0^t LT_s f ds$$

- (iii) $\mathcal{D}(L)$ is dense in C_0 .
- (iv) The resolvent R_λ defined by

$$R_\lambda f = \int_0^\infty e^{-\lambda t} T_t f dt$$

exists for $\lambda > 0$, $\|R_\lambda\| \leq \lambda^{-1}$, R_λ maps C_0 into $\mathcal{D}(L)$ and

$$(\lambda I - L)R_\lambda = R_\lambda(\lambda I - L) = I$$

- (v) From L we can recover R_λ as $(\lambda I - L)^{-1}$ and T_t can be recovered from R_λ as

$$T_t = \lim_{\lambda \rightarrow \infty} e^{t\lambda(R_\lambda - I)}$$

This prescription for recovering T_t from L although satisfactory from a theoretical perspective is not very useful in practice. Often the domain is not fully known. From the known part of the domain one has to show that the range of $\lambda I - L$ is dense. Then invert it. Equations of the form

$$\lambda I - L = f$$

need to be solved. If L is a differential operator this solving differential equations. Still the semigroup theory is useful because it provides a framework and suggests things we need to do, to construct analytically the semigroup and the process, starting from the generator.

7.3. Generators and Martingales

We start with the transition probabilities $p(t, x, dy)$ having the Feller property and construct a Markov family $\{P_x\}$ of measures. We assume that the paths are almost surely right continuous and hence the strong Markov property is valid.

THEOREM 7.2 *Let f be a bounded continuous function in $\mathcal{D}(L)$. Then for every x ,*

$$M(t) = f(x(t)) - f(x(0)) - \int_0^t (Lf)(x(s))ds$$

is a martingale with respect to $(\Omega, \mathcal{F}_t, P_x)$.

PROOF: Let us calculate

$$\begin{aligned} E^{P_x}[M(t) - M(s) \mid \mathcal{F}_s] &= E^{P_x}\left[f(x(t)) - f(x(s)) - \int_s^t (Lf)(x(s'))ds' \mid \mathcal{F}_s\right] \\ &= (T_{t-s}f)(x(s)) - f(x(s)) - \int_s^t (T_{s'-s}Lf)(x(s))ds' \\ &= \left[T_{t-s}f - f - \int_0^{t-s} T_{s'}Lf ds'\right](x(s)) \\ &= 0 \end{aligned}$$

□

REMARK 7.2. This allows us to use the stopping theorem and conclude that for reasonable stopping times

$$E^{P_x}\left[f(x(\tau)) - f(x) - \int_0^\tau (Lf)(x(s))ds\right] = 0$$

For small open sets U around x , if τ_U is the exit time from U , using the continuity of Lf at x ,

$$(Lf)(x) = \lim_{U \downarrow \{x\}} \frac{E^{P_x}[f(x(\tau_U)) - f(x)]}{E^{P_x}[\tau_U]}$$

This is referred to as Dynkin's formula for the generator.

EXERCISE 7.1. What does it say when the process is Brownian motion and $U = (x - h, x + h)$ with $h \downarrow 0$?

There are other martingales as well. For instance

$$e^{-\lambda t} f(x(t)) - \int_0^t e^{-\lambda s} (Lf - \lambda f)(x(s)) ds$$

are martingales. There are exponential martingales. If $f \in \mathcal{D}(L)$ is uniformly positive then

$$f(x(t)) \exp \left[- \int_0^t \frac{Lf}{f}(x(s)) ds \right]$$

are martingales.

7.4. Invariant Measures and Ergodic Theory

An invariant probability measure for the process is a measure μ that satisfies

$$\mu(A) = \int p(t, x, A) d\mu(x)$$

or

$$\langle T_t f, \mu \rangle = \langle f, \mu \rangle$$

for all continuous functions f and $t > 0$. One can state this as $T_t^* \mu = \mu$ for all $t > 0$ or $L^* \mu = 0$. One can also state this in an equivalent form

$$\langle Lf, \mu \rangle = 0$$

for all $f \in \mathcal{D}(L)$. For one-dimensional diffusions the adjoint equation

$$L^* \phi = 0$$

is easy to solve. It reduces to

$$\frac{1}{2} [a(x)\phi(x)]'' = [b(x)\phi(x)]'$$

or

$$\frac{1}{2} [a(x)\phi(x)]' = [b(x)\phi(x)]$$

which is easy to handle. Sometimes the semigroup T_t is self adjoint with respect to a measure μ .

$$\int (T_t f)(x) g(x) d\mu(x) = \int (T_t g)(x) f(x) d\mu(x)$$

for all f and g . If μ is a finite measure (of total mass 1) then it is an invariant probability measure. The generator L is self adjoint in $L_2(\mu)$ and there is spectral theory which can be useful. The contraction property implies that the spectrum of L is contained in the negative real axis $\lambda \leq 0$. One-dimensional diffusions are always self adjoint with respect to the invariant probability measure if it exists.

In general the set of invariant probability measures forms a convex set and the extreme points are the ergodic ones. They can be characterized in two equivalent ways. An almost invariant set A for an invariant measure μ is a set $A \in \mathbb{R}$ such that

$$p(t, x, A) = \mathbf{1}_A(x) \text{ a.e. } \mu \text{ for all } t > 0.$$

An invariant measure is ergodic if any almost invariant set has measure 0 or 1 under μ . Another characterization is to use the invariant measure to construct a stationary process on $(-\infty, \infty)$ which can be done consistently if the marginals are taken as μ . One wants the stationary process to be ergodic. If the invariant measure is unique it is ergodic. This is always the case for diffusions if $a(x) > 0$.

APPENDIX A

Measures on Polish Spaces

Once we decide to view stochastic processes as probability measures on function spaces, it becomes useful to have a framework for their study. Most of the function spaces that we come across will be complete separable metric spaces, i.e. Polish spaces.

Let X be a complete separable metric space. There is a natural σ -field, the Borel σ -field which is the smallest σ -field containing all open sets. It is also the smallest σ -field with respect to which all continuous functions are measurable. Probability measures will be defined on this σ -field.

The basic theorem about measures on Polish spaces is that probability measures are essentially supported on compact sets. More precisely we have the following theorem.

THEOREM A.1 *Given a probability measure P on X and $\epsilon > 0$, there exists a compact set K_ϵ such that $P(K_\epsilon) \geq 1 - \epsilon$.*

PROOF: The first step is to note that if X is separable metric space for any ϵ it is a countable union of sphere of radius ϵ . There are a countable set of points $\{x_i\}$ such that $\cup B(x_i, \epsilon) = X$. Let us pick $\epsilon = \frac{1}{k}$ and will have points $\{x_{i,k}\}$ such that $X = \cup B(x_{i,k}, \frac{1}{k})$ for every k . Because P is countably additive given $\epsilon > 0$ for each k there is an integer n_k such that

$$(A.1) \quad P\left[\bigcup_{i=1}^{n_k} B\left(x_{i,k}, \frac{1}{k}\right)\right] \geq 1 - \frac{\epsilon}{2^k}$$

Then

$$P\left[\bigcap_{k=1}^{\infty} \bigcup_{i=1}^{n_k} B\left(x_{i,k}, \frac{1}{k}\right)\right] \geq 1 - \epsilon$$

and the set

$$B_\epsilon = \bigcup_{i=1}^{n_k} B(x_{i,k}, \frac{1}{k})$$

is totally bounded in a complete metric space and therefore has a compact closure K_ϵ

$$P[K_\epsilon] \geq P[B_\epsilon] \geq 1 - \epsilon$$

□

REMARK A.1. In a space like $\mathbb{R}^d$ which is a countable union of compact sets, the theorem is obvious. But most function spaces do not have the property.

REMARK A.2. Completeness is important. There are probability measures on separable metric spaces such every compact set has zero measure. The restriction of Lebesgue outer measure on $[0, 1]$ to a set with inner measure 0 and outer measure 1 will do it.

It is important for proving existence of stochastic processes and to construct approximations, to have a notion of convergence of probability measures on our Polish space X . The relevant notion is one of weak convergence.

DEFINITION A.1 A sequence $\{P_n\}$ of probability measures on X converges weakly to a probability measure P if

$$\lim_{n \rightarrow \infty} \int f(x) dP_n = \int f(x) dP$$

for every bounded continuous function on X .

The following theorem gives several equivalent formulations of weak convergence, usually denoted by $P_n \Rightarrow P$.

THEOREM A.2 *The following are equivalent,*

- (i) $P_n \Rightarrow P$.
- (ii) *For every bounded uniformly continuous function $f(x)$*

$$\lim_{n \rightarrow \infty} \int f(x) dP_n = \int f(x) dP$$

- (iii) *For every closed set $C \subset X$*

$$P(C) \geq \limsup_{n \rightarrow \infty} P_n(C)$$

- (iv) *For every open set $G \subset X$*

$$P(G) \leq \liminf_{n \rightarrow \infty} P_n(G)$$

- (v) *For every set A such that $P(A^\circ) = P(\bar{A})$*

$$\lim_{n \rightarrow \infty} P_n(A) = P(A)$$

PROOF:

(i) $\Rightarrow$ (ii). Since a uniformly continuous function is continuous this is obvious.

(ii) $\Rightarrow$ (iii). Let C be a closed. Let $f_k(x) = \frac{1}{[1+d(x,C)]^k}$. Then $f_k(x)$ are uniformly continuous and $f_k(x) \downarrow \mathbf{1}_C(x)$. Therefore

$$\int f_k(x) dP = \lim_{n \rightarrow \infty} \int f_k(x) dP_n \geq \limsup_{n \rightarrow \infty} P_n(C)$$

If we let $k \rightarrow \infty$ we get (iii).

(iii) $\Leftrightarrow$ (iv). Just take complements.

(iii) and (iv) $\Rightarrow$ (v). We have

$$\begin{aligned} P(A^\circ) &\leq \liminf_{n \rightarrow \infty} P_n(A^\circ) \leq \liminf_{n \rightarrow \infty} P_n(A) \\ &\leq \limsup_{n \rightarrow \infty} P_n(A) \leq \limsup_{n \rightarrow \infty} P_n(\bar{A}) \leq P(\bar{A}) \end{aligned}$$

If $P(A^\circ) = P(\bar{A})$ all the terms are equal.

(v) $\Rightarrow$ (i). Given a continuous function f bounded by M , and $\epsilon > 0$, we split the interval $[-M, M]$ into a finite number of intervals $[a_i, a_{i+1}]$ such that $|a_i - a_{i+1}| \leq \epsilon$ and $P[x : f(x) = a_i] = 0$ for every i . Then $A_i = [x : a_i \leq f(x) < a_{i+1}]$ are continuity sets for P (i.e. $P(A^\circ) = P(\bar{A})$). Hence

$$\begin{aligned} \limsup_{n \rightarrow \infty} \left| \int f(x) dP_n - \int f(x) dP \right| &\leq 2\epsilon + \limsup_{n \rightarrow \infty} \sum_j a_j |P_n(A_j) - P(A_j)| \\ &\leq 2\epsilon \end{aligned}$$

We are done. $\square$

REMARK A.3. The measures need not be probability measures. They can be finite measures with arbitrary total mass. In (iii) and (iv) we need to add that $P_n(X) \rightarrow P(X)$. We can then normalize to get probability measures if we so wish.

The notion of uniform continuity depends on the choice of the metric where as the notion of continuity depends only on the topology. It is a fact from the theory of metric spaces that every separable metric space admits a metric in which it is totally bounded and therefore the space of uniformly continuous functions is a separable Banach space. This means there is a sequence of uniformly continuous functions $\{f_j\}$ bounded by 1 such that their linear combinations are dense in the space of all uniformly continuous functions. This allows weak convergence to be metrized.

$$(A.2) \quad D(P, Q) = \sum_{j=1}^{\infty} \left| \int f_j(x) dP - \int f_j(x) dQ \right|$$

will do it. This makes the space $\mathcal{M}$ of probability measures on X into a metric space. It is easy to see that it is separable. Probability distributions with rational masses on a finite subset of a fixed countable dense subset of X is countable and dense. Although the metric D of (A.2) is not in general complete there is an equivalent metric which is complete, when ever X is complete.

We now turn our attention to a discussion of compact subsets of $\mathcal{M}$.

THEOREM A.3 *If X is compact then $\mathcal{M}$ is compact.*

PROOF: If X is compact then the space of uniformly continuous functions is the same as the space of $C(X)$ of continuous functions and this space is separable. Given $P_n \in \mathcal{M}$ we can choose a subsequence P_{n_j} (diagonalization procedure) such that the linear functionals

$$\Lambda_{n_j}(f) = \int f(x) dP_{n_j}$$

have a limit for a countable dense subset of $C(X)$. Then it has a limit for every $f \in C(X)$ and the limit $\Lambda(f) = \lim_{j \rightarrow \infty} \Lambda_{n_j}(f)$ will have by Riesz representation theorem will be

$$\Lambda(f) = \int f(x) dP$$

for some P . Since one can verify that $\Lambda(f) \geq 0$ if $f \geq 0$ and $\Lambda(1) = 1$, P will be a probability distribution. $\square$

For noncompact spaces the following theorem of Prohorov works very well.

THEOREM A.4 *Let $\mathcal{E} \subset \mathcal{M}$ be such that for any $\epsilon > 0$ there exists a compact set $K_\epsilon \subset X$ such that*

$$P(K_\epsilon) \geq 1 - \epsilon \quad \text{for all } P \in \mathcal{E}$$

Then every sequence $\{P_n\}$ from $\mathcal{E}$ has a subsequence that converges weakly to a limit in $\mathcal{M}$.

PROOF: Let $\epsilon = \frac{1}{k}$. Assume by choosing subsequences as we need that the restriction of P_n to $K_{\frac{1}{k}}$ has a weak limit Q_k . $Q_k(X) \leq 1$ and it is easy to check that Q_k is $\uparrow$ in k . Then the limit $Q = \lim_k Q_k$ exists and it is easy to check that $P_n \rightarrow Q$. As for as integration of a bounded function is concerned there is not much action outside K_ϵ . $\square$

REMARK A.4. The above condition is also necessary if X is complete. The proof is a modification of the proof of Theorem A.1. We need to show that if $\mathcal{E} \subset \mathcal{M}$ is compact then (A.1) holds uniformly over $P \in \mathcal{E}$. Since the convergence is monotone this is just Dini's theorem.

A.1. The Space $C[0, 1]$

In the theory of stochastic processes it is important to find reasonable conditions on the finite-dimensional distributions of a stochastic process that ensures that a probability measure P can be defined on the space $C[0, 1]$ of continuous functions on the interval $[0, 1]$ such that the finite-dimensional distribution matches those of the given process. The main theorem is due to Kolmogorov.

THEOREM A.5 *If a Stochastic process $x(t)$ satisfies for some $\alpha, \beta > 0$, and $C < \infty$,*

$$(A.3) \quad E[|x(t) - x(s)|^\beta] \leq C|t - s|^{1+\alpha}$$

for all $0 \leq s \leq t \leq 1$, then there exists a measure P on $C[0, 1]$ with the same finite-dimensional distributions.

PROOF: The first proof consists of constructing successive approximations that are continuous and prove almost sure uniform convergence. At the n -th stage we define $x_n(t)$ to be equal to $x(t)$ when t is a dyadic rational of the form $\frac{j}{2^n}$, $j = 0, 1, \dots, 2^n$. We define $x_n(t)$ by interpolation to be linear

$$x_n(t) = 2^n \left(t - \frac{j}{2^n} \right) x \left(\frac{j+1}{2^n} \right) + 2^n \left(\frac{j+1}{2^n} - t \right) x \left(\frac{j}{2^n} \right)$$

in each interval $[\frac{j}{2^n}, \frac{j+1}{2^n}]$.

We can estimate the difference

$$\begin{aligned}
 & \sup_{0 \leq t \leq 1} |x_{n+1}(t) - x_n(t)| \\
 &= \sup_{1 \leq j \leq 2^n} \sup_{\frac{j-1}{2^n} \leq t \leq \frac{j}{2^n}} |x_{n+1}(t) - x_n(t)| \\
 &= \sup_{1 \leq j \leq 2^n} \left| x_{n+1}\left(\frac{2j-1}{2^{n+1}}\right) - x_n\left(\frac{2j-1}{2^{n+1}}\right) \right| \\
 &= \sup_{1 \leq j \leq 2^n} \left| x_{n+1}\left(\frac{2j-1}{2^{n+1}}\right) - \frac{1}{2} \left[x_n\left(\frac{j-1}{2^n}\right) + x_n\left(\frac{j}{2^n}\right) \right] \right| \\
 &\leq \sup_{1 \leq j \leq 2^n} \max \left[\left| x\left(\frac{j-1}{2^n}\right) - x\left(\frac{2j-1}{2^{n+1}}\right) \right|, \left| x\left(\frac{2j-1}{2^{n+1}}\right) - x\left(\frac{j}{2^n}\right) \right| \right]
 \end{aligned}$$

Therefore

$$\begin{aligned}
 & P \left[\sup_{0 \leq t \leq 1} |x_{n+1}(t) - x_n(t)| \geq 2^{-n\gamma} \right] \\
 & \leq 2^{n+1} \sup_j P \left[\left| x\left(\frac{j}{2^{n+1}}\right) - x\left(\frac{j+1}{2^{n+1}}\right) \right| \geq 2^{-n\gamma} \right] \\
 & \leq C 2^{n+1} 2^{-(n+1)(1+\alpha)} 2^{n(1+\beta)\gamma}
 \end{aligned}$$

If we choose γ such that $1 + (1 + \beta)\gamma < 1 + \alpha$, then

$$\sum_n P \left[\sup_{0 \leq t \leq 1} |x_{n+1}(t) - x_n(t)| \geq 2^{-n\gamma} \right] < \infty$$

We invoke the Borel-Cantelli lemma and conclude that with probability 1,

$$\lim_{n \rightarrow \infty} x_n(t) = x^*(t)$$

exists uniformly and $x^*(t)$ is almost surely a continuous function of t . By construction $x(t) = x^*(t)$ with probability 1 at dyadic points. Since both $x(t)$ and $x^*(t)$ are continuous in probability it is clear that they have the same finite-dimensional distributions, in fact, $P[x(t) = x^*(t)] = 1$ for all $0 \leq t \leq 1$. $\square$

REMARK A.5. With a little extra work, one can use the Hölder norm instead of the supremum norm and if $\delta < \frac{\alpha}{\beta}$ one can prove that the process $x^*(t)$ is actually supported on functions that satisfy a Hölder condition with exponent δ .

In fact, we can obtain an estimate of the form

$$P \left[\sup_{0 \leq s < t \leq 1} \frac{|x(t) - x(s)|}{|t - s|^\delta} \geq A \right] \leq C(\alpha, \beta, \delta) \frac{C}{A^\beta}$$

provided $\delta < \frac{\alpha}{\beta}$. $C(\alpha, \beta, \delta)$ depends only on α, β and δ and C is the same constant as in (A.5). In particular, a family $\mathcal{E}$ satisfying (A.5) uniformly with the same α, β and C , will be compact in $\mathcal{M}$ provided the marginal distributions at $t = 0$ form a compact family in $\mathcal{M}(\mathbb{R})$. The Ascoli-Arzelà theorem will produce the necessary compact set in $C[0, 1]$.

A.2. The Space $D[0, 1]$

The space $D[0, 1]$ consists of all functions $f(\cdot)$ on $[0, 1]$ that have left limits $f(t-0)$ for all $t > 0$ and the right limits $f(t+0)$ for all $t < 1$. It is normalized so that $f(t) = f(t+0)$ for all $t < 1$ and $f(1) = f(1-0)$. It is easy to see that such a function f is necessarily bounded. For each $\delta > 0$, there can only be a finite set of points where $|f(t+0) - f(t-0)| \geq \delta$. Consequently there can be at most a countable set where f can be discontinuous. The notion of convergence in $D[0, 1]$ that we will use is called *Skorohod topology*. He introduced several topologies and the one we will use is called the J_1 topology, But that is the only one that we will use and we will call it just Skorohod topology.

DEFINITION A.2 A sequence $f_n(\cdot)$ will converge to $f(\cdot)$ in $D[0, 1]$ in Skorohod topology if there exists a sequence of homeomorphisms, i.e. continuous one to one invertible maps $\lambda_n(\cdot)$ of $[0, 1]$ with inverses $\lambda_n^{-1}(\cdot)$ such that

$$(A.4) \quad \lim_{n \rightarrow \infty} \sup_{0 \leq t \leq 1} [|\lambda_n(t) - t| + |\lambda_n^{-1}(t) - t|] = 0$$

$$(A.5) \quad \lim_{n \rightarrow \infty} \sup_{0 \leq t \leq 1} |f_n(\lambda_n(t)) - f(t)| \rightarrow 0$$

It is clear that it is weaker than uniform convergence. Uniform convergence requires the jumps of f_n to align exactly with the jumps of f . Skorohod convergence allows the jumps to be located at points nearby that converge to the location of the jumps of f . If f is continuous then convergence to f has to be uniform. Skorohod convergence does not allow two jumps to come together in the limit and either augment or cancel each other. The location and sizes of individual jumps converge to the corresponding ones of the limit. Jumps can become small and disappear, but they can not suddenly appear in the limit. The topology is metrizable as a complete separable metric space. Just as there is a modulus continuity for $C[0, 1]$ there is a modulus continuity for $D[0, 1]$. The $C[0, 1]$ modulus continuity is based on the oscillation $\omega_f[I]$ over an open interval defined as

$$\omega_f(I) = \sup_{s, t \in I} |f(s) - f(t)| \quad \text{and} \quad \omega_f(\delta) = \sup_{\substack{I \subset [0, 1] \\ |I| < \delta}} \omega_f(I)$$

The $D[0, 1]$ modulus continuity makes allowance for one jump. If $I = (a, b)$, allowing for a jump at arbitrary c , $a < c < b$,

$$\bar{\omega}_f(I) = \inf_{c \in (a, b)} [\max\{\omega_f(a, c), \omega_f(c, b)\}]$$

and

$$\bar{\omega}_f(\delta) = \max\left\{ \sup_{\substack{I \subset [0, 1] \\ |I| < \delta}} \bar{\omega}_f(I), \omega_f([0, \delta)), \omega_f((1 - \delta, 1]) \right\}$$

One can show that for any $f \in D[0, 1]$, $\bar{\omega}_f(\delta) \rightarrow 0$ as $\delta \rightarrow 0$ and conversely if f is a bounded function such that $\bar{\omega}_f(\delta) \rightarrow 0$ as $\delta \rightarrow 0$ then f can be normalized so that it is in $D[0, 1]$. Compact subsets of $D[0, 1]$ are characterized as sets of functions with a uniform bound and the modulus of continuity going to 0 uniformly as $\delta \rightarrow 0$. This is very similar to the Ascoli-Arzelà theorem. We refer to [5] for a proof of these facts.

A stochastic process $x(t)$ is uniformly conditionally stochastically continuous if for each $\epsilon > 0$ there exists a nonrandom function $\psi_\epsilon(h) \downarrow 0$ as $h \downarrow 0$ such that for all t, h and almost all ω

$$(A.6) \quad P[|x(t+h) - x(t)| \geq \epsilon \mid \mathcal{F}_t] \leq \psi_\epsilon(h)$$

THEOREM A.6 *Let $x(t)$ be a uniformly conditionally stochastically continuous process. Then it can be realized in the space $D[0, 1]$.*

REMARK A.6. If we have a Markov process with values in $\mathbb{R}$ or $\mathbb{R}^d$ with transition probability function $p(t, x, A)$ then the uniform conditional stochastic continuity reduces to

$$\lim_{t \rightarrow 0} \sup_x p(t, x, \{y : |y - x| \geq \epsilon\}) = 0$$

REMARK A.7. If $x(t)$ is a process with stationary independent increments the condition reduces to

$$\lim_{t \rightarrow 0} P[|x(t)| \geq \epsilon] = 0$$

PROOF: The proof is carried out in several steps. The first step is an improvement of the estimate.

$$(A.7) \quad P\left[\sup_{0 \leq s \leq h} |x(t+s) - x(t)| \geq \epsilon \mid \mathcal{F}_t\right] \leq \frac{\psi_{\frac{\epsilon}{2}}(h)}{1 - \psi_{\frac{\epsilon}{2}}(h)}$$

This is really routine. There is a classical result of Lévy that states if $S_n = X_1 + X_2 + \dots + X_n$ are partial sums of independent random variables and

$$\sup_{1 \leq i < j \leq n} P[|S_i - S_j| \geq \epsilon] \leq p$$

then

$$P\left[\sup_{1 \leq j \leq n} |S_j| \geq 2\epsilon\right] \leq \frac{p}{1 - p}$$

The proof runs as follows: let $F_j = \{|S_1| < 2\epsilon, \dots, |S_{j-1}| < 2\epsilon, |S_j| \geq 2\epsilon\}$. Then

$$\begin{aligned} P\left[\sup_{1 \leq j \leq n} |S_j| \geq 2\epsilon \cap |S_n| \leq \epsilon\right] &= \sum_{j=1}^{n-1} P[F_j \cap |S_n| \leq \epsilon] \\ &\leq \sum_j P[F_j] P[|S_n - S_j| \geq \epsilon] \\ &\leq p \sum_j P[F_j] \\ (A.8) \quad &= p P\left[\sup_{1 \leq j \leq n} |S_j| \geq 2\epsilon\right] \end{aligned}$$

$$(A.9) \quad P\left[\sup_{1 \leq j \leq n} |S_j| \geq 2\epsilon \cap |S_n| > \epsilon\right] \leq P[|S_n| > \epsilon] \leq p$$

Adding (A.8) and (A.9) we get

$$P\left[\sup_{1 \leq j \leq n} |S_j| \geq 2\epsilon\right] \leq p P\left[\sup_{1 \leq j \leq n} |S_j| \geq 2\epsilon\right] + p$$

which yields the desired inequality. If X_i are not independent one can always replace probabilities by uniform conditional bounds.

$$P[F_j \cap |S_n - S_j| \geq \epsilon] \leq P[F_j] \times \sup_{\omega} P[|S_n - S_j| \geq \epsilon \mid \mathcal{F}_j]$$

This will establish (A.7). The $D[0, 1]$ modulus of continuity is the maximum of three terms two of which, oscillations near 0 and 1, are controlled by (A.3). We now try and control the oscillations after allowing for one jump.

Let us define τ_1 as the first time $|x(t) - x(0)|$ exceeds ϵ , $\tau + \tau_2$ the first time $|x(t) - x(\tau_1)|$ exceeds ϵ and so on. The oscillation of the function in any interval $(\tau_1 + \dots + \tau_{k-1}, \tau_1 + \dots + \tau_k)$ is at most 2ϵ . Just like the strong Markov property the uniform conditional stochastic continuity estimate holds for stopping times as well. Therefore τ_i are not too small.

$$P[\tau_{i+1} \leq h \mid \mathcal{F}_{\tau_1 + \dots + \tau_i}] \leq \frac{\psi_{\frac{\epsilon}{2}}(h)}{1 - \psi_{\frac{\epsilon}{2}}(h)}$$

This will provide an estimate of the form

$$E[e^{-\tau_{i+1}} \mid \mathcal{F}_{\tau_1 + \dots + \tau_i}] \leq \rho < 1$$

Let $\tau_1 + \tau_2 + \dots + \tau_k > 1$, and $\min(\tau_1, \tau_2, \dots, \tau_k) > h$. Then, any interval I of length at most h can have at most one of the points $\{\tau_1 + \tau_2 + \dots + \tau_j\}$ inside. Therefore

$$\bar{\omega}_{x(\cdot)}(h) \leq 2\epsilon$$

Clearly

$$P[\min(\tau_1, \dots, \tau_k) \leq h] \leq k \frac{\psi_{\frac{\epsilon}{2}}(h)}{1 - \psi_{\frac{\epsilon}{2}}(h)}$$

and all we need to do is control $P[\tau_1 + \dots + \tau_k \leq 1]$.

$$P[\tau_1 + \dots + \tau_k \leq 1] \leq e E[e^{-\sum_{i=1}^k \tau_i}] \leq e \rho^k$$

REMARK A.8. If a process can be realized in $D[0, 1]$ to show that is in fact supported on $C[0, 1] \subset D[0, 1]$ is often simple. We need to compute the expected number of jumps of size larger than ϵ and show that it is 0. This requires

$$\lim_{n \rightarrow \infty} \sum_{j=1}^n P \left[\left| x \left(\frac{j+1}{n} \right) - x \left(\frac{j}{n} \right) \right| \geq \epsilon \right] = 0$$

An estimate, for each $\epsilon > 0$, of the form $P[|x(t+h) - x(t)| \geq \epsilon] = o(h)$ will do. $\square$

REMARK A.9. If (A.9) holds uniformly over a family, is that family of processes a compact subset of $\mathcal{M}$? We have control over the modulus of continuity. But the modulus of continuity allows one jump in small interval. It does not control the size of the jump. Either we have to control the size of the jumps or we need to make sure that we can control the tail probabilities $P[|x(t)| \geq A]$ uniformly.

EXERCISE A.1. What is the connection to Lindeberg conditions for the central limit theorem?

APPENDIX B

Additional Remarks

These notes contain material usually covered in the second semester of a graduate probability course. There are many possible additional sources that can be used to supplement the material presented here. For the study of probability measures on function spaces, the treatment in Appendix A is very cursory and [5] is an excellent source. For the study of Markov chains when the state space is countable, the classic source is [1]. For Brownian motion and one-dimensional diffusions [2] and [4] are fine additional sources. For those interested proceeding further into theory of Diffusion processes, especially in higher dimensions, [6] can be a source. It is the hope of the author that in the not-too-distant future an additional volume in the series will appear covering some more advanced material.

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Stochastic Processes

S. R. S. VARADHAN

This is a brief introduction to stochastic processes studying certain elementary continuous-time processes. After a description of the Poisson process and related processes with independent increments as well as a brief look at Markov processes with a finite number of jumps, the author proceeds to introduce Brownian motion and to develop stochastic integrals and Itô's theory in the context of one-dimensional diffusion processes. The book ends with a brief survey of the general theory of Markov processes.

The book is based on courses given by the author at the Courant Institute and can be used as a sequel to the author's successful book *Probability Theory* in this series.



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